5G RAN

Mobility Load Balancing in Connected Mode Feature Parameter Description

Issue Draft A

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for transfer of downlink CA UEs by the following connected mode MLB functions: • UE-number-based connected mode MLB • Spectral-efficiency-based connected mode MLB • Inter-vendor UE-number-based connected mode MLB Details of this change are provided in the corresponding sections for each function. For example, the change related to UE-number-based connected mode MLB is detailed in: • **5.1.5.1.1 Conditions Used for Initial Selection** • **5.1.5.2.2 Serving Cell Determination** • **5.3.2.3 Function Impacts** • **5.4.1.1 Data Preparation** • **5.4.1.2 Using MML Commands**	Added the **NRCellMlb**.***MlbTransCA*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for intra-sector UE-number-based connected mode MLB by LampSite base stations. For details, see **9.3.3 Hardware**.	None	High-frequency TDD	DBS5900 LampSite
Added support for inter-frequency MLB for mmWave PCells. For details, see: • **9.1 Principles** • **9.2 Network Analysis** • **9.3 Requirements** • **9.4 Operation and Maintenance**	Modified parameters: • Added the **MMW_PCELL_INTER_FREQ_MLB_SW** option to the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter. • Added support for the **NRCellMlb**.***InterFreqMlbUlUeNumThld***, **NRCellMlb**.***MlbUlUeNumOffset***, and **NRCellMlb**.***ConnMlbUeNumThld*** parameters in high-frequency TDD.	High-frequency TDD	• 5900 series base stations (macro base stations) • DBS5900 LampSite

- Editorial changes

  Added descriptions of the mutually exclusive relationship between cell deployment in IAB mode and load balancing. For details, see **5.3.2.2 Mutually Exclusive Functions**, **6.3.2.2 Mutually Exclusive Functions**, **8.3.2.2 Mutually Exclusive Functions**, **10.3.2.2 Mutually Exclusive Functions**, and **7.3.2.2 Mutually Exclusive Functions**.

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FOFD-051212	Converged Load Balancing	6 Spectral-Efficiency-based Connected Mode MLB 8 Radio-Resource-based Connected Mode MLB

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
UE-number-based connected mode MLB	Supported in both SA and NSA networking. A handover involves a serving cell change in SA networking and a primary secondary cell (PSCell) change in NSA networking.	5 UE-Number-based Connected Mode MLB
Spectral-efficiency-based connected mode MLB	Supported in both SA and NSA networking. A handover involves a serving cell change in SA networking and a PSCell change in NSA networking.	6 Spectral-Efficiency-based Connected Mode MLB
5QI-specific UE-number-based connected mode MLB	Supported only in SA networking	7 5QI-specific UE-Number-based Connected Mode MLB
Radio-resource-based connected mode MLB	Supported in both SA and NSA networking. A handover involves a serving cell change in SA networking and a PSCell change in NSA networking.	8 Radio-Resource-based Connected Mode MLB
Intra-sector UE-number-based connected mode MLB (high-frequency TDD)	Supported in both SA and NSA networking. A handover involves a serving cell change in SA networking and a PSCell change in NSA networking.	9 Intra-Sector UE-Number-based Connected Mode MLB (High-Frequency TDD)

Function Name	Difference	Chapter/Section
UE-number-based MLB in frequency rotation mode	Supported in both NSA and SA networking, with the following differences: • A handover involves a serving cell change in SA networking and a PSCell change in NSA networking. • In NSA and SA hybrid networking and SA networking, handover- and blind redirection-based FR1-to-FR2 frequency selection in rotation mode is supported for SA UEs. In NSA and SA hybrid networking and NSA networking, only handover-based FR1-to-FR2 frequency selection in rotation mode is supported for NSA UEs.	**10 UE-Number-based MLB in Frequency Rotation Mode**
Inter-vendor connected mode MLB	Supported in both SA and NSA networking. A handover involves a serving cell change in SA networking and a PSCell change in NSA networking.	**11 Inter-Vendor Connected Mode MLB**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
UE-number-based connected mode MLB	Supported only in low frequency bands	**5 UE-Number-based Connected Mode MLB**
Spectral-efficiency-based connected mode MLB	Supported only in low frequency bands	**6 Spectral-Efficiency-based Connected Mode MLB**

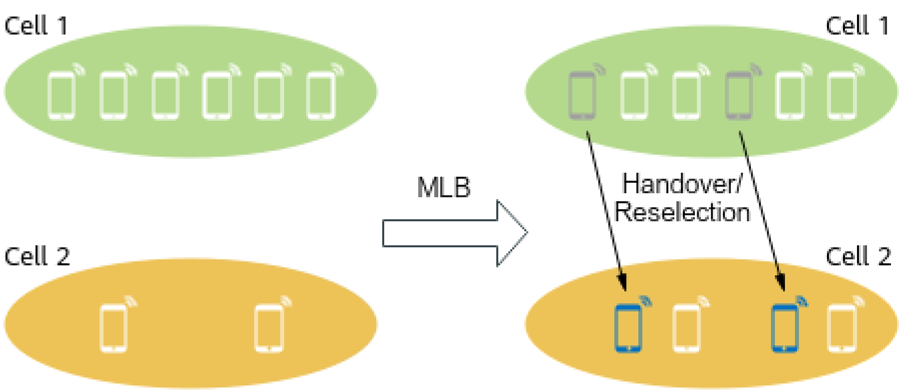
Function Name	Difference	Chapter/Section
5QI-specific UE-number-based connected mode MLB	Supported only in low frequency bands	**7 5QI-specific UE-Number-based Connected Mode MLB**
Radio-resource-based connected mode MLB	Supported only in low frequency bands	**8 Radio-Resource-based Connected Mode MLB**
Intra-sector UE-number-based connected mode MLB (high-frequency TDD)	Supported only in high frequency bands	**9 Intra-Sector UE-Number-based Connected Mode MLB (High-Frequency TDD)**
UE-number-based MLB in frequency rotation mode	Supported only in handovers or blind redirections from FR1 cells to FR2 cells	**10 UE-Number-based MLB in Frequency Rotation Mode**
Inter-vendor connected mode MLB	Supported only in low frequency bands	**11 Inter-Vendor Connected Mode MLB**

3 Overview

Among inter-frequency co-coverage cells in a 5G network, some cells may be heavily loaded while others are lightly loaded, which causes load imbalance. This imbalance will affect UE performance in high-load cells and waste resources in low-load ones.

To address this issue, mobility load balancing (MLB) is introduced. As shown in **Figure 3-1**, MLB balances cell loads by transferring UEs in connected or UEs in idle/inactive mode from high-load cells to low-load ones through handovers or reselections. This improves the resource utilization and capacity of cells involved in MLB.

Figure 3-1 MLB



NOTE

Unless otherwise specified, a handover in this document refers to a serving cell change in SA networking and a PSCell change in NSA networking.

4 General Principles

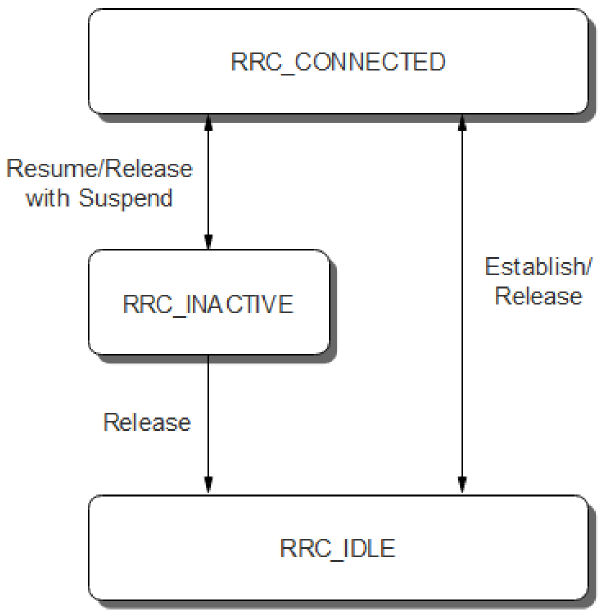
This chapter describes the general principles of MLB.

4.1 Basic Concepts

UE States

In a 5G network, a UE is in one of the following radio resource control (RRC) states: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states. **Figure 4-1** illustrates the transitions between these RRC states.

Figure 4-1 RRC states



MLB Mode

MLB can be classified into the connected mode MLB and idle/inactive mode MLB functions to work on UEs in different states:

- Connected mode MLB: This function transfers UEs in connected mode from high-load cells to low-load ones through handovers to balance cell loads. For details about handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.
- Idle/Inactive mode MLB: UEs in idle mode and inactive mode do not contribute to the cell load. This is because no RRC connection is set up between UEs in idle mode and the gNodeB and data processing for UEs in inactive mode is suspended. Idle/Inactive mode MLB transfers UEs in idle mode and inactive mode from high-load cells to low-load ones through reselections. In this way, UEs will not enter connected mode in the high-load cells, and thus will not further increase load of the high-load cells. For details about cell reselection, see *SA Mobility Management in Idle Mode*.

  NOTE

  Both idle mode MLB and inactive mode MLB are implemented through cell reselection, the principles of which are the same. This document uses idle mode MLB as an example to offer corresponding descriptions. For details about whether UEs in inactive mode support certain idle mode MLB functions, see **4.3.3 MLB Overview**.

MLB Direction

MLB involves the following directions:

- Uplink MLB: performed based on the uplink load of a cell only. Uplink MLB only transfers uplink UEs from high-load cells to low-load cells.
- Downlink MLB: performed based on the downlink load of a cell only. Downlink MLB only transfers downlink UEs from high-load cells to low-load cells.
- Uplink and downlink MLB: performed based on both the uplink and downlink loads of a cell. Uplink and downlink MLB transfer both uplink and downlink UEs from high-load cells to low-load cells.

NOTE

For the definitions of uplink and downlink loads, see **4.2 Load Evaluation Dimensions**. For the definitions of uplink and downlink UEs, see **4.2.1 Number of UEs**.

4.2 Load Evaluation Dimensions

The load of a cell is evaluated from the following dimensions:

- **4.2.1 Number of UEs**
- **4.2.2 Air Interface Load of a Cell**
- **4.2.3 Number of 5QI-specific UEs**
- **4.2.4 5QI-specific UE-Number-based Air Interface Load**
- **4.2.5 Equivalent CCE Usage**

During the MLB procedure, the gNodeB evaluates the load of a cell based on one or more load evaluation dimensions.

4.2.1 Number of UEs

The number of UEs in RRC_CONNECTED mode in a cell is used to measure the cell load:

- Number of UEs in the downlink in a cell, including:
  - Average number of UEs in RRC_CONNECTED mode in a cell (including non-CA UEs and CA UEs that treat the local cell as their PCell), which can be observed using the **N.User.RRCConn.Avg** counter
  - Average number of downlink CA UEs that treat the local cell as their SCell, which can be observed using the **N.User.CA.SCell.DL.Avg** counter
- Number of UEs in the uplink in a cell, including:
  - Average number of UEs in RRC_CONNECTED mode in a cell (including non-CA UEs and CA UEs that treat the local cell as their PCell), which can be observed using the **N.User.RRCConn.Avg** counter
  - Average number of uplink CA UEs that treat the local cell as their SCell, which can be observed using the **N.User.CA.SCell.UL.Avg** counter
  - Average number of UEs that treat the local cell as their SUL cell, which can be observed using the **N.User.SUL.UL.Avg** counter

NOTE

As UEs in idle mode and inactive mode have no impact on cell load, the number of UEs in connected mode, instead of those in idle or inactive mode, in a cell is still used to measure the cell load in idle mode MLB.

Unless otherwise stated, the number of UEs defined in this section applies throughput this document.

4.2.2 Air Interface Load of a Cell

The air interface load of a cell reflects the spectral efficiency of the cell. The air interface load of a cell is calculated as follows:

- Downlink air interface load of a cell = Number of downlink UEs in the cell/ Downlink air interface capability of the cell
- Uplink air interface load of a cell = Number of uplink UEs in the cell/Uplink air interface capability of the cell

The air interface capability of a cell is related to the load evaluation policy used for MLB:

- Static evaluation (The **NRCellMlb**.***LoadEvaluationPolicy*** parameter is set to **STATIC_EVAL**.)
  - Downlink air interface capability of a cell = (Cell bandwidth in the unit of MHz x Proportion of downlink slots in the cell x Downlink cell capability scale factor)/1000. (Downlink cell capability scale factor is specified by the **NRCellMlb**.***CellDlCapbScaleFactor*** parameter.)
  - Uplink air interface capability of a cell = (Cell bandwidth in the unit of MHz x Proportion of uplink slots in the cell x Uplink cell capability scale factor)/1000. (Uplink cell capability scale factor is specified by the **NRCellMlb**.***CellUlCapbScaleFactor*** parameter.)
- Dynamic evaluation (The **NRCellMlb**.***LoadEvaluationPolicy*** parameter is set to **DYNAMIC_EVAL**.)

– Downlink air interface capability of a cell = (Downlink cell traffic volume x Number of available PRBs for the downlink data channel/Number of PRBs actually occupied by the downlink data channel)/100000
– Uplink air interface capability of a cell = (Uplink cell traffic volume x Number of available PRBs for the uplink data channel/Number of PRBs actually occupied by the uplink data channel)/100000

The numbers of available PRBs for the uplink and downlink data channels are indicated by the values of **N.PRB.UL.Avail.Avg** and **N.PRB.DL.Avail.Avg**. However, in dynamic spectrum sharing (DSS) scenarios or other scenarios alike, the number of available PRBs in a cell dynamically changes, and the values of these two counters do not dynamically change. As a result, the load evaluation results are inaccurate. Precise load evaluation is recommended for this issue. After precise load evaluation is enabled, the gNodeB uses the values of **N.PRB.UL.Actual.Avail.Avg** and **N.PRB.DL.Actual.Avail.Avg** to indicate the number of available PRBs for the uplink and downlink data channels, making load evaluation more accurate. Precise load evaluation is controlled by the **PRECISE_LOAD_EVAL_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter.

If the number of UEs in a cell is small (less than 20) or the PRB usage of the cell is low (less than 20% in the uplink or less than 10% in the downlink) for a long time, static evaluation is recommended. In other scenarios, dynamic evaluation is recommended. When the dynamic evaluation policy is used and the number of UEs or the PRB usage is small in a short period, the gNodeB automatically switches to the static evaluation policy to calculate the air interface capability of a cell.

NOTE

- Proportion of downlink slots in a cell: NR TDD: Proportion of downlink slots in a cell = (Number of downlink-only slots + Number of self-contained slots)/(Number of downlink-only slots + Number of self-contained slots + Number of uplink-only slots). For example, if the slot configuration of a cell is 4:1, the proportion of downlink slots in the cell is 0.8 (4/5).
- Proportion of uplink slots in a cell: NR TDD: Proportion of uplink slots in a cell = Number of uplink-only slots/(Number of downlink-only slots + Number of self-contained slots + Number of uplink-only slots). For example, if the slot configuration of a cell is 4:1, the proportion of uplink slots in the cell is 0.2 (1/5).

4.2.3 Number of 5QI-specific UEs

The number of 5QI-specific UEs refers to the number of UEs running services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell.

- Number of 5QI-specific downlink UEs: Number of UEs running downlink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell
- Number of 5QI-specific uplink UEs: Number of UEs running uplink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell

NOTE

If inter-gNodeB MLB is triggered, the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter must be set to **YES** on the gNodeB serving the inter-gNodeB neighboring cells for the 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter of the serving cell.

4.2.4 5QI-specific UE-Number-based Air Interface Load

- 5QI-specific UE-number-based downlink air interface load = Number of UEs running downlink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell/Downlink air interface capability of a cell

  Downlink air interface capability of a cell = Downlink cell traffic volume x Number of available PRBs on the downlink data channel/Number of PRBs occupied by the downlink data channel
- 5QI-specific UE-number-based uplink air interface load = Number of UEs running uplink services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell/Uplink air interface capability of a cell

  Uplink air interface capability of a cell = Uplink cell traffic volume x Number of available PRBs on the uplink data channel/Number of PRBs occupied by the uplink data channel

4.2.5 Equivalent CCE Usage

The equivalent CCE usage is calculated as follows:

- Downlink equivalent CCE usage = **N.CCE.DLDCI.Used.Equivalent**/(**N.CCE.Avail.Equivalent** - **N.CCE.ULDCI.Avail.Equivalent**)

  That is: Downlink equivalent CCE usage = Number of PDCCH CCEs whose power is equivalent to the reference power and that are used by the downlink DCI/(Number of available PDCCH CCEs with power equivalent to the reference power - Number of PDCCH CCEs whose power is equivalent to the reference power and that are available for the uplink DCI)
- Uplink equivalent CCE usage = **N.CCE.ULDCI.Used.Equivalent**/**N.CCE.ULDCI.Avail.Equivalent**

  That is: Uplink equivalent CCE usage = Number of PDCCH CCEs whose power is equivalent to the reference power and that are used by the uplink DCI/Number of PDCCH CCEs whose power is equivalent to the reference power and that are available for the uplink DCI

4.3 MLB Procedure

4.3.1 Connected Mode MLB

The connected mode MLB procedure varies depending on the scenario:

- Low-frequency scenarios: See **4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)**.
- High-frequency TDD scenarios: See **4.3.1.2 Connected Mode MLB (in High-Frequency TDD Scenarios)**.
- From low-frequency TDD to high-frequency TDD scenarios: See **4.3.1.3 Connected Mode MLB (From Low-Frequency to High-Frequency TDD Scenarios)**.

4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)

Connected mode MLB in low-frequency scenarios is controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the

NRCellAlgoSwitch.***MlbAlgoSwitch*** parameter, with the **NRCellMlb**.***MlbTriggerMode*** parameter specifying the MLB triggering mode. During an evaluation period specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter, the gNodeB determines whether a cell enters or exits the MLB state in the next period based on the cell load. Different connected mode MLB functions use different conditions to determine whether a cell enters or exits the MLB state. For details, see the principles of the corresponding functions.

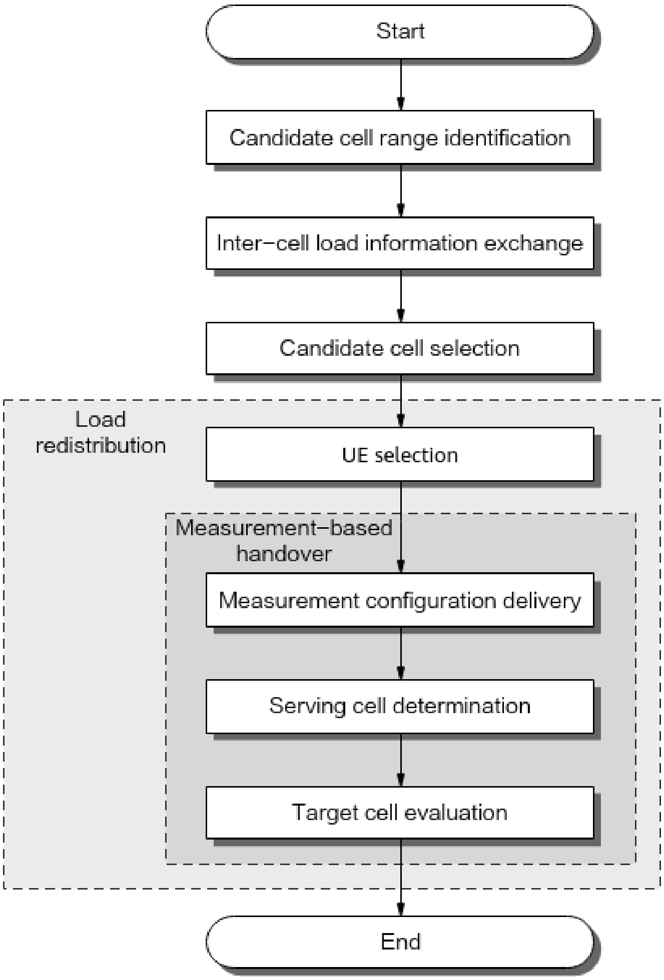
The MLB direction is configured using the **NRCellMlb**.***MlbDirection*** parameter.

- If the **NRCellMlb**.***MlbDirection*** parameter is set to **DOWNLINK**, downlink MLB is supported.
- If the **NRCellMlb**.***MlbDirection*** parameter is set to **UPLINK**, uplink MLB is supported.
- If the **NRCellMlb**.***MlbDirection*** parameter is set to **DUAL**, both downlink and uplink MLB are supported.

If the gNodeB determines that a cell meets the conditions for entering or exiting the MLB state in the current MLB evaluation period, the cell enters or exits the MLB state in the next MLB evaluation period.

After a cell enters the MLB state, connected mode MLB is performed according to the procedure as shown in **Figure 4-2**. Different connected mode MLB functions have different procedures. For details, see the principles of the corresponding functions.

Figure 4-2 Basic connected mode MLB procedure



The basic connected mode MLB procedure includes the following steps:

1. Candidate cell range identification: The gNodeB identifies the candidate cell range for UE handovers
2. Inter-cell load information exchange: The gNodeB triggers load information exchange between the serving cell and candidate cells, including intra- and inter-base-station load information exchange.
3. Candidate cell selection: The gNodeB compares the load information of the serving cell with that of all candidate cells, and it proceeds to load redistribution if there is any candidate cell that meets the load redistribution conditions.
4. Load redistribution:
   a. UE selection: UEs for load redistribution are selected based on certain conditions.
   b. Measurement-based handovers:
      i. Measurement configuration delivery: As for each selected UE, the gNodeB delivers the frequencies to be measured and the information related to event A4 or A5 to the UE. The UE performs measurements and sends a measurement report to the gNodeB when the event entering conditions are met.

ii. Serving cell determination: The gNodeB checks whether the reported cells are within the candidate cell range for load redistribution and determines whether to instruct the UE to perform a handover.

iii. Target cell evaluation: The target cell evaluates whether to allow MLB-based incoming handovers. If yes, the UE is handed over to the target cell, and the MLB procedure ends.

NOTE

Any step fails within an MLB evaluation period will cause the MLB procedure to fail for this period. And another round of procedure cannot be started until the next period begins.

In low frequency scenarios, UE-number-based connected mode MLB is the basis for connected mode MLB, and other functions are extended variations based on it. Table 4-1 describes the differences between the MLB procedure of other functions and that of this basic one. In the table, "√" indicates that this step for this function is the same as that for UE-number-based connected mode MLB, "×" indicates that this step for this function is different from that for UE-number-based connected mode MLB, and "-" indicates that this step is not involved in this function.

Table 4-1 Differences between the MLB procedure of other functions and that of UE-number-based connected mode MLB

Function Name	Entering and Exiting MLB	Candidate Cell Range Identification	Inter-Cell Load Information Exchange	Candidate Cell Selection	UE Selection	Handover Measurement	Serving Cell Determination	Target Cell Evaluation
Spectral-efficiency-based connected mode MLB	√	√	√	×	×	√	×	√
5QI-specific UE-number-based connected mode MLB	×	×	×	×	×	√	×	-
Radio-resource-based connected mode MLB	×	√	×	×	×	√	×	√

4.3.1.2 Connected Mode MLB (in High-Frequency TDD Scenarios)

In high-frequency TDD scenarios, only intra-sector UE-number-based connected mode MLB is supported. This function is controlled by the

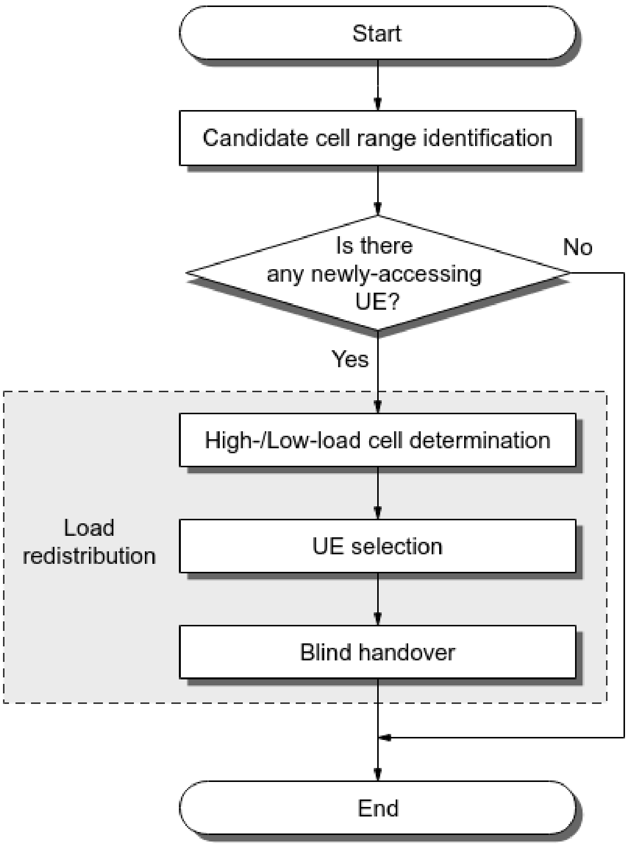
INTER_FREQ_CONNECTED_MLB_SW option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter and requires that the **NRCellMlb**.***MlbTriggerMode*** parameter be set to **INTRA_SECTOR_CELL_USER_NUM**. During an evaluation period specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter, the gNodeB determines whether a cell enters or exits the MLB state in the next period based on the number of UEs in connected mode in a sector.

Intra-sector UE-number-based connected mode MLB supports only uplink MLB. Therefore, the **NRCellMlb**.***MlbDirection*** parameter must be set to **UPLINK**.

If the gNodeB determines that a cell meets the conditions for entering or exiting the MLB state in the current MLB evaluation period, the cell enters or exits the MLB state in the next MLB evaluation period.

After a cell enters the MLB state, connected mode MLB is performed according to the procedure as shown in **Figure 4-3**.

Figure 4-3 Procedure for intra-sector UE-number-based connected mode MLB



The basic procedure for intra-sector UE-number-based connected mode MLB includes the following steps:

1. Candidate cell range identification: The gNodeB identifies the candidate cell range for UE handovers.
2. Determination of whether there is any newly-accessing UE: When a new UE accesses a cell in a sector, load redistribution is triggered.
3. Load redistribution:

a. High-/Low-load cell determination: The gNodeB determines high-load and low-load cells in the sector.
b. UE selection: The gNodeB selects UEs to be handed over from high-load cells in the sector.
c. Blind handover: The selected UEs are blindly handed over to low-load cells in the sector.

For details about the procedure for intra-sector UE-number-based connected mode MLB, see **9.1 Principles**.

4.3.1.3 Connected Mode MLB (From Low-Frequency to High-Frequency TDD Scenarios)

During handovers or blind redirections from low-frequency (FR1) cells to high-frequency (FR2) cells, if one FR2 cell in a sector is selected as the target cell for most UEs, the load of this cell is higher than that of other cells in the sector. As a result, load imbalance occurs between the cells. UE-number-based MLB in frequency rotation mode selects different frequencies in rotation mode for UEs to perform handovers or blind redirections. In this way, UEs are evenly distributed among cells in the same sector after they are transferred from a low-frequency cell to high-frequency cells, achieving inter-cell load balancing.

For details about the procedure for UE-number-based MLB in frequency rotation mode, see **10.1 Principles**.

4.3.2 Idle Mode MLB

For details about the procedure for idle mode MLB, see *Mobility Load Balancing in Idle Mode*.

4.3.3 MLB Overview

MLB mainly provides the following functions:

- Connected mode MLB (as shown in **Table 4-2**)

  In low-frequency scenarios, load evaluation is performed twice during connected mode MLB.

  – First load evaluation: Before entering or exiting MLB, the gNodeB evaluates the load of the serving cell and determines whether to enter or exit MLB based on the relationship between the load of the serving cell and the corresponding threshold.
  – Second load evaluation: During candidate cell selection, the gNodeB evaluates the load of the serving cell and all candidate cells, selects the target candidate cell for load redistribution based on the load difference, and calculates the maximum number of UEs allowed for load redistribution.
- Idle mode MLB (see *Mobility Load Balancing in Idle Mode*)

Table 4-2 Connected mode MLB overview

MLB Function	RAT	Dimension of the First Load Evaluation	Dimension of the Second Load Evaluation	MLB Procedure	Support Inter-Vendor Connected Mode MLB or Not
5 UE-Number-based Connected Mode MLB	Low-frequency TDD	Number of UEs	Number of UEs	**4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)**	Yes
6 Spectral-Efficiency-based Connected Mode MLB	Low-frequency TDD	Number of UEs	Air interface load of a cell	**4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)**	No
7 5QI-specific UE-Number-based Connected Mode MLB	Low-frequency TDD	Number of 5QI-specific UEs	5QI-specific UE-number-based air interface load	**4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)**	No
8 Radio-Resource-based Connected Mode MLB	Low-frequency TDD	Cell air interface load and equivalent CCE usage	Cell air interface load and equivalent CCE usage	**4.3.1.1 Connected Mode MLB (in Low-Frequency Scenarios)**	No

MLB Function	RAT	Dimension of the First Load Evaluation	Dimension of the Second Load Evaluation	MLB Procedure	Support Inter-Vendor Connected Mode MLB or Not
9 Intra-Sector UE-Number-based Connected Mode MLB (High-Frequency TDD)	High-frequency TDD	/	/	**4.3.1.2 Connected Mode MLB (in High-Frequency TDD Scenarios)**	No
10 UE-Number-based MLB in Frequency Rotation Mode	Low-frequency and high-frequency TDD	/	/	**4.3.1.3 Connected Mode MLB (From Low-Frequency to High-Frequency TDD Scenarios)**	No

NOTE

Inter-vendor connected mode MLB refers to the scenario where one of the base stations is provided by another vendor during inter-base-station MLB. Inter-vendor connected mode MLB differs from intra-vendor connected mode MLB in certain procedures. For details, see **11 Inter-Vendor Connected Mode MLB**.

5 UE-Number-based Connected Mode MLB

5.1 Principles

The UE-number-based connected mode MLB function uses the number of UEs in a cell as an indicator of the cell load. With this function, when cells enter the MLB state, UEs in connected mode are handed over from high-load cells to low-load ones to achieve load balancing. UE-number-based connected mode MLB is enabled if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected and the **NRCellMlb**.***MlbTriggerMode*** parameter is set to **CELL_USER_NUM**.

It is recommended that this function be enabled when all of the following conditions are met:

- The same proportions of small-packet and large-packet services are independently running in inter-frequency co-coverage cells.
- Most UEs are motionless or move slowly.
- The source and target cells are intra-base-station cells. Alternatively, the cells are served by Huawei base stations, and an Xn interface has been set up between the base stations.
- Inter-frequency cells involved in MLB are configured with the same bandwidth.

If connected mode MLB is required between cells served by Huawei base stations and non-Huawei base stations, inter-vendor connected mode MLB can be enabled. For details, see **11 Inter-Vendor Connected Mode MLB**.

5.1.1 Entering and Exiting MLB

After this function is enabled, the gNodeB determines whether a cell enters or exits the MLB state in the next MLB evaluation period based on the cell load within the current period.

Based on the direction, MLB includes downlink MLB, uplink MLB, as well as downlink and uplink MLB.

- Downlink MLB

- Entering condition: The number of UEs in the downlink is greater than or equal to the sum of the inter-frequency MLB downlink UE number threshold and MLB downlink UE number offset throughout an MLB evaluation period.
- Exiting condition: The number of UEs in the downlink is less than the inter-frequency MLB downlink UE number threshold throughout an MLB evaluation period.

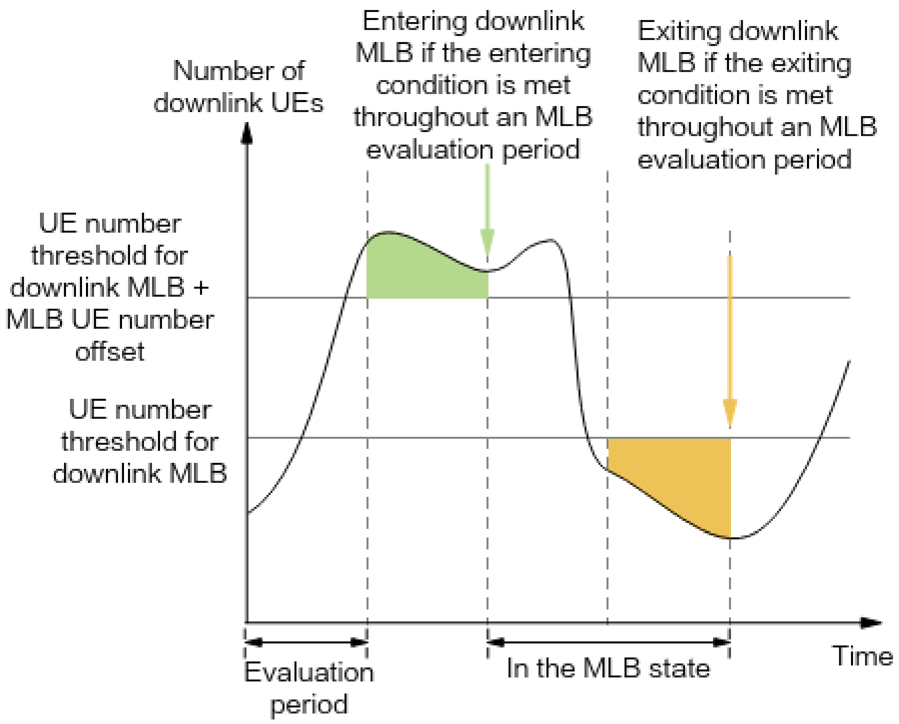
- Uplink MLB
  - Entering condition: The number of UEs in the uplink is greater than or equal to the sum of the inter-frequency MLB uplink UE number threshold and MLB uplink UE number offset throughout an MLB evaluation period.
  - Exiting condition: The number of UEs in the uplink is less than the inter-frequency MLB uplink UE number threshold throughout an MLB evaluation period.
- Uplink and downlink MLB
  - Entering condition: The entering condition for the uplink or downlink MLB state is met.
  - Exiting condition: The exiting conditions for both the uplink and downlink MLB states are met.

The following describes the parameters involved in the conditions for entering and exiting the MLB state.

- The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.
- The MLB direction is configured using the **NRCellMlb**.***MlbDirection*** parameter.
  - **DOWNLINK**: indicates that downlink MLB is enabled.
  - **UPLINK**: indicates that uplink MLB is enabled.
  - **DUAL**: indicates that uplink and downlink MLB are enabled.
- The UE number threshold for inter-frequency downlink MLB is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The offset to the number of UEs for downlink MLB is specified by the **NRCellMlb**.***MlbUeNumOffset*** parameter.
- The UE number threshold for inter-frequency uplink MLB is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The offset to the number of UEs for uplink MLB is specified by the **NRCellMlb**.***MlbUlUeNumOffset*** parameter.

If the gNodeB determines that a cell meets the conditions for entering or exiting MLB in the current MLB evaluation period, the cell enters or exits MLB in the next MLB evaluation period. **Figure 5-1** uses downlink MLB as an example and shows the conditions for entering and exiting downlink MLB.

Figure 5-1 Conditions for entering and exiting downlink MLB



NOTE

For details about the definition of the MLB direction, see **MLB Direction** in **4.1 Basic Concepts**. For details about the definitions of the number of UEs in the downlink and uplink, see **4.2.1 Number of UEs**.

5.1.2 Candidate Cell Range Identification

The gNodeB selects candidate cells for handovers. A neighboring cell that meets all of the following conditions can be selected as a candidate cell:

- The cell works on a frequency different from that of the serving cell, and is configured as a neighboring cell of the serving cell through the **NRCellRelation** MO.
- The cell allows load information exchange. That is, the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the cell.
- Any of the following conditions is met:
  - The serving cell is an SA cell, the neighboring cell is an SA cell, and SA UEs can be handed over to the neighboring cell.
  - The serving cell is an NSA cell, the neighboring cell is an NSA cell, and NSA UEs can be handed over to the neighboring cell.
  - The serving cell is a hybrid NSA-SA cell, and UEs can be handed over to the neighboring cell.

NOTE

- The SA/NSA type of the serving cell and its neighboring cells is specified by the **NRDUCellOp**.***NrNetworkingOption*** parameter. A cell is considered to be in SA or NSA networking only when the networking option of all operators of the cell is SA or NSA, respectively.
- The SA/NSA type of an external neighboring cell is specified by the **NRExternalNCell**.***NrNetworkingOption*** parameter. A cell is considered to be in SA or NSA networking only when the networking option of all operators of the cell is SA or NSA, respectively.
- Whether a neighboring cell allows incoming handovers of SA UEs and NSA UEs is specified by the **NRCellRelation**.***NoHoFlag*** parameter.

- The cell is in the active state, which is indicated by the **NR Cell State** parameter in the **DSP NRCELL** command output.

NOTE

If inter-frequency handover collaboration is enabled by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter, cells already in the connected mode MLB state will not be selected as target cells for non-MLB-based inter-frequency handovers. For details about inter-frequency handover collaboration, see *SA Mobility Management in Connected Mode*.

The total number of neighboring cells that can be selected as candidate cells is subject to whether the MLB neighboring cell specification improvement function is enabled. This function is controlled by the **MLB_NCELL_SPEC_EXTEND_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.

- When this function is disabled, a maximum of 50 cells that meet the specific conditions can be selected as candidate cells.
- When this function is enabled, all cells that meet the specific conditions can be selected as candidate cells.

5.1.3 Inter-Cell Load Information Exchange

The serving cell exchanges load information with candidate cells.

- The gNodeB triggers intra-base-station load information exchanges between the serving cell and the candidate cells by default. For details, see **5.1.3.1 Intra-gNodeB Inter-Cell Load Information Exchange**.
- If there are candidate inter-gNodeB cells, the gNodeB triggers inter-gNodeB inter-cell load information exchanges. For details, see **5.1.3.2 Inter-gNodeB Inter-Cell Load Information Exchange**.

5.1.3.1 Intra-gNodeB Inter-Cell Load Information Exchange

The exchanged load information includes:

- Number of UEs in a cell. For details, see **4.2.1 Number of UEs**.
- Proportion of UEs in RRC_CONNECTED mode in a cell, which is equal to the number of UEs in a cell divided by the maximum number of UEs in RRC_CONNECTED mode in the cell. For details about the maximum number of UEs in RRC_CONNECTED mode in a cell, see "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.

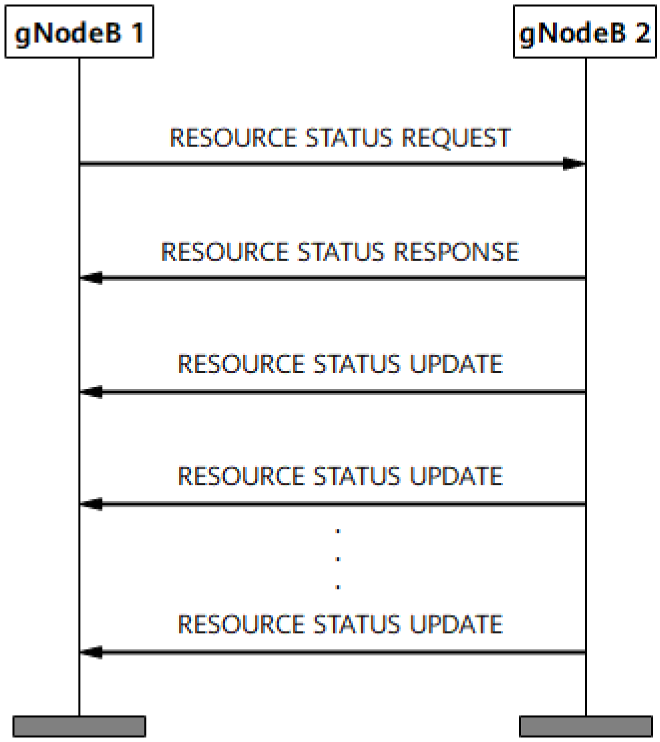
- Indication of whether the cell is in the high-load state. A cell is considered in the high-load state if the proportion of UEs in RRC_CONNECTED mode in the uplink or downlink is greater than or equal to the value of **NRCellMlb**.***MlbHoAdmissionThld***.
- Indication of whether the cell is in the interference state. A cell is in the interference state when it meets the conditions for starting uplink-interference-based inter-frequency handover. For details about uplink-interference-based inter-frequency handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.
- Uplink cell bandwidth
- Downlink cell bandwidth

5.1.3.2 Inter-gNodeB Inter-Cell Load Information Exchange

The inter-gNodeB inter-cell load information exchange procedure is performed using the method defined in section 8.4.10 "Resource Status Reporting Initiation" in 3GPP TS 38.423 V16.4.0, as shown in **Figure 5-2**. The exchanged load information includes:

- Number of UEs in a cell. For details, see **4.2.1 Number of UEs**.
- Proportion of UEs in RRC_CONNECTED mode in a cell, which is equal to the number of UEs in a cell divided by the maximum number of UEs in RRC_CONNECTED mode in the cell. For details about the maximum number of UEs in RRC_CONNECTED mode in a cell, see "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- Indication of whether the cell is in the high-load state. A cell is considered in the high-load state if the proportion of UEs in RRC_CONNECTED mode in the uplink or downlink is greater than or equal to the value of **NRCellMlb**.***MlbHoAdmissionThld***.
- Indication of whether the cell is in the interference state
- Uplink cell bandwidth
- Downlink cell bandwidth

Figure 5-2 Inter-gNodeB inter-cell load information exchange procedure



NOTE

During inter-gNodeB inter-cell load information exchange, Xn self-setup can be triggered when there is no Xn interface between gNodeBs to support later MLB procedures, in all SA, NSA, or NSA and SA hybrid networking. The **XN_SON_SETUP_SW** option of the **gNBXnSonConfig**.***XnSonConfigSwitch*** parameter specifies whether to enable Xn self-setup. For details, see *NG and Xn Self-Management*.

5.1.4 Candidate Cell Selection

The gNodeB compares the load information of the serving cell with that of all candidate cells.

If at least one candidate cell meets all of the following conditions, the gNodeB considers that there are candidate cells available for load redistribution and proceeds to it. Otherwise, the gNodeB considers that there are no candidate cells available for load redistribution and does not perform load redistribution in the current period.

- The load difference between the serving cell and the candidate cell is greater than the load difference threshold.
  - For downlink MLB

    Condition: The downlink load difference between the serving cell and the candidate cell is greater than the downlink load difference threshold.

    Downlink load difference between the serving cell and the candidate cell = (Number of UEs in the downlink in the serving cell – Number of UEs in the downlink in the candidate cell)/Number of UEs in the downlink in the serving cell. (The downlink load difference threshold is specified by the **NRCellMlb**.***UeNumDiffThld*** parameter.)

- For uplink MLB

  Condition: The uplink load difference between the serving cell and the candidate cell is greater than the uplink load difference threshold.

  Uplink load difference between the serving cell and the candidate cell = (Number of UEs in the uplink in the serving cell – Number of UEs in the uplink in the candidate cell)/Number of UEs in the uplink in the serving cell. (The uplink load difference threshold is specified by the **NRCellMlb**.***UlUeNumDiffThld*** parameter.)

- For uplink and downlink MLB

  Any of the preceding load difference conditions is met.

- The candidate cell is not in the high-load state. A cell is considered in the high-load state if the proportion of UEs in connected mode in the uplink or downlink is greater than or equal to the value of **NRCellMlb**.***MlbHoAdmissionThld***.
- The difference in the number of UEs meets the following requirements:
  - Downlink load balancing: The number of downlink UEs in the local cell minus that in the candidate cell is greater than five.
  - Uplink load balancing: The number of uplink UEs in the local cell minus that in the candidate cell is greater than five.
  - Uplink/Downlink load balancing: Either of the preceding requirements is met.
- The candidate cell is not in the interference state. A cell is in the interference state when it meets the conditions for starting uplink-interference-based inter-frequency handover. For details about uplink-interference-based inter-frequency handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

5.1.5 Load Redistribution

Load redistribution is a process in which some UEs in connected mode in the serving cell are handed over to candidate cells that meet certain signal quality requirements.

5.1.5.1 UE Selection

The procedure for selecting a UE to be handed over is as follows:

1. UEs that meet certain conditions are initially selected. For details about the conditions, see **5.1.5.1.1 Conditions Used for Initial Selection**.
2. The priorities of the initially selected UEs are determined according to UE selection policies based on radio bearer attributes. For details about UE priorities, see **5.1.5.1.2 UE Priority Determination**.
3. The upper limit (N_{max}) of the total number of UEs that can be involved in load redistribution is calculated in each MLB evaluation period. For details about the calculation, see **5.1.5.1.3 Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution**.
4. The initially selected UEs are selected for measurement-based handovers in descending order of their priorities until the upper limit (N_{max}) is reached. If multiple UEs have the same priority, the gNodeB randomly selects some of them.

5.1.5.1.1 Conditions Used for Initial Selection

Each of the selected UEs must meet all of the following conditions:

- The UE is in connected mode.
- The UE type meets all the following requirements:
  - The UE is not running a voice over NR (VoNR) service.
  - The UE is not a super uplink UE.
  - The UE is not an SUL UE.
  - For an SA UE, the **SA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected. For an NSA UE, the **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected.
- The UE meets one of the following conditions in terms of the CA state:
  - The UE is not in the CA state.
  - The UE is a downlink CA UE (configured with downlink SCC(s) only, but not configured with any uplink SCC) and has no activated SCC. In this case, the serving cell must meet all of the following conditions:
    - The **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA** for the serving cell.
    - The **NRCellMlb**.***MlbDirection*** parameter is set to either **DUAL** and the MLB procedure is triggered only by uplink MLB meeting its entering condition, or **UPLINK**.
- The UE supports MLB. UEs in any of the following scenarios do not support MLB:
  - The RAT/Frequency Selection Priority (RFSP) of the UE is configured on the core network, it matches the **gNBRfspConfig**.***RfspIndex*** parameter configured on the gNodeB, and the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter is deselected.
  - The RFSP of the UE is not configured on the core network, and the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter corresponding to the **gNBRfspConfig**.***RfspIndex*** parameter with the value of **0** is deselected for the operator.
- The UE does not have any bearer of a QoS class identifier (QCI) for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer**.***MobilityFunInd*** parameter is selected (indicating that unnecessary inter-frequency handovers are prohibited).

  NOTE

  When the setting of the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig**.***MlbAlgoSwitch*** parameter is changed:

  - For UEs that have accessed the network, the base station evaluates whether the UEs support MLB based on the original setting of the switch unless a bearer change or handover occurs.
  - For UEs that newly access the network, the base station evaluates whether the UEs support MLB based on the new setting of the switch.
- If the UE is handed over to the serving cell for MLB, the UE has been in the serving cell for at least 120s.

- The UE's last failure to be handed over to a candidate cell for MLB must be at least 120s ago.

5.1.5.1.2 UE Priority Determination

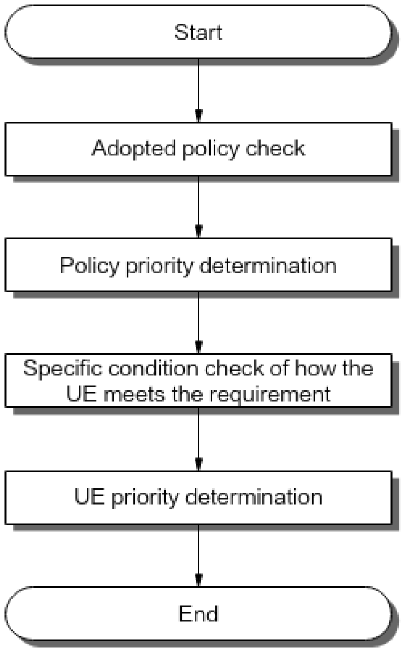
The priority of a UE is determined by checking whether the UE meets specific requirements. (For details, see **Specific Condition Check of How the UE Meets the Requirement**.) Such requirements are called the UE selection policies based on radio bearer attributes. Huawei provides five of these:

- UE selection policy based on the allocation and retention priority (ARP) threshold
- UE selection policy based on the QCI threshold
- UE selection policy based on the QCI-specific switch
- UE selection policy based on the PRB usage threshold
- UE selection policy based on the far point threshold

Huawei supports one or more policies to be used to determine the priority of a UE based on the network plan. **Figure 5-3** shows the procedure for determining the UE priority, which includes the following steps:

- Check the policies to be adopted based on the settings of the UE selection policy based on radio bearer attributes. For details, see **Adopted Policy Check**.
- Check the priorities of policies if multiple selection policies are adopted. A higher priority of a policy indicates a higher probability of selecting a UE that meets the policy requirement. For details, see **Policy Priority Determination**.
- Evaluate UEs one by one to check if each of them meets the requirements specified in the adopted UE selection policies based on radio bearer attributes. For details, see **Specific Condition Check of How the UE Meets the Requirement**. If a UE meets the requirements of more than one policy, the priority of the UE needs to be further determined. The gNodeB does not select UEs that do not meet the requirements of all of the adopted policies.
- Determine the UE priority based on whether the UE meets the specific requirements. For details, see **UE Priority Determination**.

Figure 5-3 UE priority determination



Adopted Policy Check

There are five UE selection policies based on radio bearer attributes, which are listed in descending order of default priority as follows:

- UE selection policy based on the ARP threshold: configured using the **NRCellMlb**.***ArpThldBasedUeSelPolicyPri*** parameter
- UE selection policy based on the QCI threshold: configured using the **NRCellMlb**.***QciThldBasedUeSelPolicyPri*** parameter
- UE selection policy based on the QCI-specific switch: configured using the **NRCellMlb**.***QciSwBasedUeSelPolicyPri*** parameter
- UE selection policy based on the PRB usage threshold: configured using the **NRCellMlb**.***PrbUsageThldUeSelPolicyPri*** parameter.
- UE selection policy based on the far point threshold: configured using the **NRCellMlb**.***FarPointThldUeSelPolicyPri*** parameter

The preceding parameters can be set to values ranging from **0** to **255**.

- The value **0** indicates that this policy is not considered during UE priority determination.
- The value **255** indicates that only this policy is considered during UE priority determination. If multiple parameters are set to **255**, only the policy with the highest default priority among them is considered.
- A value ranging from **1** to **254** indicates that this policy is considered during UE priority determination if no other policy is set to **255**.

If the parameters corresponding to all policies are set to **0**, UEs are not prioritized through the UE selection priority based on radio bearer attributes. In this case, all UEs have the same priority.

If multiple policies need to be considered, the default priorities of these policies become invalid. The priorities of these policies are to be further determined. For details, see **Policy Priority Determination**.

Policy Priority Determination

If multiple policies are adopted, the specific priority of each policy needs to be determined. A higher priority of a policy indicates a higher probability of selecting a UE that meets the policy requirement.

The adopted policies are sequenced in descending order based on the configured values of the parameters corresponding to the policies, so as to determine the priorities of the policies. Policies with the same parameter value have the same priority.

For example, when the selection policy parameters are configured as described in **Table 5-1**, the priorities of the policies are as follows: policy 2 > policy 5 = policy 4 > policy 1. And policy 3 is not adopted.

Table 5-1 Configurations and parameter value meanings for UE-selection-policy-based priorities

No.	Policy	Parameter	Configured Value
Policy 1	UE selection policy based on the ARP threshold	**NRCellMlb**.***ArpThldBasedUeSelPolicyPri***	50
Policy 2	UE selection policy based on the QCI threshold	**NRCellMlb**.***QciThldBasedUeSelPolicyPri***	200
Policy 3	UE selection policy based on the QCI-specific switch	**NRCellMlb**.***QciSwBasedUeSelPolicyPri***	0
Policy 4	UE selection policy based on the PRB usage threshold	**NRCellMlb**.***PrbUsageThldUeSelPolicyPri***	150
Policy 5	UE selection policy based on the far point threshold	**NRCellMlb**.***FarPointThldUeSelPolicyPri***	150

Specific Condition Check of How the UE Meets the Requirement

Each UE selection policy based on radio bearer attributes corresponds to a requirement as shown in **Table 5-2**. The gNodeB checks whether each UE meets the requirement.

Table 5-2 Criteria in UE selection policies based on radio bearer attributes

Policy	Requirement
UE selection policy based on the ARP threshold	The ARP values of all services of the UE are greater than or equal to the value of the **NRCellMlb**.***MlbUeSelectionArpThld*** parameter.
UE selection policy based on the QCI threshold	The MLB triggered by threshold switch is selected for at least the QCI of one of the services running on the UE. In this case, the gNodeB selects the UE that meets at least one of the following conditions throughout an MLB detection period and hands over the UE to a low-load cell. ● When the cell is in the downlink MLB state or uplink and downlink MLB state, the UE running services of the specific QCI is insufficiently scheduled in the downlink. That is, the downlink buffered data volume of the UE is continuously greater than the downlink buffered data volume threshold for MLB and the downlink UE throughput is continuously less than the downlink throughput threshold for MLB. ● When the cell is in the uplink MLB state or uplink and downlink MLB state, the UE running services of the specific QCI is insufficiently scheduled. That is, the uplink buffered data volume of the UE indicated by the buffer status report (BSR) is continuously greater than the uplink BSR-indicated data volume threshold for MLB and the uplink UE throughput is continuously less than the uplink throughput threshold for MLB. The involved parameters are as follows: ● The MLB triggered by threshold switch is set through the **MLB_TRIG_BY_THLD** option of the **NRCellQciBearer**.***MlbAlgoSwitch*** parameter. ● The MLB detection period is specified by the **NRCellQciBearer**.***MlbDetectTime*** parameter. ● The downlink buffered data volume threshold for MLB is specified by the **NRCellQciBearer**.***MlbDlBfrSize*** parameter. ● The downlink throughput threshold for MLB is specified by the **NRCellQciBearer**.***MlbDlThptThld*** parameter. ● The uplink BSR-indicated data volume threshold for MLB is specified by the **NRCellQciBearer**.***MlbUlBsrDataSize*** parameter. ● The uplink throughput threshold for MLB is specified by the **NRCellQciBearer**.***MlbUlThptThld*** parameter.
UE selection policy based on the QCI-specific switch	The **MLB_FILTER_SW** option of the **NRCellQciBearer**.***MlbAlgoSwitch*** parameter is deselected for all QCIs of services of the UE.

Policy	Requirement
UE selection policy based on the PRB usage threshold	● Downlink MLB: Downlink PRB usage of the UE > **NRCellMlb**.***MlbUeSelDlPrbUsageThld*** ● Uplink MLB: Uplink PRB usage of the UE > **NRCellMlb**.***MlbUeSelUlPrbUsageThld*** ● Uplink and downlink MLB: Either of the preceding conditions is met.
UE selection policy based on the far point threshold	● If the **NRCellMlb**.***MlbUeSelDistancePolicy*** parameter is set to **SEL_FAR_POINT_UE**, UEs at the cell edge are selected for load transfer. The SRS RSRP value of the selected UEs must be less than or equal to the **NRCellMlb**.***MlbUeSelectionRsrpThld*** parameter value. ● If the **NRCellMlb**.***MlbUeSelDistancePolicy*** parameter is set to **SEL_NEAR_POINT_UE**, UEs in the cell center are selected for load transfer. The SRS RSRP value of the selected UEs must be greater than the **NRCellMlb**.***MlbUeSelectionRsrpThld*** parameter value.

UE Priority Determination

After the UEs that meet the requirements for at least one policy are selected, the priorities of the selected UEs are further determined.

- A UE that meets the requirements of more policies has a higher priority.
- For UEs that meet the requirements of the same number of policies, the base station checks the priorities of these policies. A UE that meets the requirements of a higher-priority policy has a higher priority.

For example, Table 5-3 lists the configured priority values of each policy and the compliance of five UEs with each of the policies. In the table, "√" indicates that the UE meets the requirements of the policy, "×" indicates that the UE does not meet the requirements of the policy, and "-" indicates that it is not necessary to check whether the UE meets the requirements of the policy. The priorities of the five UEs in descending order are as follows: UE 1 > UE 2 = UE 3 > UE 5 > UE 4.

Table 5-3 UE priority determination

Policy	Parameter	Configured Value	UE 1	UE 2	UE 3	UE 4	UE 5
UE selection policy based on the QCI threshold	**NRCellMlb**.***QciThldBasedUeSelPolicyPri***	200	√	√	√	√	√

Policy	Parameter	Configured Value	UE 1	UE 2	UE 3	UE 4	UE 5
UE selection policy based on the far point threshold	**NRCellMlb**.***FarPointThldUeSelPolicyPri***	150	√	×	√	×	√
UE selection policy based on the PRB usage threshold	**NRCellMlb**.***PrbUsageThldUeSelPolicyPri***	150	√	√	×	×	×
UE selection policy based on the ARP threshold	**NRCellMlb**.***ArpThldBasedUeSelPolicyPri***	50	√	√	√	√	×
UE selection policy based on the QCI-specific switch	**NRCellMlb**.***QciSwBasedUeSelPolicyPri***	0	-	-	-	-	-

5.1.5.1.3 Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution

After the priorities of the selected UEs are determined, the upper limit (N_{max}) of the total number of UEs involved in load redistribution needs to be calculated. The gNodeB selects UEs in descending order of UE priority for measurement-based handovers, and the number of selected UEs cannot exceed the upper limit (N_{max}).

In downlink MLB, the upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period takes the minimum of the following three values:

- [Current number of UEs in the serving cell – (Downlink UE number threshold for inter-frequency MLB – 1)] x *T*
- MLB UE number judge factor x (Number of UEs in the downlink in the serving cell – Minimum number of UEs in the downlink in the candidate cell) x *T*
- UE number upper limit for connected mode MLB

In uplink MLB, the upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period takes the minimum of the following three values:

- [Current number of UEs in the serving cell – (Uplink UE number threshold for inter-frequency MLB – 1)] x *T*
- MLB UE number judge factor x (Number of UEs in the uplink in the serving cell – Minimum number of UEs in the uplink in the candidate cell) x *T*

- UE number upper limit for connected mode MLB

The involved parameters are as follows:

- The value of T depends on the MLB evaluation period.
  - If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is greater than or equal to 10s, *T* is 1.
  - If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is less than 10s, *T* is the value of **NRCellMlb**.***MlbJudgePeriod*** divided by 10.
- The downlink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The uplink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The MLB UE number judge factor is specified by the **NRCellMlb**.***MlbUeNumJudgeFactor*** parameter.
- The UE number upper limit for connected mode MLB is specified by the **NRCellMlb**.***ConnMlbUeNumThld*** parameter.

5.1.5.2 Measurement-based Handover

Measurement-based handovers are performed for a selected UE. The gNodeB delivers the operating frequencies of candidate cells available for load redistribution and the information related to event A4 or A5 to such a UE, and the UE performs measurements. When the event entering conditions are met, the UE sends a measurement report to the gNodeB. After receiving the measurement report, the gNodeB in the serving cell checks whether the reported cells are within the candidate cell range for load redistribution and determines whether to instruct the UE to perform a handover. During the handover, the target cell also evaluates whether to allow this UE to access its network.

For details about the basic procedure of measurement-based handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. This section describes only the parts that have been adapted to accommodate MLB, including measurement configuration delivery, serving cell determination, and target cell determination.

5.1.5.2.1 Measurement Configuration Delivery

During MLB, the gNodeB filters and selects frequencies and performs measurement event configurations when delivering measurement configurations to UEs.

- Frequency filtering requirements: If the minimum bandwidth supported by the UE is greater than the bandwidth of the candidate cell in the uplink or downlink, the operating frequency of the candidate cell will be filtered out and excluded from the measurement configuration delivered to the UE.
- Frequency selection policies:
  - Fairness policy (The **NRCellMlb**.***FreqSelectionPolicy*** parameter is set to **FAIR_POLICY**)
    - If there is only one candidate cell or all candidate cells operate on the same frequency, the gNodeB delivers the measurement configuration of only this frequency.

- If there are multiple candidate cells on different frequencies, the gNodeB delivers the measurement configurations of these neighboring frequencies sorted in ascending order of the **NRCellFreqRelation**.***SsbFreqPos*** parameter value. After receiving measurement reports, the gNodeB instructs the UE to initiate a handover to the target cell on a first-come, first-selected basis.

– Frequency-priority-based policy (The **NRCellMlb**.***FreqSelectionPolicy*** parameter is set to **FREQ_PRIORITY**): When this policy is used, the gNodeB filters out the cells on frequencies for which the **NRCellFreqRelation**.***MlbFreqPriority*** parameter is set to **0**. The remaining cells are candidate cells.

- If no cell remains after filtering, the gNodeB does not deliver the measurement configuration in the current MLB evaluation period.
- If there is only one candidate cell or all candidate cells operate on the same frequency, the gNodeB delivers the measurement configuration of only this frequency.
- If there are multiple candidate cells on different frequencies, the gNodeB delivers the measurement configurations of these neighboring frequencies by frequency priority specified by the **NRCellFreqRelation**.***MlbFreqPriority*** parameter. After receiving a measurement report, the gNodeB waits for a maximum of 2s if none of the operating frequencies of the cells in the measurement report is the highest-priority one in the frequency set. The gNodeB then selects the cell on the highest-priority frequency among those in the received reports as the target cell for a handover.

- Measurement event configuration:

– If the **NRCellMobilityConfig**.***InterFreqMeasEventConfig*** parameter is set to **EVENT_A4** for a cell, the gNodeB uses event A4 (indicating the signal quality of a neighboring cell exceeds a specific threshold) as the measurement event in the handover execution phase.

- The threshold for event A4 is specified by the **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** parameter.
- The time-to-trigger and hysteresis for event A4 are specified by the **NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig*** and **NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst*** parameters, respectively.

– If the **NRCellMobilityConfig**.***InterFreqMeasEventConfig*** parameter is set to **EVENT_A5** for a cell, the gNodeB uses event A5 (indicating the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2) as the measurement event in the handover execution phase.

- Threshold 1 for event A5 is always –43 dBm, and threshold 2 is specified by the **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** parameter.
- The time-to-trigger and hysteresis for event A5 are specified by the **NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig*** and

NRCellInterFHoMeaGrp.***InterFreqA4A5Hyst*** parameters, respectively.

5.1.5.2.2 Serving Cell Determination

After receiving the measurement report, the gNodeB determines whether the reported cells are within the candidate cell range for load redistribution. For details about the conditions involved in determination, see **5.1.4 Candidate Cell Selection**. If the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA** and the selected UE is a CA UE, the gNodeB also checks whether a candidate cell's CA configuration remains applicable to the CA UE following a handover to the candidate cell. If the configuration is no longer applicable, the candidate cell will not be selected as the target cell of the CA UE's handover.

The UE is instructed to perform a handover depending on the determination result.

- If none of the reported cells is a candidate cell for load redistribution, no handover is performed.
- If only one cell in the measurement report is a candidate cell for load redistribution, the UE is handed over to this cell.
- If multiple cells in the measurement report are candidate cells for load redistribution, the handover policy to these cells is controlled by the **ONLY_STRONGEST_CELL_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.
  - If this option is selected, the gNodeB instructs the UE to perform a handover only to the cell with the highest signal quality, and handovers to other cells will not be performed once this handover fails.
  - If this option is deselected, a handover to the cell with the highest signal quality is attempted. If the handover fails, a handover to the cell with the second highest signal quality is attempted, until the handover succeeds or handovers to all candidate cells for load redistribution have been attempted.

5.1.5.2.3 Target Cell Determination

The target cell evaluates whether to allow MLB-based incoming handovers.

1. If the proportion of UEs in the uplink or downlink in the cell is greater than or equal to the value of **NRCellMlb**.***MlbHoAdmissionThld***, then the cell rejects the incoming handovers with a cause value "No Radio Resources Available in Target Cell". Otherwise, the cell allows the incoming handovers.
2. A UE that is handed over to the cell will not be selected as a candidate UE for MLB within 120s.

NOTE

- Proportion of UEs in the uplink in the cell = Number of UEs in the uplink in a cell/ Maximum number of UEs in the uplink that the cell can serve
- Proportion of UEs in the downlink in the cell = Number of UEs in the downlink in a cell/ Maximum number of UEs in the downlink that the cell can serve

5.2 Network Analysis

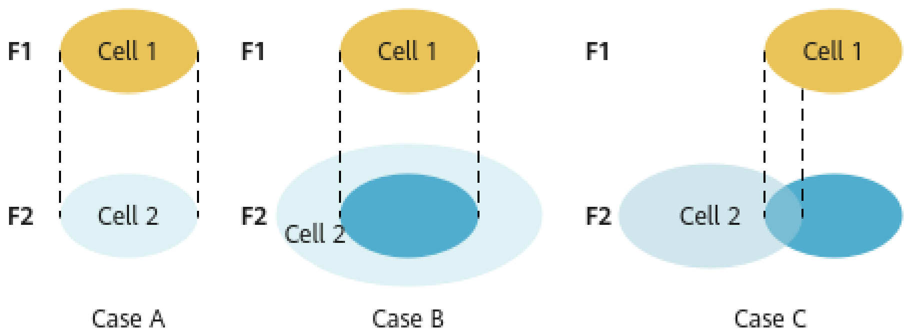
5.2.1 Benefits

UE-number-based connected mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains.

The following must also be considered when using this function:

- When UEs in the inter-frequency co-coverage cells are high-mobility UEs, this function brings neither positive gains nor negative impacts.
- The gains of this function will not be as high as expected when UEs differ greatly in their band support capabilities or there are restrictions on the serving PLMNs of UEs due to UE subscription.
- The gains of this function will not be as high as expected when cells on multiple frequencies differ greatly in their coverage capabilities, as shown in **Figure 5-4**.
  - Case A: When two inter-frequency cells (cell 1 and cell 2) cover the exact same area, the load balancing transfer between the two cells is the most efficient and effective.
  - Case B: When the coverage area of cell 2 completely includes that of cell 1, the load balancing transfer from cell 1 to cell 2 is the most efficient and effective, but the load balancing transfer from cell 2 to cell 1 cannot achieve the optimal efficiency and gains. The larger the coverage difference between the two cells, the lower the efficiency and the fewer the gains of load balancing transfer from cell 2 to cell 1.
  - Case C: When the coverage areas of cell 1 and cell 2 overlap partially, the optimal efficiency and gains cannot be achieved regardless of whether load balancing transfer is from cell 1 to cell 2 or from cell 2 to cell 1. The larger the coverage difference between the two cells, the lower the efficiency and the fewer the gains of load balancing transfer.

Figure 5-4 Inter-cell coverage overlapping



5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

If (1) the bandwidths of inter-frequency cells involved in MLB are different, (2) this function is enabled, and (3) UEs in cells with large bandwidths are handed over to cells with small bandwidths, the uplink and downlink user-perceived rates of the handed-over UEs will be affected.

Successful MLB-based incoming handovers to a target cell may increase the PRB usage of the target cell. Excessively high PRB usage may indicate that the target cell is heavily loaded. The PRB usage of the target cell may further rise if handovers of CA UEs are permitted (by setting the **NRCellMlb**.***MlbTransCA*** parameter to **DL_CA** for the serving cell).

After this function is enabled, measurement gaps are set up for inter-frequency measurements by candidate UEs for MLB. Gap-assisted measurements will cause a performance loss in terms of the traffic volume.

When both MLB and unnecessary handovers except MLB-based handovers take effect, MLB-oriented UE selection policies are required. For example, the base station can differentiate between the UEs that MLB selects for load transfer and UEs for which the unnecessary handovers are performed based on the RFSPs or QCIs, preventing ping-pong handovers between cells.

If large-packet and small-packet services account for different proportions in co-coverage inter-frequency cells, the number of UEs cannot correctly reflect the cell load. In this case, enabling MLB may cause an overload in a cell with a large number of large-packet services.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MLB UE selection policy	**NRCellMlb**.***MlbUeSelectionPolicy***	*Mobility Load Balancing in Connected Mode*	UE-number-based connected mode MLB can be enabled only when at least one of the **SA_USER** and **NSA_USER** options of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	Connected mode MLB (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) cannot be enabled for cells deployed in IAB mode.

5.3.2.3 Function Impacts

- UEs with any of the following listed functions in effect will not be further selected for MLB. However, during an MLB procedure, any of these functions can take effect for a UE already selected for a handover, and if that occurs, the current MLB procedure will continue without interruption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB can select CA UEs only when the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA**, and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with uplink intra-band CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB can select CA UEs only when the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA**, and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with uplink intra-FR inter-band CA in effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	When selecting UEs to transfer, the gNodeB excludes those running VoNR services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW** or **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When selecting UEs to be handed over, the gNodeB does not select UEs with super uplink enabled. This MLB function is not recommended in cells with a high penetration rate of UEs with super uplink enabled, because it cannot achieve the expected effects in such cells.
Low-frequency TDD	UL and DL decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*	When selecting UEs to transfer, the gNodeB excludes those with SUL in effect. This MLB function is not recommended in cells with a high penetration of SUL UEs, as it cannot deliver the expected results in such cells.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When selecting UEs to transfer, the gNodeB excludes those with the low latency and high reliability function enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter LTE: **NSA_LTE_FDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	After the "NSA carrier combination selection based on user experience" function takes effect, UE-number-based connected mode MLB does not take effect for NSA UEs in the cell.

- Impact relationships with other handover functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and frequency-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD High-frequency TDD	Operator-specific-priority-based inter-frequency handover	**OP_DED_PRI_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and operator-specific-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Service-based inter-frequency handover	**SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and service-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	If both coverage-based inter-frequency handover and connected mode MLB are enabled, coverage-based inter-frequency handovers are triggered by event A5, and the value of **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** is less than or equal to the value of **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***, then the MLB-based handover success rate will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both UE-number-based connected mode MLB and experience-based smart carrier selection are enabled. If a handover is triggered for experience-based smart carrier selection, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with UE-number-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, cells in the MLB state will be filtered out, avoiding ping-pong handovers.
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarriersSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both UE-number-based connected mode MLB and multi-frequency beam coordination are enabled. If a handover is triggered for multi-frequency beam coordination, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	**ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both UE-number-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If inter-frequency handovers are triggered based on energy efficiency, cells in the MLB state will not be selected as the target cells, preventing ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Network slice admission control	**ADMISSION_BASED_ON_NS_ID_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	Assume that MLB is enabled and UEs in a network slice are selected for MLB. If the allowed number of UEs in a target cell is limited, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce the probability of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam**.***ResNotAvailForSliceTimer*** parameter must be set to a value greater than **20**. In this case, no UEs in the network slice will be handed over to the target cell within the specified period.
Low-frequency TDD	Uplink-experience-based inter-frequency handover	**UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink-experience-based inter-frequency handover does not take effect for RedCap UEs that are handed over to the local cell through connected mode MLB.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*CA Experience Improvement*	Assume that both UE-number-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.
Low-frequency TDD	Load-based inter-RAT mobility from NG-RAN to E-UTRAN	**LOAD_MOBILITY_TO_EUTRAN_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	If both load-based inter-RAT mobility from NG-RAN to E-UTRAN and UE-number-based connected mode MLB are enabled, UE-number-based connected mode MLB will preferentially take effect if the conditions for both functions are met at the same time. If the conditions for these two functions are met sequentially, the functions are triggered sequentially.

- Impact relationships with other functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter	*ANR*	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the neighboring cell.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

The serving cell has at least one inter-frequency co-coverage neighboring cell.

During inter-base-station MLB, the base stations involved must be provided by Huawei and equipped with the Xn interface.

An **NRCellFreqRelation** MO must be already configured.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Select this option.
MLB Trigger Mode	**NRCellMlb**.***MlbTriggerMode***	None	Set this parameter to **CELL_USER_NUM**.

The following parameters are also involved in MLB:

- Parameters such as the MLB evaluation period, MLB direction, as well as UE number threshold and offset must be configured for cells to enter or exit the MLB state. For details, see **Table 5-5**.
- Parameters such as the MLB handover indicator and no handover indicator must be configured for candidate cell range determination. For details, see **Table 5-6**.
- Parameters such as the MLB handover admission threshold and the Xn self-setup switch must be configured for inter-cell load information exchange. For details, see **Table 5-7**.
- The UE number difference threshold must be configured for candidate cell selection. For details, see **Table 5-8**.
- Parameters such as the UE selection policy, UE selection policy based on radio bearer attributes, and switches and thresholds for each policy must be configured for UE selection. For details, see **Table 5-9**.
- Parameters such as the frequency selection policy, thresholds and hysteresis for related events, and traffic priorities must be configured for measurement configuration delivery. For details, see **Table 5-10**.
- Parameters such as the MLB handover admission threshold and strongest-cell handover switch must be configured for the serving cell determination and target cell determination. For details, see **Table 5-11**.

Table 5-5 Parameters involved in the entering and exiting MLB procedure

Parameter Name	Parameter ID	Option	Setting Notes
MLB Judge Period	**NRCellMlb**.***MlbJudgePeriod***	None	Use the recommended value.
MLB Direction	**NRCellMlb**.***MlbDirection***	None	Set this parameter based on the network plan.
Inter-frequency MLB UE Number Threshold	**NRCellMlb**.***InterFreqMlbUeNumThld***	None	Set this parameter based on the network plan.
MLB UE Number Offset	**NRCellMlb**.***MlbUeNumOffset***	None	Set this parameter based on the network plan.
Inter-frequency MLB UL UE Number Thld	**NRCellMlb**.***InterFreqMlbUlUeNumThld***	None	Set this parameter based on the network plan.
MLB UL UE Number Offset	**NRCellMlb**.***MlbUlUeNumOffset***	None	Set this parameter based on the network plan.

<table>
<tr><th>Parameter Name</th><th>Parameter ID</th><th>Option</th><th>Setting Notes</th></tr>
<tr><td colspan="4">Note: For details about how to configure the UE number threshold and offset for inter-frequency MLB, see the following:
● NRCellMlb.InterFreqMlbUeNumThld: Recommended value = Reference value of the UE number threshold (downlink) x 52/60
● NRCellMlb.MlbUeNumOffset: Recommended value = Reference value of the UE number threshold (downlink) x 8/60
● NRCellMlb.InterFreqMlbUlUeNumThld: Recommended value = Reference value of the UE number threshold (uplink) x 52/60
● NRCellMlb.MlbUlUeNumOffset: Recommended value = Reference value of the UE number threshold (uplink) x 8/60
If the busy-hour PRB usage of every cell that is planned to be involved in MLB is greater than or equal to 60%:
● Reference value of the UE number threshold (downlink) = Number of UEs in the downlink in the cell when Downlink Resource Block Utilizing Rate (DU) is equal to 60%
● Reference value of the UE number threshold (uplink) = Number of UEs in the uplink in the cell when Uplink Resource Block Utilizing Rate (DU) is equal to 60%
If the busy-hour PRB usage of the serving cell involved in MLB is always less than 60%, and there are other cells whose PRB usage is greater than or equal to 60%, then the cell with the highest PRB usage among the other cells is selected as the reference cell. The calculation is as follows: Reference value of the UE number threshold (uplink/downlink) = Number of UEs in the reference cell when the PRB usage (uplink/downlink) is 60% x Air interface capability of the serving cell (uplink/downlink)/Air interface capability of the reference cell (uplink/downlink)
● Uplink air interface capability = N.ThpVol.UL.Cell/(Uplink Resource Block Utilizing Rate (DU)/100000)
● Downlink air interface capability = N.ThpVol.DL.Cell/(Downlink Resource Block Utilizing Rate (DU)/100000)
If the busy-hour PRB usage of every cell is less than 60%, the uplink and downlink UE number thresholds are set to their default values for the cell with the highest uplink and downlink bandwidths. The reference values of the UE number thresholds for cells with other bandwidths are calculated as follows:
● Reference value of the UE number threshold for another cell (uplink) = (Default value of the NRCellMlb.InterFreqMlbUlUeNumThld parameter + Default value of the NRCellMlb.MlbUlUeNumOffset parameter) x (Uplink bandwidth of the cell/Maximum uplink cell bandwidth)
● Reference value of the UE number threshold for another cell (downlink) = (Default value of the NRCellMlb.InterFreqMlbUeNumThld parameter + Default value of the NRCellMlb.MlbUeNumOffset parameter) x (Downlink bandwidth of the cell/Maximum downlink cell bandwidth)</td></tr>
</table>

Table 5-6 Parameters involved in candidate cell range determination

Parameter Name	Parameter ID	Option	Setting Notes
MLB-Based Handover Flag	**NRCellRelation**.***MlbHoFlag***	None	Set this parameter based on the network plan. Only the cell whose **NRCellRelation**.***MlbHoFlag*** is set to **TRUE** can be considered as a candidate cell, that is, determined to be within the candidate cell range.
No Handover Flag	**NRCellRelation**.***NoHoFlag***	None	Set this parameter based on the network plan.
MLB Enhancement Algorithm Switch	**NRCellMlb**.***MlbAlgoEnhSwitch***	MLB_NCELL_SPEC_EXTEND_SW	It is recommended that this option be selected when the local cell has more than 50 neighboring cells for MLB.

Table 5-7 Parameters involved in inter-cell load information exchange

Parameter Name	Parameter ID	Option	Setting Notes
MLB HO Admission Thld	**NRCellMlb**.***MlbHoAdmissionThld***	None	Set this parameter based on the network plan.

Table 5-8 Parameters involved in candidate cell selection

Parameter Name	Parameter ID	Option	Setting Notes
UE Number Difference Threshold	**NRCellMlb**.***UeNumDiffThld***	None	Set this parameter based on the network plan.
UL UE Number Difference Threshold	**NRCellMlb**.***UlUeNumDiffThld***	None	Set this parameter based on the network plan.

Table 5-9 Parameters involved in UE selection

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch[a]	**gNBRfspConfig**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Set this parameter based on the network plan. To disable MLB for UEs with a specific RFSP, deselect the **INTER_FREQ_CONNECTED_MLB_SW** option.
MLB Support Transfer CA User	**NRCellMlb**.***MlbTransCA***	None	If downlink CA UEs need to be transferred during uplink MLB, you are advised to set this parameter to **DL_CA**. In other scenarios, you are advised to set this parameter to **NULL**.
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	SA_USER	Set this parameter based on the network plan.
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	NSA_USER	Set this parameter based on the network plan.
Mobility Function Indication	**gNBQciBearer**.***MobilityFunInd***	INTER_FREQ_HO_SUPPRESS_SW	Set this parameter based on the network plan. If a UE has a bearer with a specific QCI for which the **INTER_FREQ_HO_SUPPRESS_SW** option is selected, the base station does not select the UE for handover.

Parameter Name	Parameter ID	Option	Setting Notes
Connected MLB UE Num Upper Limit	**NRCellMlb**.***ConnMlbUeNumThld***	None	Set this parameter based on the network plan. If the **NRCellMlb**.***MlbJudgePeriod*** parameter is set to a value less than **10** (in the unit of second), it is recommended that the **NRCellMlb**.***ConnMlbUeNumThld*** parameter value be less than or equal to the **NRCellMlb**.***MlbJudgePeriod*** parameter value.
ARP Thld Based UE Selection Policy Pri	**NRCellMlb**.***ArpThldBasedUeSelPolicy-Pri***	None	Set this parameter based on the network plan.
MLB UE Selection ARP Threshold	**NRCellMlb**.***MlbUeSelectionArpThld***	None	Set this parameter based on the network plan.
QCI Thld Based UE Selection Policy Pri	**NRCellMlb**.***QciThldBasedUeSelPolicy-Pri***	None	Set this parameter based on the network plan.
MLB Algorithm Switch	**NRCellQciBearer**.***MlbAlgoSwitch***	MLB_TRIG_BY_THLD	Set this parameter based on the network plan. Select the **MLB_TRIG_BY_THLD** option if UEs need to be transferred for MLB when they meet QCI criterion 1 in the QCI-threshold-based UE selection policy described in **5.1.5.1.2 UE Priority Determination**.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellQciBearer**.***MlbAlgoSwitch***	MLB_FILTER_SW	Set this parameter based on the network plan. To disable MLB for a specific QCI, select the **MLB_FILTER_SW** option of the **NRCellQciBearer**.***MlbAlgoSwitch*** parameter for the QCI.
MLB Detection Time	**NRCellQciBearer**.***MlbDetectTime***	None	Use the recommended value.
MLB DL Buffer Size	**NRCellQciBearer**.***MlbDlBfrSize***	None	Set this parameter based on the network plan.
MLB DL Thpt Threshold	**NRCellQciBearer**.***MlbDlThptThld***	None	Set this parameter based on the network plan.
MLB UL BSR Data Size	**NRCellQciBearer**.***MlbUlBsrDataSize***	None	Set this parameter based on the network plan.
MLB UL Thpt Threshold	**NRCellQciBearer**.***MlbUlThptThld***	None	Set this parameter based on the network plan.
QCI Sw Based UE Selection Policy Pri	**NRCellMlb**.***QciSwBasedUeSelPolicyPri***	None	Set this parameter based on the network plan.
PRB Usage Thld Based UE Sel Policy Pri	**NRCellMlb**.***PrbUsageThldUeSelPolicy-Pri***	None	Set this parameter based on the network plan.
Far Point Thld Based UE Sel Policy Pri	**NRCellMlb**.***FarPointThldUeSelPolicy-Pri***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
MLB UE Selection Distance Policy	**NRCellMlb**.***MlbUeSelDistancePolicy***	None	Set this parameter based on the network plan. You are advised to set this parameter to **SEL_FAR_POINT_UE** or **SEL_NEAR_POINT_UE** so that UEs at the cell edge or UEs in the cell center can be selected, respectively.
MLB UE Selection UL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelUlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE Selection DL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelDlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE Selection RSRP Threshold	**NRCellMlb**.***MlbUeSelectionRsrpThld***	None	Set this parameter based on the network plan.
MLB UE Number Judge Factor	**NRCellMlb**.***MlbUeNumJudgeFactor***	None	Set this parameter based on the network plan.
a: Assume that a UE initiates an access request. The gNodeB regards that this UE always does not support MLB after the access if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter is deselected for the RFSP of the UE. The gNodeB regards that this UE always supports MLB after the access if either of the following conditions is met: (1) The **INTER_FREQ_CONNECTED_MLB_SW** option is selected for the RFSP of the UE. (2) An RFSP is configured for the UE on the core network, but the index of this RFSP is not specified using the **gNBRfspConfig**.***RfspIndex*** parameter on the gNodeB.			

Table 5-10 Parameters involved in measurement configuration delivery

Parameter Name	Parameter ID	Option	Setting Notes
Inter-Freq Measurement Event Configuration	**NRCellMobilityConfig**.***InterFreqMeasEventConfig***	None	Set this parameter based on the network plan.
MLB-based Inter-Frequency RSRP Threshold	**NRCellInterFHoMeaGrp.*InterFreqMlbA4RsrpThld***	None	Set this parameter based on the network plan. The threshold for event A4 or threshold 2 for event A5 for each type of inter-frequency handover must be higher than the threshold for event A2 for coverage-based inter-frequency handover to ensure that a coverage-based inter-frequency measurement is not triggered immediately after a UE is handed over to the target frequency. This helps decrease the probability of ping-pong handovers.
Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig***	None	Set this parameter based on the network plan.
Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Frequency Priority	**NRCellFreqRelation**.***MlbFreqPriority***	None	Set this parameter based on the network plan. When the **NRCellMlb**.***FreqSelectionPolicy*** parameter is set to **FREQ_PRIORITY** and the **NRCellFreqRelation**.***MlbFreqPriority*** parameter is configured for non-serving frequencies involved in MLB, the number of handovers will increase, and the average uplink and downlink UE throughputs will fluctuate. The fluctuation depends on the load of the cell to which the UE is handed over. When the **NRCellFreqRelation**.***MlbFreqPriority*** parameter is set to **0** for a frequency, this frequency will not be selected as a target frequency for MLB-based handovers.

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Selection Policy	**NRCellMlb**.***FreqSelectionPolicy***	None	This parameter takes effect only when **NRCellMlb**.***MlbTriggerMode*** is set to **RADIO_RESOURCE**, **CELL_USER_NUM**, **SPEC_EFF_BASED_USER_NUM**, or **SPECIFIED_5QI_USER_NUM**. ● If the **NRCellFreqRelation**.***MlbFreqPriority*** parameter is set to **0**, it is recommended that the **NRCellMlb**.***FreqSelectionPolicy*** parameter be set to **FAIR_POLICY**. ● If the **NRCellFreqRelation**.***MlbFreqPriority*** parameter is set to a non-zero value, it is recommended that the **NRCellMlb**.***FreqSelectionPolicy*** parameter be set to **FREQ_PRIORITY**.

Table 5-11 Parameters involved in serving cell determination and target cell determination

Parameter Name	Parameter ID	Option	Setting Notes
MLB HO Admission Thld	**NRCellMlb**.***MlbHoAdmissionThld***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Enhancement Algorithm Switch	**NRCellMlb.*MlbAlgoEnhSwitch***	ONLY_STRONGEST_CELL_SW	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-1;
//Setting the MLB trigger mode
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM;
```

Optimization Command Examples

```
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the serving cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Disabling MLB for UEs with a specific RFSP
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
//Setting the UE selection policy of MLB for the cell
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM,
MlbUeSelectionPolicy=SA_USER-1&NSA_USER-0;
//Setting MlbTransCA to DL_CA if downlink CA UEs need to be transferred during uplink MLB
MOD NRCELLMLB: NrCellId=0, MlbTransCA=DL_CA;
//Setting the ArpThldBasedUeSelPolicyPri and MlbUeSelectionArpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, ArpThldBasedUeSelPolicyPri=210,
MlbUeSelectionArpThld=1;
//Setting the QciThldBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, QciThldBasedUeSelPolicyPri=220;
//Selecting the MLB_TRIG_BY_THLD option
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_TRIG_BY_THLD-1;
//Setting the MlbDetectTime parameter
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDetectTime=5;
//Setting the MlbDlBfrSize and MlbDlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDlBfrSize=20, MlbDlThptThld=30000;
```



```
//Setting the MlbUlBsrDataSize and MlbUlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbUlBsrDataSize=10, MlbUlThptThld=30000;
//Setting the QciSwBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, QciSwBasedUeSelPolicyPri=230;
//Turning on MLB_FILTER_SW for the QCI without MLB involved
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_FILTER_SW-1;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Setting the PrbUsageThldUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, PrbUsageThldUeSelPolicyPri=240;
//Setting the FarPointThldUeSelPolicyPri and MlbUeSelectionRsrpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, FarPointThldUeSelPolicyPri=250,
MlbUeSelDistancePolicy=SEL_FAR_POINT_UE, MlbUeSelectionRsrpThld=-111;
//Setting the MlbUeSelDlPrbUsageThld and MlbUeSelUlPrbUsageThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, MlbUeSelDlPrbUsageThld=2,
MlbUeSelUlPrbUsageThld=2;
//Setting the FreqSelectionPolicy parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=CELL_USER_NUM, FreqSelectionPolicy=FREQ_PRIORITY;
//Setting the InterFreqMeasEventConfig parameter for MLB
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A4;
//Setting the InterFreqMlbA4RsrpThld parameter
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqMlbA4RsrpThld=-102;
//Setting the InterFreqA4A5TimeToTrig and InterFreqA4A5Hyst parameters
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
//Setting the MLB frequency priority (based on the assumption that SsbFreqPos is set to 8000 in TDD
scenarios)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, MlbFreqPriority=1;
//Turning on the MLB neighboring cell specification extension switch
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=MLB_NCELL_SPEC_EXTEND_SW-1;
//Turning on ONLY_STRONGEST_CELL_SW
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=ONLY_STRONGEST_CELL_SW-1;
//Turning on INTER_FREQ_HO_SUPPRESS_SW for a specific QCI of UEs so that UEs will not be selected for
handover
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Setting MlbUeNumJudgeFactor
MOD NRCELLMLB: NrCellId=0, MlbUeNumJudgeFactor=50;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

After this function is enabled, if the cell load meets the conditions for triggering load redistribution described in **5.1.5 Load Redistribution**, the following are true:

- If the **SA_USER** option of the **NRCellMlb.*MlbUeSelectionPolicy*** parameter is selected and the **N.HO.InterFreq.Mlb.PrepAttOut** counter value is not 0, this function has taken effect.

- If the **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Att** counter value is not 0, this function has taken effect.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. If BIT0 or BIT1 of the binary number converted from the decimal value of **MLB Status** is **1**, MLB has taken effect.

5.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful MLB-based inter-frequency outgoing handovers.

Observe the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counter to obtain the number of successful MLB-based inter-frequency PSCell changes in the LTE-NR NSA DC scenario.

Observe the **N.InterFreq.Mlb.Num** counter to obtain the number of times a cell enters the uplink or downlink MLB state.

Observe the **N.InterFreq.Mlb.Dur** counter to obtain the duration in which a cell is in the uplink or downlink MLB state.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. Observe **MLB HO Out Succ User Num** to obtain the number of UEs that are successfully handed over from a cell.

6 Spectral-Efficiency-based Connected Mode MLB

6.1 Principles

Spectral-efficiency-based connected mode MLB uses the number of UEs in a cell as the cell load indicator to determine whether the cell should enter or exit MLB. For cells in the MLB state, their air interface loads are evaluated. If the air interface load is unbalanced among cells, UEs in connected mode are handed over from high-load cells to low-load cells to achieve a load balance. Spectral-efficiency-based connected mode MLB is enabled if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected and the **NRCellMlb**.***MlbTriggerMode*** parameter is set to **SPEC_EFF_BASED_USER_NUM**.

It is recommended that this function be enabled when all of the following conditions are met:

- The same proportions of small-packet and large-packet services are independently running in inter-frequency co-coverage cells.
- Most UEs are motionless or move slowly.
- The source and target cells are intra-base-station cells. Alternatively, the cells are served by Huawei base stations, and an Xn interface has been set up between the base stations.

6.1.1 Entering and Exiting MLB

After spectral-efficiency-based connected mode MLB is enabled, the gNodeB determines whether a cell enters or exits MLB in the next MLB evaluation period based on the cell load within the current period specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter. The determination method is the same as that for UE-number-based connected mode MLB. For details, see **5.1.1 Entering and Exiting MLB**.

6.1.2 Candidate Cell Range Identification

The gNodeB selects candidate cells for UE handovers. Candidate cell range identification here is the same as that for UE-number-based connected mode MLB. For details, see **5.1.2 Candidate Cell Range Identification**.

6.1.3 Inter-Cell Load Information Exchange

Inter-cell load information exchange for spectral-efficiency-based connected mode MLB is the same as that for UE-number-based connected mode MLB. For details, see **5.1.3 Inter-Cell Load Information Exchange**.

6.1.4 Candidate Cell Selection

The gNodeB compares the load information of the serving cell with that of all candidate cells.

If at least one candidate cell meets all the following conditions, the gNodeB considers that there are candidate cells available for load redistribution and proceeds to it. Otherwise, the gNodeB considers that there is no candidate cell available for load redistribution and does not execute load redistribution in the current period.

- The load difference between the serving cell and the candidate cell is greater than the load difference threshold.
  - For downlink MLB (The **NRCellMlb**.***MlbDirection*** parameter is set to **DOWNLINK**.)

    Condition: The downlink load difference between the serving cell and the candidate cell is greater than the downlink load difference threshold (specified by the **NRCellMlb**.***UeNumDiffThld*** parameter). Downlink load difference between the serving cell and the candidate cell = (Downlink air interface load of the serving cell – Downlink air interface load of the candidate cell)/Downlink air interface load of the serving cell.
  - For uplink MLB (The **NRCellMlb**.***MlbDirection*** parameter is set to **UPLINK**.)

    Condition: The uplink load difference between the serving cell and the candidate cell is greater than the uplink load difference threshold (specified by the **NRCellMlb**.***UlUeNumDiffThld*** parameter). Uplink load difference between the serving cell and the candidate cell = (Uplink air interface load of the serving cell – Uplink air interface load of the candidate cell)/Uplink air interface load of the serving cell.
  - For uplink and downlink MLB (The **NRCellMlb**.***MlbDirection*** parameter is set to **DUAL**.)

    Any of the preceding load difference conditions is met.

  NOTE

  For details about the definitions of the uplink air interface load and downlink air interface load of a cell, see **4.2.2 Air Interface Load of a Cell**.
- The candidate cell is not in the high-load state.
- When the number of UEs in the local cell is less than 20:
  - Downlink load balancing: The number of downlink UEs in the local cell minus that in the candidate cell is greater than five.

- Uplink load balancing: The number of uplink UEs in the local cell minus that in the candidate cell is greater than five.
- Uplink/Downlink load balancing: Either of the preceding requirements is met.

- The candidate cell is not in the interference state. A cell is in the interference state when it meets the conditions for starting uplink-interference-based inter-frequency handover. For details about uplink-interference-based inter-frequency handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

6.1.5 Load Redistribution

Load redistribution is a process in which some UEs in connected mode in the serving cell are handed over to candidate cells that meet certain signal quality requirements.

6.1.5.1 UE Selection

The procedure for selecting a UE to be handed over is as follows:

1. According to basic conditions that UEs must meet, initial selection is performed. The basic conditions that UEs must meet are the same as those for UE-number-based connected mode MLB. For details, see **5.1.5.1.1 Conditions Used for Initial Selection**.
2. The priorities of the initially selected UEs are determined according to UE selection policies based on radio bearer attributes. UE priority determination is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.1.2 UE Priority Determination**.
3. The upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period is calculated. For details about the calculation, see **Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution**.
4. The initially-selected UEs are then selected in descending order of UE priorities for measurement-based handovers, and the number of selected UEs cannot exceed N_{max}. If UEs have the same priority, the gNodeB randomly selects one from them.

Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution

In spectral-efficiency-based connected mode MLB, the upper limit of the total number of UEs involved in load redistribution is calculated as follows:

- In downlink MLB, the upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period takes the minimum of the following three values:
  - [Current number of UEs in the serving cell – (Downlink UE number threshold for inter-frequency MLB – 1)] x *T*
  - MLB UE number judge factor x (Downlink air interface load in the serving cell – Minimum of the downlink air interface loads in the candidate cells) x *T* x Downlink air interface capability of the serving cell

  – UE number upper limit for connected mode MLB

- In uplink MLB, the upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period takes the minimum of the following three values:
  – [Current number of UEs in the serving cell – (Uplink UE number threshold for inter-frequency MLB – 1)] x *T*
  – MLB UE number judge factor x (Uplink air interface load in the serving cell – Minimum of the uplink air interface loads in the candidate cells) x *T* x Uplink air interface capability of the serving cell
  – UE number upper limit for connected mode MLB

The involved parameters are as follows:

- The value of T depends on the MLB evaluation period.
  – If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is greater than or equal to 10s, *T* is 1.
  – If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is less than 10s, *T* is the value of **NRCellMlb**.***MlbJudgePeriod*** divided by 10.
- The downlink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The uplink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The MLB UE number judge factor is specified by the **NRCellMlb**.***MlbUeNumJudgeFactor*** parameter.
- The UE number upper limit for connected mode MLB is specified by the **NRCellMlb**.***ConnMlbUeNumThld*** parameter.

NOTE

For details about the definitions of the uplink and downlink air interface load and uplink and downlink air interface capability of a cell, see **4.2.2 Air Interface Load of a Cell**.

6.1.5.2 Measurement-based Handover

Measurement-based handovers are performed for a selected UE. The gNodeB delivers the operating frequencies of candidate cells available for load redistribution and the information related to event A4 or A5 to such a UE, and the UE performs measurements. When the event entering conditions are met, the UE sends a measurement report to the gNodeB. After receiving the measurement report, the gNodeB in the serving cell checks whether the reported cells are within the candidate cell range for load redistribution and determines whether to instruct the UE to perform a handover. During the handover, the target cell also evaluates whether to allow this UE to access its network.

For details about the basic procedure of measurement-based handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. This section describes only the parts that have been adapted to accommodate MLB, including measurement configuration delivery, serving cell determination, and target cell determination.

6.1.5.2.1 Measurement Configuration Delivery

This procedure is the same as that for UE-number-based connected mode MLB. For details, see **6.1.5.2.1 Measurement Configuration Delivery**.

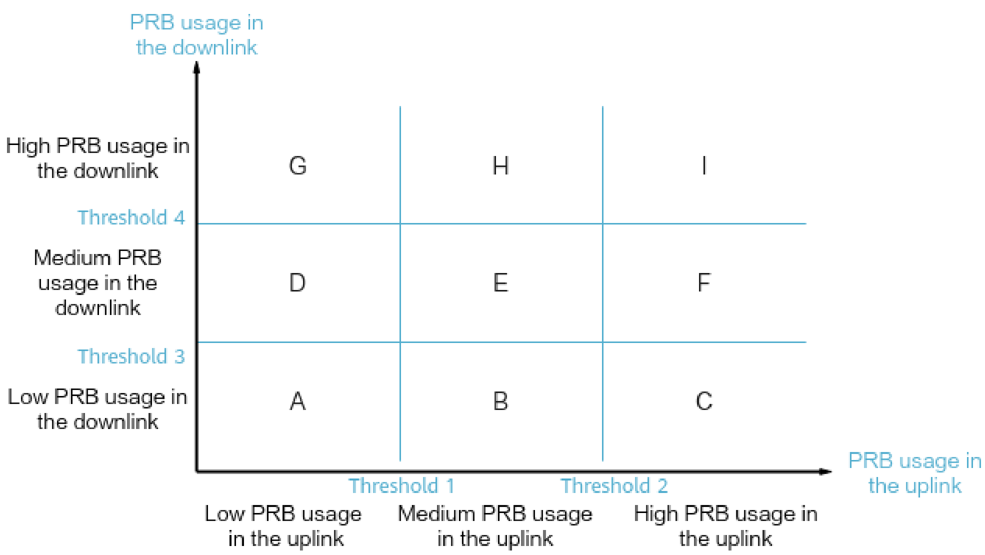
6.1.5.2.2 Serving Cell Determination

The gNodeB determines whether the reported cells are candidate cells that can be used for load redistribution. For details about the conditions involved in determination, see **6.1.4 Candidate Cell Selection**. If the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA** and the selected UE is a CA UE, the gNodeB also checks whether a candidate cell's CA configuration remains applicable to the CA UE following a handover to the candidate cell. If the configuration is no longer applicable, the candidate cell will not be selected as the target cell of the CA UE's handover.

If none of the reported cells is a candidate cell for load redistribution, no handover is performed. Otherwise, candidate UEs are classified by PRB usage, and candidate cells in the measurement reports are classified by load difference. Then the base station matches the candidate UEs and cells. Once the match between a UE and a cell succeeds, a handover of the UE to the cell can be initiated.

1. As shown in **Figure 6-1**, there are nine types of candidate UEs, categorized from type A to type I, based on PRB usage. For example, H-type UEs are those with medium PRB usage in the uplink and with high PRB usage in the downlink.

Figure 6-1 Classification of candidate UEs



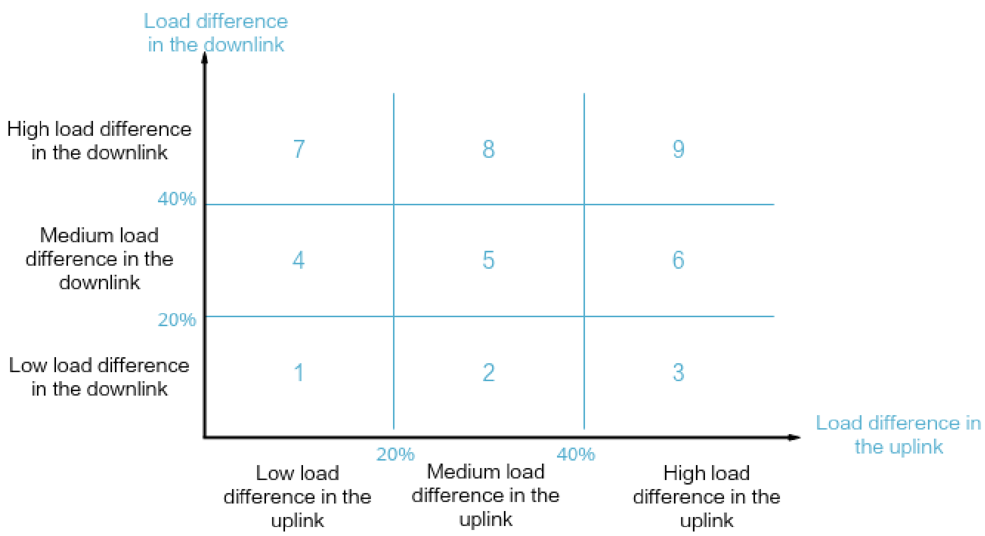
Specifically:

– Threshold 1 = **NRCellMlb**.***MlbUeSelUlPrbUsageThld*** + **NRCellMlb**.***MlbUeUlPrbLowThldOfs***/10
– Threshold 2 = **NRCellMlb**.***MlbUeSelUlPrbUsageThld*** + **NRCellMlb**.***MlbUeUlPrbHighThldOfs***/10
– Threshold 3 = **NRCellMlb**.***MlbUeSelDlPrbUsageThld*** + **NRCellMlb**.***MlbUeDlPrbLowThldOfs***/10

- Threshold 4 = **NRCellMlb**.***MlbUeSelDlPrbUsageThld*** + **NRCellMlb**.***MlbUeDlPrbHighThldOfs***/10
- When the uplink PRB usage of a UE is equal to threshold 1 or 2, the UE is classified as a medium-PRB-usage UE in the uplink. When the downlink PRB usage of a UE is equal to threshold 3 or 4, the UE is classified as a medium-PRB-usage UE in the downlink.

2. As shown in **Figure 6-2**, there are nine types of candidate cells in a measurement report, categorized from type 1 to type 9, based on load difference between the serving cell and the candidate cells. For example, 8-type cells are those with medium load difference in the uplink and with high load difference in the downlink. For details about the definition of the load difference, see **6.1.4 Candidate Cell Selection**.

Figure 6-2 Classification of candidate cells

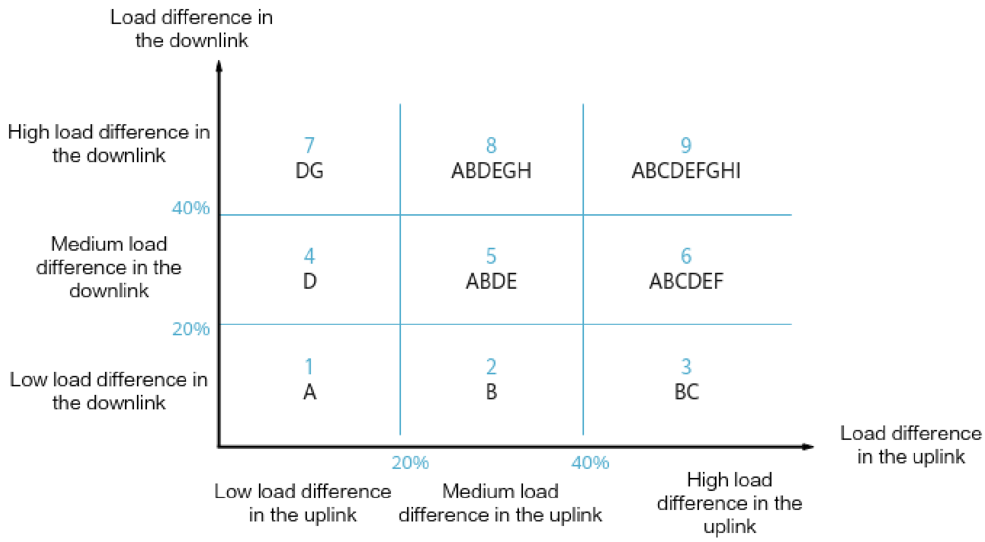


Specifically:

- When the load difference is equal to 20%, the candidate cell is a medium-load-difference cell. When the load difference is equal to 40%, the candidate cell is a high-load-difference cell.
- When the load difference is less than or equal to 0, the candidate cell is a low-load-difference cell.

3. The candidate cells in the measurement report are matched with the candidate UEs according to the principles described in **Figure 6-3**. For example, if the candidate cell is a 3-type cell (uplink load difference ≥ 40%; downlink load difference < 20%), the following types of UEs are matched with it:
   - B type: UEs with medium PRB usage in the uplink and with low PRB usage in the downlink
   - C type: UEs with high PRB usage in the uplink and with low PRB usage in the downlink

Figure 6-3 Principles for matching candidate cells with candidate UEs



NOTE

- If the **NRCellMlb**.***MlbDirection*** parameter is set to **DOWNLINK** or **UPLINK**, classification and matching for UEs and candidate cells are performed based on the situation either in the downlink or in the uplink. For example, assume that only downlink MLB is performed. In this case, if the candidate cell is a medium-load-difference cell in the downlink, it is a 4/5/6-type candidate cell according to **Figure 6-2**. According to **Figure 6-3**, the matched candidate UEs can be the type-A to type-F ones (the union set of UE types that can be matched with 4/5/6-type candidate cells). The corresponding UE types in **Figure 6-1** are medium- and low-PRB-usage UEs in the downlink.
- If the **NRCellMlb**.***MlbDirection*** parameter is set to **DUAL**, classification and matching for UEs and candidate cells are performed based on the situation both in the downlink and in the uplink.

After the serving cell determination is completed, the gNodeB selects a handover policy based on the setting of the **ONLY_STRONGEST_CELL_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.

- If this option is selected, the gNodeB instructs the UE to perform a handover only to the cell with the highest signal quality, and handovers to other cells will not be performed once this handover fails.
- If this option is deselected, a handover is attempted to the cell with the highest signal quality. When the handover fails, a handover to the cell with the next highest signal quality is attempted, until the handover succeeds or handovers to all candidate cells for load redistribution have been attempted.

6.1.5.2.3 Target Cell Determination

This procedure is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.2.3 Target Cell Determination**.

6.2 Network Analysis

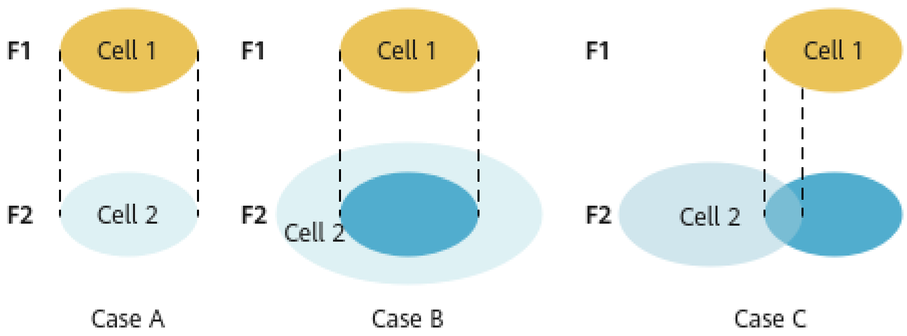
6.2.1 Benefits

Spectral-efficiency-based connected mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains. Spectral-efficiency-based connected mode MLB transfers UEs in connected mode from high-load cells to low-load cells to achieve load balance between cells.

The following must also be considered when using this function:

- When UEs in the inter-frequency co-coverage cells are high-mobility UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- The gains of this function will not be as high as expected when UEs differ greatly in their band support capabilities or there are restrictions on the serving PLMNs of UEs due to UE subscription.
- The gains of this function will not be as high as expected when cells on multiple frequencies differ greatly in their coverage capabilities, as shown in **Figure 6-4**.
  - Case A: When two inter-frequency cells (cell 1 and cell 2) cover the exact same area, the load balancing transfer between the two cells is the most efficient and effective.
  - Case B: When the coverage area of cell 2 completely includes that of cell 1, the load balancing transfer from cell 1 to cell 2 is the most efficient and effective, but the load balancing transfer from cell 2 to cell 1 cannot achieve the optimal efficiency and gains. The larger the coverage difference between the two cells, the lower the efficiency and the less the gains of load balancing transfer from cell 2 to cell 1.
  - Case C: When the coverage areas of cell 1 and cell 2 overlap partially, the optimal efficiency and gains cannot be achieved regardless of whether load balancing transfer is from cell 1 to cell 2 or from cell 2 to cell 1. The larger the coverage difference between the two cells, the less efficient and effective the load balancing transfer is.

Figure 6-4 Inter-cell coverage overlapping



6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

Successful MLB-based incoming handovers to a target cell may increase the PRB usage of the target cell. Excessively high PRB usage may indicate that the target cell is heavily loaded. The PRB usage of the target cell may further rise if handovers of CA UEs are permitted (by setting the **NRCellMlb**.***MlbTransCA*** parameter to **DL_CA** for the serving cell).

After this function is enabled, measurement gaps are set up for inter-frequency measurements by candidate UEs for MLB. Gap-assisted measurements will cause a performance loss in terms of the traffic volume.

When both MLB and unnecessary handovers except MLB-based handovers take effect, MLB-oriented UE selection policies are required. For example, the base station can differentiate between the UEs that MLB selects for load transfer and UEs for which the unnecessary handovers are performed based on the RFSPs or QCIs, preventing ping-pong handovers between cells.

If large-packet and small-packet services account for different proportions in co-coverage inter-frequency cells, the number of UEs cannot correctly reflect the cell load. In this case, enabling MLB may cause an overload in a cell with a large number of large-packet services.

The MLB frequency priorities specified by the **NRCellFreqRelation**.***MlbFreqPriority*** parameter must be set to the same value for inter-frequency cells involved in MLB. Otherwise, the number of handovers will increase significantly, and the average uplink and downlink UE throughputs will fluctuate. The fluctuation depends on the load of the cell to which UEs are handed over.

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051212	Converged Load Balancing	NR0S000CLB00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MLB UE selection policy	**NRCellMlb**.***MlbUeSelectionPolicy***	*Mobility Load Balancing in Connected Mode*	Spectral-efficiency-based connected mode MLB can be enabled only when at least one of the **SA_USER** and **NSA_USER** options of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected.

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	Connected mode MLB (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) cannot be enabled for cells deployed in IAB mode.

6.3.2.3 Function Impacts

- UEs with any of the following listed functions in effect will not be further selected for MLB. However, during an MLB procedure, any of these functions can take effect for a UE already selected for a handover, and if that occurs, the current MLB procedure will continue without interruption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB can select CA UEs only when the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA**, and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with uplink intra-band CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB can select CA UEs only when the **NRCellMlb**.***MlbTransCA*** parameter is set to **DL_CA**, and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with uplink intra-FR inter-band CA in effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	When selecting UEs to transfer, the gNodeB excludes those running VoNR services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW** or **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When selecting UEs to be handed over, the gNodeB does not select UEs with super uplink enabled. This MLB function is not recommended in cells with a high penetration rate of UEs with super uplink enabled, because it cannot achieve the expected effects in such cells.
Low-frequency TDD	UL and DL decoupling	**NRDUCellAlgoSwitch.*UlDlDecouplingSwitch***	*UL and DL Decoupling*	When selecting UEs to transfer, the gNodeB excludes those with SUL in effect. This MLB function is not recommended in cells with a high penetration of SUL UEs, as it cannot deliver the expected results in such cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter LTE: **NSA_LTE_FDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	After the "NSA carrier combination selection based on user experience" function takes effect, spectral-efficiency-based connected mode MLB does not take effect for NSA UEs in the cell.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When selecting UEs to transfer, the gNodeB excludes those with the low latency and high reliability function enabled.

- Impact relationships with other handover functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and frequency-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD High-frequency TDD	Operator-specific-priority-based inter-frequency handover	**OP_DED_PRI_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and operator-specific-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Service-based inter-frequency handover	**SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and service-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	If both coverage-based inter-frequency handover and connected mode MLB are enabled, coverage-based inter-frequency handovers are triggered by event A5, and the value of **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** is less than or equal to the value of **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***, then the MLB-based handover success rate will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both spectral-efficiency-based connected mode MLB and experience-based smart carrier selection are enabled. If a handover is triggered for experience-based smart carrier selection, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with spectral-efficiency-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. Cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarriersSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both spectral-efficiency-based connected mode MLB and multi-frequency beam coordination are enabled. If a handover is triggered for multi-frequency beam coordination, cells in the MLB state will be filtered out, avoiding ping-pong handovers. When both spectral-efficiency-based connected mode MLB and multi-frequency beam coordination are enabled, the gains of multi-frequency beam coordination decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	**ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both spectral-efficiency-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If an inter-frequency handover is triggered based on energy efficiency, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Network slice admission control	**ADMISSION_BASED_ON_NS_ID_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	Assume that MLB is enabled and UEs in a network slice are selected for MLB. If the allowed number of UEs in a target cell is limited, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce the probability of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam**.***ResNotAvailForSliceTimer*** parameter must be set to a value greater than **20**. In this case, no UEs in the network slice will be handed over to the target cell within the specified period.
Low-frequency TDD	Uplink-experience-based inter-frequency handover	**UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink-experience-based inter-frequency handover does not take effect for RedCap UEs that are handed over to the local cell through connected mode MLB.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*CA Experience Improvement*	Assume that both spectral-efficiency-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.

- Impact relationships with other functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter	*ANR*	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the neighboring cell.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Networking

The serving cell has at least one inter-frequency co-coverage neighboring cell.

During inter-base-station MLB, the base stations involved must be provided by Huawei and equipped with the Xn interface.

An **NRCellFreqRelation** MO must be already configured.

6.3.5 Others

None

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Select this option.
MLB Trigger Mode	**NRCellMlb**.***MlbTriggerMode***	None	Set this parameter to **SPEC_EFF_BASED_USER_NUM**.

The following parameters are also involved in MLB:

- Same as those for UE-number-based connected mode MLB. Parameters such as the MLB evaluation period, MLB direction, as well as UE number threshold and offset must be configured for cells to enter and exit the MLB state. For details, see **Table 5-5**.

- Same as those for UE-number-based connected mode MLB. Parameters such as the MLB handover indicator and no handover indicator must be configured for candidate cell range determination. For details, see **Table 5-6**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the MLB handover admission threshold and the Xn self-setup switch must be configured for inter-cell load information exchange. For details, see **Table 5-7**.
- The UE number difference threshold and the parameters related to the air interface capability of a cell must be configured for candidate cell selection. For details, see **Table 6-2**.
- Parameters such as the UE selection policy, UE selection policy based on radio bearer attributes, and switches and thresholds for each policy must be configured for UE selection. For details, see **Table 6-2**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the frequency selection policy, thresholds and hysteresis for related events, and traffic priorities must be configured for measurement configuration delivery. For details, see **Table 5-10**.
- Parameters such as the MLB handover admission threshold, PRB usage threshold for UEs, strongest-cell handover switch must be configured for the serving cell determination and target cell determination. For details, see **Table 6-4**.

Table 6-2 Parameters involved in candidate cell selection

Parameter Name	Parameter ID	Option	Setting Notes
UE Number Difference Threshold	**NRCellMlb.*UeNumDiffThld***	None	Set this parameter based on the network plan.
UL UE Number Difference Threshold	**NRCellMlb.*UlUeNumDiffThld***	None	Set this parameter based on the network plan.
Load Evaluation Policy	**NRCellMlb.*LoadEvaluationPolicy***	None	The value **DYNAMIC_EVAL** is recommended.
Cell DL Capability Scale Factor	**NRCellMlb.*CellDlCapbScaleFactor***	None	Set this parameter based on the network plan.
Cell UL Capability Scale Factor	**NRCellMlb.*CellUlCapbScaleFactor***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Enhanced Switch	**NRDUCellDlSch**.***DlSchAlgoEnhSwitch***	PRECISE_LOAD_EVAL_SW	It is recommended that this option be selected when the number of available PRBs in the cell dynamically changes.

Table 6-3 Parameters involved in UE selection

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch[a]	**gNBRfspConfig**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Set this parameter based on the network plan. To disable MLB for UEs with a specific RFSP, deselect the **INTER_FREQ_CONNECTED_MLB_SW** option.
MLB Support Transfer CA User	**NRCellMlb**.***MlbTransCA***	None	If downlink CA UEs need to be transferred during uplink MLB, you are advised to set this parameter to **DL_CA**. In other scenarios, you are advised to set this parameter to **NULL**.
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	SA_USER	Set this parameter based on the network plan.
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	NSA_USER	Set this parameter based on the network plan.
Mobility Function Indication	**gNBQciBearer**.***MobilityFunInd***	INTER_FREQ_HO_SUPPRESS_SW	Set this parameter based on the network plan. If a UE has a bearer with a specific QCI for which the **INTER_FREQ_HO_SUPPRESS_SW** option is selected, the base station does not select the UE for handover.

Parameter Name	Parameter ID	Option	Setting Notes
Connected MLB UE Num Upper Limit	**NRCellMlb**.***ConnMlbUeNumThld***	None	Set this parameter based on the network plan. If the **NRCellMlb**.***MlbJudgePeriod*** parameter is set to a value less than **10** (in the unit of second), it is recommended that the **NRCellMlb**.***ConnMlbUeNumThld*** parameter value be less than or equal to the **NRCellMlb**.***MlbJudgePeriod*** parameter value.
ARP Thld Based UE Selection Policy Pri	**NRCellMlb**.***ArpThldBasedUeSelPolicy-Pri***	None	Set this parameter based on the network plan.
MLB UE Selection ARP Threshold	**NRCellMlb**.***MlbUeSelectionArpThld***	None	Set this parameter based on the network plan.
QCI Thld Based UE Selection Policy Pri	**NRCellMlb**.***QciThldBasedUeSelPolicy-Pri***	None	Set this parameter based on the network plan.
MLB Algorithm Switch	**NRCellQciBearer**.***MlbAlgoSwitch***	MLB_TRIG_BY_THLD	Set this parameter based on the network plan. Select the **MLB_TRIG_BY_THLD** option if UEs need to be transferred for MLB when they meet QCI criterion 1 in the QCI-threshold-based UE selection policy described in **6.1.5.1 UE Selection**.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellQciBearer**.***MlbAlgoSwitch***	MLB_FILTER_SW	Set this parameter based on the network plan. To disable MLB for a specific QCI, select the **MLB_FILTER_SW** option of the **NRCellQciBearer**.***MlbAlgoSwitch*** parameter for the QCI.
MLB Detection Time	**NRCellQciBearer**.***MlbDetectTime***	None	Use the recommended value.
MLB DL Buffer Size	**NRCellQciBearer**.***MlbDlBfrSize***	None	Set this parameter based on the network plan.
MLB DL Thpt Threshold	**NRCellQciBearer**.***MlbDlThptThld***	None	Set this parameter based on the network plan.
MLB UL BSR Data Size	**NRCellQciBearer**.***MlbUlBsrDataSize***	None	Set this parameter based on the network plan.
MLB UL Thpt Threshold	**NRCellQciBearer**.***MlbUlThptThld***	None	Set this parameter based on the network plan.
QCI Sw Based UE Selection Policy Pri	**NRCellMlb**.***QciSwBasedUeSelPolicyPri***	None	Set this parameter based on the network plan.
PRB Usage Thld Based UE Sel Policy Pri	**NRCellMlb**.***PrbUsageThldUeSelPolicy-Pri***	None	When spectral-efficiency-based connected mode MLB is enabled, it is recommended that this parameter be set to a non-zero value to ensure the transfer efficiency.
Far Point Thld Based UE Sel Policy Pri	**NRCellMlb**.***FarPointThldUeSelPolicy-Pri***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
MLB UE Selection Distance Policy	**NRCellMlb**.***MlbUeSelDistancePolicy***	None	Set this parameter based on the network plan. You are advised to set this parameter to **SEL_FAR_POINT_UE** or **SEL_NEAR_POINT_UE** so that UEs at the cell edge or UEs in the cell center can be selected, respectively.
MLB UE Selection UL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelUlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE Selection DL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelDlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE Selection RSRP Threshold	**NRCellMlb**.***MlbUeSelectionRsrpThld***	None	Set this parameter based on the network plan.
MLB UE Number Judge Factor	**NRCellMlb**.***MlbUeNumJudgeFactor***	None	Set this parameter based on the network plan.
a: Assume that a UE initiates an access request. The gNodeB regards that this UE always does not support MLB after the access if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter is deselected for the RFSP of the UE. The gNodeB regards that this UE always supports MLB after the access if either of the following conditions is met: (1) The **INTER_FREQ_CONNECTED_MLB_SW** option is selected for the RFSP of the UE. (2) An RFSP is configured for the UE on the core network, but the index of this RFSP is not specified using the **gNBRfspConfig**.***RfspIndex*** parameter on the gNodeB.			

Table 6-4 Parameters involved in serving cell determination and target cell determination

Parameter Name	Parameter ID	Option	Setting Notes
MLB UE Selection UL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelUlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE Selection DL PRB Usage Threshold	**NRCellMlb**.***MlbUeSelDlPrbUsageThld***	None	Set this parameter based on the network plan.
MLB UE DL PRB High Thld Offset	**NRCellMlb**.***MlbUeDlPrbHighThldOfs***	None	Set this parameter based on the network plan.
MLB UE DL PRB Low Thld Offset	**NRCellMlb**.***MlbUeDlPrbLowThldOfs***	None	Set this parameter based on the network plan.
MLB UE UL PRB High Thld Offset	**NRCellMlb**.***MlbUeUlPrbHighThldOfs***	None	Set this parameter based on the network plan.
MLB UE UL PRB Low Thld Offset	**NRCellMlb**.***MlbUeUlPrbLowThldOfs***	None	Set this parameter based on the network plan.
MLB HO Admission Thld	**NRCellMlb**.***MlbHoAdmissionThld***	None	Set this parameter based on the network plan.
MLB Enhancement Algorithm Switch	**NRCellMlb**.***MlbAlgoEnhSwitch***	ONLY_STRONGEST_CELL_SW	Set this parameter based on the network plan.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-1;
//Setting the cell capability scale factors and the following parameters: MlbTriggerMode and
LoadEvaluationPolicy
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
LoadEvaluationPolicy=DYNAMIC_EVAL, CellDlCapbScaleFactor=1, CellUlCapbScaleFactor=1;
```

Optimization Command Examples

```
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the serving cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Disabling MLB for UEs with a specific RFSP
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
//Setting the UE selection policy of MLB for the cell
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
MlbUeSelectionPolicy=SA_USER-1&NSA_USER-0;
//Setting MlbTransCA to DL_CA if downlink CA UEs need to be transferred during uplink MLB
MOD NRCELLMLB: NrCellId=0, MlbTransCA=DL_CA;
//Setting the ArpThldBasedUeSelPolicyPri and MlbUeSelectionArpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
ArpThldBasedUeSelPolicyPri=210, MlbUeSelectionArpThld=1;
//Setting the QciThldBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
QciThldBasedUeSelPolicyPri=220;
//Selecting the MLB_TRIG_BY_THLD option
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_TRIG_BY_THLD-1;
//Setting the MlbDetectTime parameter
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDetectTime=5;
//Setting the MlbDlBfrSize and MlbDlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDlBfrSize=20, MlbDlThptThld=30000;
//Setting the MlbUlBsrDataSize and MlbUlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbUlBsrDataSize=10, MlbUlThptThld=30000;
//Setting the QciSwBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
QciSwBasedUeSelPolicyPri=230;
//Turning on MLB_FILTER_SW for the QCI without MLB involved
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_FILTER_SW-1;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Setting the PrbUsageThldUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
PrbUsageThldUeSelPolicyPri=240;
//Setting the FarPointThldUeSelPolicyPri and MlbUeSelectionRsrpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
FarPointThldUeSelPolicyPri=250, MlbUeSelDistancePolicy=SEL_FAR_POINT_UE,
MlbUeSelectionRsrpThld=-111;
//Setting the MlbUeSelDlPrbUsageThld and MlbUeSelUlPrbUsageThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
MlbUeSelDlPrbUsageThld=2, MlbUeSelUlPrbUsageThld=2;
//Setting the FreqSelectionPolicy parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPEC_EFF_BASED_USER_NUM,
FreqSelectionPolicy=FREQ_PRIORITY;
//Setting the InterFreqMeasEventConfig parameter for MLB
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A4;
//Setting the InterFreqMlbA4RsrpThld parameter
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqMlbA4RsrpThld=-102;
//Setting the InterFreqA4A5TimeToTrig and InterFreqA4A5Hyst parameters
```

```
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
//Setting the offsets to PRB thresholds for UE selection in MLB
MOD NRCELLMLB: NrCellId=0, MlbUeDlPrbHighThldOfs=30, MlbUeDlPrbLowThldOfs=10,
MlbUeUlPrbHighThldOfs=30, MlbUeUlPrbLowThldOfs=10;
//Setting the MLB frequency priority (based on the assumption that SsbFreqPos is set to 8000 in TDD
scenarios)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, MlbFreqPriority=1;
//Turning on ONLY_STRONGEST_CELL_SW
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=ONLY_STRONGEST_CELL_SW-1;
//Turning on INTER_FREQ_HO_SUPPRESS_SW for a specific QCI of UEs so that UEs will not be selected for
handover
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Setting MlbUeNumJudgeFactor
MOD NRCELLMLB: NrCellId=0, MlbUeNumJudgeFactor=50;
//Enabling precise load evaluation when the number of available PRBs in the cell dynamically changes
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoEnhSwitch=PRECISE_LOAD_EVAL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

After this function is enabled, if the cell load meets the conditions for triggering load redistribution described in **6.1.5 Load Redistribution**, the following are true:

- If the **SA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and the **N.HO.InterFreq.Mlb.PrepAttOut** counter value is not 0, this function has taken effect.
- If the **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Att** counter value is not 0, this function has taken effect.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. If BIT0 or BIT1 of the binary number converted from the decimal value of **MLB Status** is **1**, MLB has taken effect.

6.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful MLB-based inter-frequency outgoing handovers.

Observe the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counter to obtain the number of successful MLB-based inter-frequency PSCell changes in the LTE-NR NSA DC scenario.

Observe the **N.InterFreq.Mlb.Num** counter to obtain the number of times a cell enters the uplink or downlink MLB state.

Observe the **N.InterFreq.Mlb.Dur** counter to obtain the duration in which a cell is in the uplink or downlink MLB state.

7 5QI-specific UE-Number-based Connected Mode MLB

7.1 Principles

In 5QI-specific UE-number-based connected mode MLB, before entering or exiting MLB, the number of UEs running services with a specified 5QI is used as an indicator to indicate the cell load. After entering or exiting MLB, when the 5QI-specific UE-number-based air interface load is unbalanced among cells, those UEs running services with a specified 5QI are handed over from a high-load cell to low-load cells to achieve a load balance. 5QI-specific UE-number-based connected mode MLB is enabled if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected and the **NRCellMlb**.***MlbTriggerMode*** parameter is set to **SPECIFIED_5QI_USER_NUM**. The **NRCellMlb**.***MlbUe5qi*** parameter specifies the 5QI and UEs running services of this 5QI can be selected for MLB.

7.1.1 Entering and Exiting MLB

After this function is enabled, the gNodeB determines whether a cell enters or exits the MLB state in the next MLB evaluation period based on the cell load within the current period. When the cell enters the MLB state, it will instruct some UEs to hand over to other low-load cells.

Based on the MLB direction, MLB is divided into downlink MLB and uplink MLB.

Downlink MLB

- Entering condition: Throughout an MLB evaluation period, the number of UEs running downlink services with a specified 5QI is greater than or equal to the sum of the UE number threshold for inter-frequency downlink MLB and offset to the number of UEs for downlink MLB.
- Exiting condition: Throughout an MLB evaluation period, the number of UEs running downlink services with a specified 5QI is less than the UE number threshold for inter-frequency downlink MLB.

Uplink MLB

- Entering condition: Throughout an MLB evaluation period, the number of UEs running uplink services with a specified 5QI is greater than or equal to the sum of the UE number threshold for inter-frequency uplink MLB and offset to the number of UEs for uplink MLB.
- Exiting condition: Throughout an MLB evaluation period, the number of UEs running uplink services with a specified 5QI is less than the UE number threshold for inter-frequency uplink MLB.

The following describes the parameters involved in the conditions for entering and exiting the MLB state.

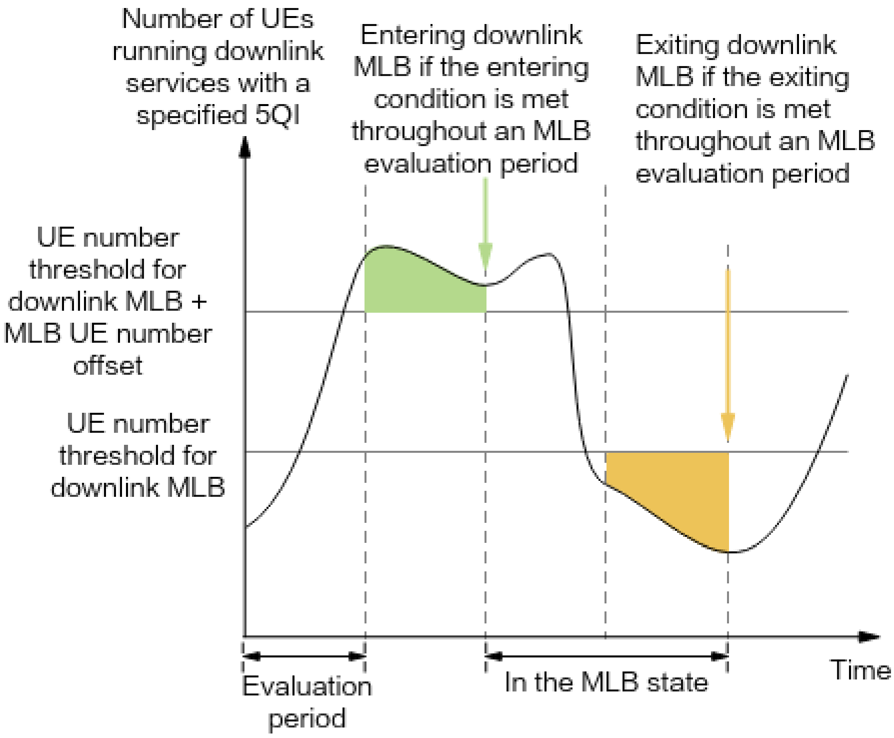
- The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.
- The MLB direction is configured using the **NRCellMlb**.***MlbDirection*** parameter.
  - **DOWNLINK**: indicates that downlink MLB is enabled.
  - **UPLINK**: indicates that uplink MLB is enabled.
- The downlink UE number threshold is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The offset to the number of UEs for downlink MLB is specified by the **NRCellMlb**.***MlbUeNumOffset*** parameter.
- The uplink UE number threshold is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The offset to the number of UEs for uplink MLB is specified by the **NRCellMlb**.***MlbUlUeNumOffset*** parameter.

NOTE

The number of 5QI-specific UEs is the number of UEs running services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter. For details about the definitions of the number of 5QI-specific UEs in the downlink and in the uplink, see **4.2.3 Number of 5QI-specific UEs**.

If the gNodeB determines that a cell meets the conditions for entering or exiting MLB in the current MLB evaluation period, the cell enters or exits MLB in the next MLB evaluation period. **Figure 7-1** uses downlink MLB as an example and shows the conditions for entering and exiting downlink MLB.

Figure 7-1 Conditions for entering and exiting downlink MLB



NOTE

When the gNodeB measures the number of UEs in connected mode running services with a 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter in a cell, the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter must be set to **YES** for the 5QI.

7.1.2 Candidate Cell Range Identification

The gNodeB selects candidate cells for handovers. Neighboring cells each of which meets all of the following conditions can be selected as candidate cells:

- The cell is configured as a neighboring cell of the serving cell through the **NRCellRelation** MO.
- The cell operates on a different frequency than the serving cell.
- The cell allows load information exchange. That is, the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the cell.
- The serving cell and the neighboring cell are both SA cells, and SA UEs can be handed over to the neighboring cell.

  NOTE

  - The SA/NSA type of the serving cell and its neighboring cells is specified by the **NRDUCellOp**.***NrNetworkingOption*** parameter. A cell is considered to be in SA networking only when the networking option of all operators of the cell is SA.
  - The SA/NSA type of an external neighboring cell is specified by the **NRExternalNCell**.***NrNetworkingOption*** parameter. A cell is considered to be in SA networking only when the networking option of all operators of the cell is SA.
  - Whether a neighboring cell allows incoming handovers of SA UEs is specified by the **NRCellRelation**.***NoHoFlag*** parameter.

- The cell allows incoming handovers. That is, the **NRCellRelation**.***NoHoFlag*** parameter is set to **PERMIT_HO** for the cell.
- The cell is in the active state, which is indicated by the **NR Cell State** parameter in the **DSP NRCELL** command output.

NOTE

If the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected, cells already in the connected mode MLB state will not be selected as target cells for non-MLB-based inter-frequency handovers. For details, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

The total number of neighboring cells that can be selected as candidate cells is subject to whether the MLB neighboring cell specification improvement function is enabled. This function is controlled by the **MLB_NCELL_SPEC_EXTEND_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.

- When this function is disabled, a maximum of 50 cells that meet the specific conditions can be selected as candidate cells.
- When this function is enabled, all cells that meet the specific conditions can be selected as candidate cells.

7.1.3 Inter-Cell Load Information Exchange

The serving cell exchanges load information with candidate cells.

- The gNodeB triggers intra-base-station load information exchanges between the serving cell and the candidate cells by default. For details, see **5.1.3.1 Intra-gNodeB Inter-Cell Load Information Exchange**.
- If there are candidate inter-gNodeB cells, the gNodeB triggers inter-gNodeB inter-cell load information exchanges. For details, see **5.1.3.2 Inter-gNodeB Inter-Cell Load Information Exchange**.

7.1.3.1 Intra-gNodeB Inter-Cell Load Information Exchange

The exchanged load information includes:

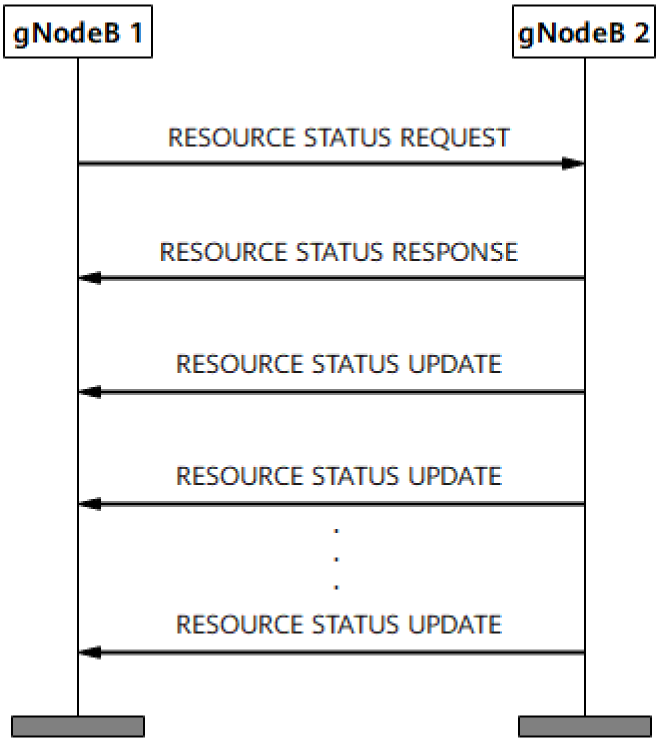
- Proportion of UEs in RRC_CONNECTED mode in a cell, which is equal to the number of UEs in a cell divided by the maximum number of UEs in RRC_CONNECTED mode in the cell. For details about the maximum number of UEs in RRC_CONNECTED mode in a cell, see "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- 5QI-specific UE-number-based air interface load of a cell. For details, see **4.2.4 5QI-specific UE-Number-based Air Interface Load**.
- Uplink cell bandwidth
- Downlink cell bandwidth

7.1.3.2 Inter-gNodeB Inter-Cell Load Information Exchange

The inter-gNodeB inter-cell load information exchange procedure is performed using the method defined in section 8.4.10 "Resource Status Reporting Initiation" in 3GPP TS 38.423 V16.4.0, as shown in **Figure 7-2**. The exchanged load information includes:

- Proportion of UEs in RRC_CONNECTED mode in a cell, which is equal to the number of UEs in a cell divided by the maximum number of UEs in RRC_CONNECTED mode in the cell. For details about the maximum number of UEs in RRC_CONNECTED mode in a cell, see "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- 5QI-specific UE-number-based air interface load of a cell. For details, see **4.2.4 5QI-specific UE-Number-based Air Interface Load**.
- Uplink cell bandwidth
- Downlink cell bandwidth

Figure 7-2 Inter-gNodeB inter-cell load information exchange procedure



NOTE

During inter-gNodeB inter-cell load information exchange, Xn self-setup can be triggered when there is no Xn interface between gNodeBs to support later MLB procedures, in all SA, NSA, or NSA and SA hybrid networking. The **XN_SON_SETUP_SW** option of the **gNBXnSonConfig**.***XnSonConfigSwitch*** parameter specifies whether to enable Xn self-setup. For details, see *NG and Xn Self-Management*.

7.1.4 Candidate Cell Selection

The gNodeB compares the load information of the serving cell with that of all candidate cells.

If at least one candidate cell meets all the following conditions, the gNodeB considers that there are candidate cells available for load redistribution and proceeds to it. Otherwise, the gNodeB considers that there is no candidate cell available for load redistribution and does not execute load redistribution in the current period.

- The load difference between the serving cell and the candidate cell is greater than the load difference threshold.
  - For downlink MLB

    Condition: The 5QI-specific UE-number-based load difference between the serving cell and the candidate cell is greater than the downlink load difference threshold (specified by the **NRCellMlb**.***UeNumDiffThld*** parameter).

    5QI-specific UE-number-based downlink air interface load difference between the serving cell and the candidate cell = (5QI-specific UE-number-based downlink air interface load of the serving cell – 5QI-specific UE-number-based downlink air interface load of the candidate cell)/5QI-specific UE-number-based downlink air interface load of the serving cell
  - For uplink MLB

    Condition: The 5QI-specific UE-number-based load difference between the serving cell and the candidate cell is greater than the uplink load difference threshold (specified by the **NRCellMlb**.***UlUeNumDiffThld*** parameter).

    5QI-specific UE-number-based uplink air interface load difference between the serving cell and the candidate cell = (5QI-specific UE-number-based uplink air interface load of the serving cell – 5QI-specific UE-number-based uplink air interface load of the candidate cell)/5QI-specific UE-number-based uplink air interface load of the serving cell

    NOTE

    For the definitions of the 5QI-specific UE-number-based uplink and downlink air interface loads, see **4.2.4 5QI-specific UE-Number-based Air Interface Load**.
- The proportion of UEs in connected mode in the candidate cell is less than 90%.

7.1.5 Load Redistribution

Load redistribution is a process in which some UEs in connected mode in the serving cell are handed over to candidate cells that meet certain signal quality requirements.

7.1.5.1 UE Selection

Each of the selected UEs must meet all of the following conditions:

- The UE is in connected mode.
- The 5QI of services that the UE is running is equal to the value of **NRCellMlb**.***MlbUe5qi***.
- The UE is not running a VoNR service. For details, see *VoNR*.
- The UE is an SA UE.
- The UE is not in the CA state.
- The UE is not a Cloud VR UE. (The **gNBQciBearer**.***ServiceType*** parameter is not set to **EMBB_CLOUD_VR**.)
- The UE supports MLB. UEs in any of the following scenarios do not support MLB:

  - The RAT/Frequency Selection Priority (RFSP) of the UE is configured on the core network, it matches the **gNBRfspConfig**.***RfspIndex*** parameter configured on the gNodeB, and the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter is deselected.
  - The RFSP of the UE is not configured on the core network, and the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** parameter corresponding to the **gNBRfspConfig**.***RfspIndex*** parameter with the value of **0** is deselected for the operator.
- A UE that fails to be handed over to a candidate cell for MLB can be selected again 120s after the handover is triggered, if any.
- The UE does not have any bearer of a QCI for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer**.***MobilityFunInd*** parameter is selected (indicating that unnecessary inter-frequency handovers are prohibited).

If the number of UEs selected in each evaluation period is greater than or equal to the **NRCellMlb**.***ConnMlbUeNumThld*** parameter, *N* UEs can be selected for load redistribution. The maximum value of *N* is specified by the **NRCellMlb**.***ConnMlbUeNumThld*** parameter.

7.1.5.2 Measurement-based Handover

Measurement-based handovers are performed for a selected UE. The gNodeB delivers the operating frequencies of candidate cells available for load redistribution and the information related to event A4 or A5 to the UE for measurements. The UE performs measurements and sends a measurement report to the gNodeB when the event entering conditions are met. After receiving the measurement report, the gNodeB in the serving cell checks whether the reported cells are within the candidate cell range for load redistribution and determines whether to instruct the UE to perform a handover.

For details about the basic procedure of measurement-based handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. This chapter describes only the parts that are different from the basic procedure, including measurement configuration delivery and serving cell determination.

7.1.5.2.1 Measurement Configuration Delivery

This procedure is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.2.1 Measurement Configuration Delivery**.

7.1.5.2.2 Serving Cell Determination

The gNodeB determines whether the reported cells are candidate cells that can be used for load redistribution.

- If none of the reported cells is a candidate cell for load redistribution, no handover is performed.
- If only one cell in the measurement report is a candidate cell for load redistribution, a handover to this cell is directly performed for the UE.
- Assume that multiple cells in the measurement report are candidate cells for load redistribution:

- If the **ONLY_STRONGEST_CELL_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter is selected, the gNodeB instructs the UE to perform a handover only to the cell with the highest signal quality, and handovers to other cells will not be performed once this handover fails.
- If the **ONLY_STRONGEST_CELL_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter is deselected, a handover is attempted to the cell with the highest signal quality. When the handover fails, a handover to the cell with the next highest signal quality is attempted, until the handover succeeds or handovers to all candidate cells for load redistribution have been attempted.

7.2 Network Analysis

7.2.1 Benefits

5QI-specific UE-number-based connected mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains.

The following must also be considered when using this function:

- If the numbers of UEs among inter-frequency co-coverage cells are not independent, for example, when a large number of UEs perform CA in two cells, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- When UEs in the inter-frequency co-coverage cells are high-mobility UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- If the traffic volume of services of a specified 5QI (specified by **NRCellMlb**.***MlbUe5qi***) is small, the gains of this function will be reduced.
- The gains of this function will not be as high as expected when cells on multiple frequencies differ greatly in their coverage capabilities, UEs differ greatly in their band support capabilities, or there are restrictions on the serving PLMNs of UEs due to UE subscription.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

After this function is enabled, measurement gaps are set up for inter-frequency measurements by candidate UEs for MLB. Gap-assisted measurements will cause a performance loss in terms of the traffic volume.

After this function is enabled, if the number of UEs running services with a 5QI (specified by the **NRCellMlb**.***MlbUe5qi*** parameter) in a cell is small, ping-pong handovers may occur between cells in the following scenarios, affecting user experience:

- In downlink MLB, the sum of the **NRCellMlb**.***InterFreqMlbUeNumThld*** and **NRCellMlb**.***MlbUeNumOffset*** parameter values is small.

- In uplink MLB, the sum of the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** and **NRCellMlb**.***MlbUlUeNumOffset*** parameter values is small.
- The **NRCellMlb**.***ConnMlbUeNumThld*** parameter value is large.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*
Low-frequency TDD	Hyper Cell	**HYPER_CELL** option of the **NRDUCell**.***NrDuCellNetworkingMode*** parameter	*Hyper Cell*
Low-frequency TDD	Cell Combination	**HYPER_CELL_COMBINE_MODE** option of the **NRDUCell**.***NrDuCellNetworkingMode*** parameter	*Cell Combination*
Low-frequency TDD	Idle mode MLB	**INTER_FREQ_IDLE_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter	*Mobility Load Balancing in Idle Mode*

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	SUL and NR FDD co-deployment	**FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*
Low-frequency TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None

7.3.2.3 Function Impacts

- UEs with any of the following listed functions in effect will not be further selected for MLB. However, during an MLB procedure, any of these functions can take effect for a UE already selected for a handover, and if that occurs, the current MLB procedure will continue without interruption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	When selecting UEs to transfer, the gNodeB excludes those running VoNR services.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When selecting UEs to transfer, the gNodeB excludes those with the low latency and high reliability function enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter LTE: **NSA_LTE_FDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	After the "NSA carrier combination selection based on user experience" function takes effect, 5QI-specific UE-number-based connected mode MLB does not take effect for NSA UEs in the cell.

- Impact relationships with other handover functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	If both coverage-based inter-frequency handover and connected mode MLB are enabled, coverage-based inter-frequency handovers are triggered by event A5, and the value of **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** is less than or equal to the value of **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***, a ping-pong effect between coverage-based handovers and MLB-based handovers will occur.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both 5QI-specific UE-number-based connected mode MLB and experience-based smart carrier selection are enabled. If a handover is triggered for experience-based smart carrier selection, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with 5QI-specific UE-number-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarriersSwitch*** parameter	*Multi-Frequency Smart Selection*	• If multi-frequency beam coordination and 5QI-specific UE-number-based connected mode MLB are both enabled for a cell, multi-frequency beam coordination does not take effect. • Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both 5QI-specific UE-number-based connected mode MLB and multi-frequency beam coordination are enabled. If a handover is triggered for multi-frequency beam coordination, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	**ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both 5QI-specific UE-number-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If an inter-frequency handover is triggered based on energy efficiency, cells in the MLB state will be filtered out, avoiding ping-pong handovers.
Low-frequency TDD	Uplink-experience-based inter-frequency handover	**UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink-experience-based inter-frequency handover does not take effect for RedCap UEs that are handed over to the local cell through connected mode MLB.

- Impact relationships with other functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter	*ANR*	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the neighboring cell.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

There must be at least two frequencies available on the live network.

An **NRCellFreqRelation** MO must be already configured.

7.3.5 Cells

If the **NRDUCell**.***DuplexMode*** parameter is not set to **CELL_TDD**, the **NRCellMlb**.***MlbTriggerMode*** parameter cannot be set to **SPECIFIED_5QI_USER_NUM**.

7.3.6 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Select this option.
MLB Trigger Mode	**NRCellMlb**.***MlbTriggerMode***	None	Set this parameter to **SPECIFIED_5QI_USER_NUM**.

Table 7-2 describes other parameters that are also involved in MLB:

Table 7-2 Parameters involved in 5QI-specific UE-number-based connected mode MLB

Parameter Name	Parameter ID	Option	Setting Notes
MLB UE 5QI	**NRCellMlb**.***MlbUe5qi***	None	Set this parameter based on the network plan.
Counter Object Subscription Indicator	**gNB5qiConfig**.***CounterObjSubscrip-tionInd***	None	Set the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter to **YES** for the 5QI specified by the **NRCellMlb**.***MlbUe5qi*** parameter.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Judge Period	**NRCellMlb**.***MlbJudgePeriod***	None	Set this parameter to its recommended value.
MLB Direction	**NRCellMlb**.***MlbDirection***	None	Set this parameter based on the network plan.
Inter-frequency MLB UE Number Threshold	**NRCellMlb**.***InterFreqMlbUeNumThld***	None	Set this parameter based on the network plan.
MLB UE Number Offset	**NRCellMlb**.***MlbUeNumOffset***	None	Set this parameter based on the network plan.
Inter-frequency MLB UL UE Number Thld	**NRCellMlb**.***InterFreqMlbUlUeNumThld***	None	Set this parameter based on the network plan.
MLB UL UE Number Offset	**NRCellMlb**.***MlbUlUeNumOffset***	None	Set this parameter based on the network plan.
MLB-Based Handover Flag	**NRCellRelation**.***MlbHoFlag***	None	Set this parameter based on the network plan.
No Handover Flag	**NRCellRelation**.***NoHoFlag***	None	Set this parameter based on the network plan.
UE Number Difference Threshold	**NRCellMlb**.***UeNumDiffThld***	None	Set this parameter based on the network plan.
UL UE Number Difference Threshold	**NRCellMlb**.***UlUeNumDiffThld***	None	Set this parameter based on the network plan.
Connected MLB UE Num Upper Limit	**NRCellMlb**.***ConnMlbUeNumThld***	None	It is recommended that this parameter be set to a value greater than or equal to **20**.

Parameter Name	Parameter ID	Option	Setting Notes
MLB-based Inter-Frequency RSRP Threshold	**NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld***	None	Set this parameter based on the network plan. The event A4 threshold or event A5 threshold 2 for each type of inter-frequency handover must be higher than the event A2 threshold for coverage-based inter-frequency handover to ensure that a coverage-based inter-frequency measurement is not triggered immediately after a UE is handed over to the target frequency. This helps decrease the probability of ping-pong handovers.
Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig***	None	Set this parameter based on the network plan.
Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst***	None	Set this parameter based on the network plan.
MLB Enhancement Algorithm Switch	**NRCellMlb**.***MlbAlgoEnhSwitch***	ONLY_STRONGEST_CELL_SW	Set this parameter based on the network plan.
MLB Enhancement Algorithm Switch	**NRCellMlb**.***MlbAlgoEnhSwitch***	MLB_NCELL_SPEC_EXTEND_SW	It is recommended that this option be selected when the local cell has more than 50 neighboring cells for MLB.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch[a]	**gNBRfspConfig.*MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Set this parameter based on the network plan. To disable MLB for UEs with a specific RFSP, deselect the **INTER_FREQ_CONNECTED_MLB_SW** option.
a: If a UE initiates an access request, the following are true: (1) If the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig.*MlbAlgoSwitch*** is deselected for the RFSP of the UE, the gNodeB regards that this UE does not support MLB after the access. (2) If the **INTER_FREQ_CONNECTED_MLB_SW** option is selected or the **gNBRfspConfig**.***RfspIndex*** parameter is not configured for the RFSP of the UE, the gNodeB regards that this UE always supports MLB after the access.			

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-1;
//Setting the MLB trigger mode
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPECIFIED_5QI_USER_NUM;
```

Optimization Command Examples

```
//Setting the 5QI, of which UEs running services can be selected for MLB
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=SPECIFIED_5QI_USER_NUM, MlbUe5qi=6;
//Setting the CounterObjSubscriptionInd parameter for a specific 5QI when UEs running services with the 5QI can be selected for MLB
MOD GNB5QICONFIG: Nr5qi=6, CounterObjSubscriptionInd=YES;
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MLB direction for the cell
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Enabling load information exchange with the serving cell for a neighboring cell
```

```
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE, NoHoFlag=PERMIT_HO;
//In inter-gNodeB MLB scenarios, setting the CounterObjSubscriptionInd parameter on the gNodeB serving the inter-gNodeB neighboring cells for a specific 5QI when UEs running services with this 5QI can be selected for MLB in the local cell where MLB is triggered
MOD GNB5QICONFIG: Nr5qi=6, CounterObjSubscriptionInd=YES;
//Disabling MLB for UEs with a specific RFSP
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Setting the InterFreqMlbA4RsrpThld parameter
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqMlbA4RsrpThld=-102;
//Setting the InterFreqA4A5TimeToTrig and InterFreqA4A5Hyst parameters
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
//Setting the MLB frequency priority (based on the assumption that SsbFreqPos is set to 8000 in TDD scenarios)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, MlbFreqPriority=1;
//Turning on the MLB neighboring cell specification extension switch
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=MLB_NCELL_SPEC_EXTEND_SW-1;
//Turning on ONLY_STRONGEST_CELL_SW
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=ONLY_STRONGEST_CELL_SW-1;
//Turning on INTER_FREQ_HO_SUPPRESS_SW for a specific QCI of UEs so that UEs will not be selected for handover
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

After this function is enabled, if the cell load meets the conditions for triggering load redistribution described in **7.1.5 Load Redistribution** and the value of **N.HO.InterFreq.Mlb.PrepAttOut** is not zero, this function has taken effect.

7.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful MLB-based inter-frequency outgoing handovers.

Observe the **N.InterFreq.Mlb.Num** counter to obtain the number of times a cell enters the uplink or downlink MLB state.

Observe the **N.InterFreq.Mlb.Dur** counter to obtain the duration in which a cell is in the uplink or downlink MLB state.

8 Radio-Resource-based Connected Mode MLB

8.1 Principles

Radio-resource-based connected mode MLB uses the number of UEs or equivalent CCE usage in a cell as the cell load indicator to determine whether the cell should enter or exit MLB. For cells in the MLB state, their air interface loads or equivalent CCE usage is evaluated. If the air interface load or equivalent CCE usage is unbalanced among cells, UEs in connected mode are handed over from high-load cells to low-load cells to achieve a load balance. Radio-resource-based connected mode MLB is enabled if the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected and the **NRCellMlb**.***MlbTriggerMode*** parameter is set to **RADIO_RESOURCE**.

It is recommended that this function be enabled when all of the following conditions are met:

- The same proportions of small-packet and large-packet services are independently running in inter-frequency co-coverage cells.
- Most UEs are motionless or move slowly.
- The source and target cells are intra-base-station cells. Alternatively, the cells are served by Huawei base stations, and an Xn interface has been set up between the base stations.

8.1.1 Entering and Exiting MLB

After this function is enabled, the gNodeB determines whether a cell enters or exits the MLB state in the next MLB evaluation period based on the cell load within the current period. If the gNodeB determines that a cell meets the conditions for entering or exiting MLB in the current MLB evaluation period, the cell enters or exits MLB in the next MLB evaluation period.

Based on the direction, MLB includes downlink MLB, uplink MLB, as well as downlink and uplink MLB. Conditions for entering and exiting MLB are described as follows, which vary depending the MLB direction. For details about the definitions of the numbers of uplink and downlink UEs and equivalent CCE usage in a cell, see **4.2 Load Evaluation Dimensions**. The following describes the parameters involved in the conditions for entering and exiting the MLB state.

- The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.
- The MLB direction is configured using the **NRCellMlb**.***MlbDirection*** parameter.
  - **DOWNLINK**: indicates that downlink MLB is enabled.
  - **UPLINK**: indicates that uplink MLB is enabled.
  - **DUAL**: indicates that uplink and downlink MLB are enabled.
- The UE number threshold for inter-frequency downlink MLB is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The offset to the downlink MLB UE number is specified by the **NRCellMlb**.***MlbUeNumOffset*** parameter.
- The CCE usage threshold for downlink MLB is specified by the **NRCellMlb**.***MlbDlCceUsageThld*** parameter.
- The UE number threshold for inter-frequency uplink MLB is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The offset to the uplink MLB UE number is specified by the **NRCellMlb**.***MlbUlUeNumOffset*** parameter.
- The CCE usage threshold for uplink MLB is specified by the **NRCellMlb**.***MlbUlCceUsageThld*** parameter.

Downlink MLB

The conditions for entering and exiting the downlink MLB state are as follows:

- Entering occurs when any of the following conditions is met:
  - The number of UEs in the downlink is greater than or equal to the sum of the inter-frequency MLB downlink UE number threshold and MLB downlink UE number offset throughout an MLB evaluation period.
  - The conditions for the equivalent CCE usage to take effect are met, and the downlink equivalent CCE usage is greater than or equal to the MLB downlink CCE usage threshold.

    The equivalent CCE usage takes effect when all of the following conditions are met:

    - The adaptively selected number of PDCCH symbols for the serving cell has reached the **NRDUCellPdcch**.***UePdcchAdaptMaxSymNum*** parameter value, or the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the NRDUCellPdcch.***PdcchAlgoExtSwitch*** parameter is deselected.
    - The **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter is deselected, or this option is selected and the function has taken effect.
- Exiting occurs when all of the following conditions are met:
  - The number of UEs in the downlink is less than the inter-frequency MLB downlink UE number threshold throughout an MLB evaluation period.

- The conditions for the equivalent CCE usage to become invalid are met, or the downlink equivalent CCE usage is less than the MLB downlink CCE usage threshold minus 15%.

  The equivalent CCE usage becomes invalid when one of the following conditions is met:

  - The **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter is selected for the serving cell, and the adaptively selected number of PDCCH symbols has not reached the **NRDUCellPdcch**.***UePdcchAdaptMaxSymNum*** parameter value.
  - The **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter is selected for the serving cell but the function does not take effect. For details about DCI extension upon uplink congestion, see *Uplink Scheduling*.

Uplink MLB

The conditions for entering and exiting the uplink MLB state are as follows:

- Entering occurs when any of the following conditions is met:
  - The number of UEs in the uplink is greater than or equal to the sum of the inter-frequency MLB uplink UE number threshold and MLB uplink UE number offset throughout an MLB evaluation period.
  - The conditions for the equivalent CCE usage to take effect are met, and the uplink equivalent CCE usage is greater than or equal to the MLB uplink CCE usage threshold. Uplink equivalent CCE usage equals to **N.CCE.ULDCI.Used.Equivalent** divided by **N.CCE.ULDCI.Avail.Equivalent**.

  The conditions for the equivalent CCE usage to take effect are the same as those in downlink MLB.
- Exiting occurs when all of the following conditions are met:
  - The number of UEs in the uplink is less than the inter-frequency MLB uplink UE number threshold throughout an MLB evaluation period.
  - The conditions for the equivalent CCE usage to become invalid are met, or the uplink equivalent CCE usage is less than the MLB uplink CCE usage threshold minus 15%.

  The conditions for the equivalent CCE usage to become invalid are the same as those in downlink MLB.

Downlink and Uplink MLB

The conditions for entering and exiting the uplink and downlink MLB state are as follows:

- Entering condition: The entering condition for the uplink or downlink MLB state is met.
- Exiting condition: The exiting conditions for both the uplink and downlink MLB states are met.

If the gNodeB determines that a cell meets the conditions for entering or exiting MLB in the current MLB evaluation period, the cell enters or exits MLB in the next MLB evaluation period.

8.1.2 Candidate Cell Range Identification

The gNodeB selects candidate cells for UE handovers. Candidate cell range identification here is the same as that for UE-number-based connected mode MLB. For details, see **5.1.2 Candidate Cell Range Identification**.

8.1.3 Inter-Cell Load Information Exchange

Compared with the inter-cell load information exchange of UE-number-based connected mode MLB, radio-resource-based connected mode MLB exchange the following additional information:

- Uplink equivalent CCE usage of a cell
- Downlink equivalent CCE usage of a cell

For details about the load information exchanged in UE-number-based connected mode MLB, see **5.1.3 Inter-Cell Load Information Exchange**.

8.1.4 Candidate Cell Selection

The gNodeB compares the load information of the serving cell with that of all candidate cells.

If any of the following conditions is met, the gNodeB considers that there are candidate cells available for load redistribution and proceeds to it. Otherwise, the gNodeB considers that there is no candidate cell available for load redistribution and does not execute load redistribution in the current period.

- The load difference between the serving cell and at least one candidate cell is greater than the load difference threshold.
  - For downlink MLB

    Condition: The downlink load difference between the serving cell and the candidate cell is greater than the downlink load difference threshold.

    Downlink load difference between the serving cell and the candidate cell = (Downlink air interface load of the serving cell – Downlink air interface load of the candidate cell)/Downlink air interface load of the serving cell. (The downlink load difference threshold is specified by the **NRCellMlb**.***UeNumDiffThld*** parameter.)
  - For uplink MLB

    Condition: The uplink load difference between the serving cell and the candidate cell is greater than the uplink load difference threshold.

    Uplink load difference between the serving cell and the candidate cell = (Uplink air interface load of the serving cell – Uplink air interface load of the candidate cell)/Uplink air interface load of the serving cell. (The uplink load difference threshold is specified by the **NRCellMlb**.***UlUeNumDiffThld*** parameter.)
  - For uplink and downlink MLB

    Any of the preceding load difference conditions is met.

NOTE

For details about the definitions of the uplink air interface load and downlink air interface load of a cell, see **4.2.2 Air Interface Load of a Cell**.

- The load difference between the serving cell and each of the candidate cells is less than or equal to the load difference threshold, and the equivalent-CCE-usage-based load difference between the serving cell and at least one candidate cell is greater than the equivalent-CCE-usage-based load difference threshold (10%).
  - For downlink MLB

    Condition: The downlink equivalent-CCE-usage-based load difference between the serving cell and a candidate cell is greater than 10%.

    Downlink equivalent-CCE-usage-based load difference between the serving cell and a candidate cell = Downlink equivalent CCE usage of the serving cell – Downlink equivalent CCE usage of the candidate cell,
  - For uplink MLB

    Condition: The uplink equivalent-CCE-usage-based load difference between the serving cell and a candidate cell is greater than 10%.

    Uplink equivalent-CCE-usage-based load difference between the serving cell and a candidate cell = Uplink equivalent CCE usage of the serving cell – Uplink equivalent CCE usage of the candidate cell,
  - For uplink and downlink MLB

    Any of the preceding load difference conditions is met.

  NOTE

  For the definitions of downlink equivalent CCE usage and uplink equivalent CCE usage of a cell, see **4.2.5 Equivalent CCE Usage**.
- The candidate cell is not in the high-load state.
- The candidate cell is not in the interference state. A cell is in the interference state when it meets the conditions for starting uplink-interference-based inter-frequency handover. For details about uplink-interference-based inter-frequency handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

8.1.5 Load Redistribution

Load redistribution is a process in which some UEs in connected mode in the serving cell are handed over to candidate cells that meet certain signal quality requirements.

8.1.5.1 UE Selection

The procedure for selecting a UE to be handed over is as follows:

1. Initial selection is performed according to basic conditions that UEs must meet. The initial selection procedure is basically the same as that for UE-number-based connected mode MLB. The difference is that radio-resource-based connected mode MLB is implemented only for non-CA UEs; that is, the **NRCellMlb**.***MlbTransCA*** parameter must be set to **NULL** for radio-resource-based connected mode MLB. For the basic conditions that UEs must meet in UE-number-based connected mode MLB, see **5.1.5.1.1 Conditions Used for Initial Selection**.
2. The priorities of the initially selected UEs are determined according to UE selection policies based on radio bearer attributes. UE priority determination

is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.1.2 UE Priority Determination**.

3. The upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period is calculated. For details about the calculation, see **Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution**.
4. The initially-selected UEs are then selected in descending order of UE priorities for measurement-based handovers, and the number of selected UEs cannot exceed N_{max}. If UEs have the same priority, the gNodeB randomly selects one from them.

Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution

In downlink MLB, the upper limit (N_{max}) of the number of UEs involved in load redistribution within each MLB evaluation period is calculated as follows:

- If the triggering variable is the UE-number-based load, N_{max} takes the minimum value among the following three values:
  – [Current number of UEs in the serving cell – (Downlink UE number threshold for inter-frequency MLB – 1)] x *T*
  – MLB UE number judge factor x (Downlink air interface load in the serving cell – Minimum of the downlink air interface loads in the candidate cells) x Downlink air interface capability of the cell x *T*
  – Connected MLB UE Num Upper Limit
- If the triggering variable is the equivalent-CCE-usage-based load, N_{max} takes the minimum value among the following three values:
  – [Current number of UEs in the serving cell – (Downlink UE number threshold for inter-frequency MLB – 1)] x *T*
  – MLB UE number judge factor x (Downlink equivalent CCE usage of the serving cell – Minimum of the downlink equivalent CCE usage among candidate cells)/Downlink equivalent CCE usage of the serving cell x Current number of UEs in the serving cell x *T*
  – Connected MLB UE Num Upper Limit

In uplink MLB, the upper limit (N_{max}) of the number of UEs involved in load redistribution within each MLB evaluation period is calculated as follows:

- If the triggering variable is the UE-number-based load, N_{max} takes the minimum value among the following three values:
  – [Current number of UEs in the serving cell – (Uplink UE number threshold for inter-frequency MLB – 1)] x *T*
  – MLB UE number judge factor x (Uplink air interface load in the serving cell – Minimum of the uplink air interface loads in the candidate cells) x Uplink air interface capability of the cell x *T*
  – Connected MLB UE Num Upper Limit
- If the triggering variable is the equivalent-CCE-usage-based load, N_{max} takes the minimum value among the following three values:

- [Current number of UEs in the serving cell – (Uplink UE number threshold for inter-frequency MLB – 1)] x *T*
- MLB UE number judge factor x (Uplink equivalent CCE usage of the serving cell – Minimum of the uplink equivalent CCE usage among candidate cells)/Uplink equivalent CCE usage of the serving cell x Current number of UEs in the serving cell x *T*
- Connected MLB UE Num Upper Limit

The involved parameters are as follows:

- The value of T depends on the MLB evaluation period.
  - If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is greater than or equal to 10s, *T* is 1.
  - If the value of the **NRCellMlb**.***MlbJudgePeriod*** parameter is less than 10s, *T* is the value of **NRCellMlb**.***MlbJudgePeriod*** divided by 10.
- The downlink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter.
- The uplink UE number threshold for inter-frequency MLB is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The MLB UE number judge factor is specified by the **NRCellMlb**.***MlbUeNumJudgeFactor*** parameter.
- The UE number upper limit for connected mode MLB is specified by the **NRCellMlb**.***ConnMlbUeNumThld*** parameter.

NOTE

For details about the definitions of the uplink and downlink air interface load and uplink and downlink air interface capability of a cell, see **4.2.2 Air Interface Load of a Cell**.

8.1.5.2 Measurement-based Handover

Measurement-based handovers are performed for a selected UE. The gNodeB delivers the operating frequencies of candidate cells available for load redistribution and the information related to event A4 or A5 to such a UE, and the UE performs measurements. When the event entering conditions are met, the UE sends a measurement report to the gNodeB. After receiving the measurement report, the gNodeB in the serving cell checks whether the reported cells are within the candidate cell range for load redistribution and determines whether to instruct the UE to perform a handover. During the handover, the target cell also evaluates whether to allow this UE to access its network.

For details about the basic procedure of measurement-based handover, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. This section describes only the parts that have been adapted to accommodate MLB, including measurement configuration delivery, serving cell determination, and target cell determination.

8.1.5.2.1 Measurement Configuration Delivery

This procedure is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.2.1 Measurement Configuration Delivery**.

8.1.5.2.2 Serving Cell Determination

The gNodeB determines whether the reported cells are candidate cells that can be used for load redistribution. For details about the conditions, see **8.1.4 Candidate Cell Selection**.

- If none of the reported cells is a candidate cell for load redistribution, no handover is performed.
- If the triggering variable for MLB is the number of UEs in connected mode, and candidate cells are selected based on the load difference of the number of UEs, candidate UEs are classified based on PRB usage, and candidate cells in the measurement reports are classified based on load difference. Then the base station matches the candidate UEs and cells. Once the match between a UE and a cell succeeds, a handover of the UE to the cell can be initiated. For details about the matching process, see **6.1.5.2.2 Serving Cell Determination**.

8.1.5.2.3 Target Cell Determination

This procedure is the same as that for UE-number-based connected mode MLB. For details, see **5.1.5.2.1 Measurement Configuration Delivery**.

8.2 Network Analysis

8.2.1 Benefits

Radio-resource-based connected mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains. Radio-resource-based connected mode MLB transfers UEs in connected mode from high-load cells to low-load cells to achieve a load balance between cells.

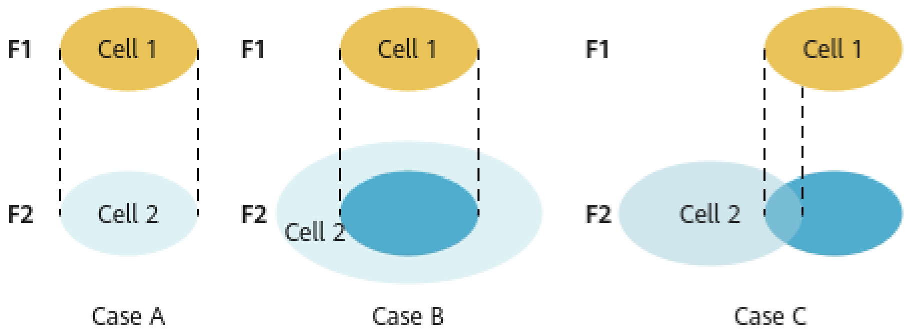
The following must also be considered when using this function:

- If the numbers of UEs in inter-frequency co-coverage cells are not independent, for example, when CA or SUL between two cells is performed for a large number of UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- When UEs in the inter-frequency co-coverage cells are high-mobility UEs, this function brings neither positive gains nor negative impacts. Therefore, it is recommended that this function be disabled.
- The gains of this function will not be as high as expected when UEs differ greatly in their band support capabilities or there are restrictions on the serving PLMNs of UEs due to UE subscription.
- The gains of this function will not be as high as expected when cells on multiple frequencies differ greatly in their coverage capabilities, as shown in **Figure 8-1**.
  - Case A: When two inter-frequency cells (cell 1 and cell 2) cover the exact same area, the load balancing transfer between the two cells is the most efficient and effective.
  - Case B: When the coverage area of cell 2 completely includes that of cell 1, the load balancing transfer from cell 1 to cell 2 is the most efficient

and effective, but the load balancing transfer from cell 2 to cell 1 cannot achieve the optimal efficiency and gains. The larger the coverage difference between the two cells, the lower the efficiency and the less the gains of load balancing transfer from cell 2 to cell 1.

– Case C: When the coverage areas of cell 1 and cell 2 overlap partially, the optimal efficiency and gains cannot be achieved regardless of whether load balancing transfer is from cell 1 to cell 2 or from cell 2 to cell 1. The larger the coverage difference between the two cells, the less efficient and effective the load balancing transfer is.

Figure 8-1 Coverage overlapping between cells



8.2.2 Impacts

The impacts between this function and other functions are described in **8.3.2.3 Function Impacts**.

After this function is enabled, measurement gaps are set up for inter-frequency measurements by candidate UEs for MLB. Gap-assisted measurements will cause a performance loss in terms of the traffic volume.

When both MLB and unnecessary handovers except MLB-based handovers take effect, MLB-oriented UE selection policies are required. For example, the base station can differentiate between the UEs that MLB selects for load transfer and UEs for which the unnecessary handovers are performed based on the RFSPs or QCIs, preventing ping-pong handovers between cells.

If large-packet and small-packet services account for different proportions in co-coverage inter-frequency cells, the number of UEs cannot correctly reflect the cell load. In this case, enabling MLB may cause an overload in a cell with a large number of large-packet services.

The MLB frequency priorities specified by the **NRCellFreqRelation**.***MlbFreqPriority*** parameter must be set to the same value for inter-frequency cells involved in MLB. Otherwise, the number of handovers will increase significantly, and the average uplink and downlink UE throughputs may fluctuate. The fluctuation depends on the load of the cell to which UEs are handed over.

8.3 Requirements

8.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051212	Converged Load Balancing	NR0S000CLB00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MLB UE selection policy	**NRCellMlb**.***MlbUeSelectionPolicy***	*Mobility Load Balancing in Connected Mode*	Radio-resource-based connected mode MLB can be enabled only when at least one of the **SA_USER** and **NSA_USER** options of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected.

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None

8.3.2.3 Function Impacts

- UEs with any of the following listed functions in effect will not be further selected for MLB. However, during an MLB procedure, any of these functions can take effect for a UE already selected for a handover, and if that occurs, the current MLB procedure will continue without interruption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When selecting UEs to transfer, the gNodeB excludes those with CA in effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	When selecting UEs to transfer, the gNodeB excludes those running VoNR services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW** or **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When selecting UEs to be handed over, the gNodeB does not select UEs with super uplink enabled. This MLB function is not recommended in cells with a high penetration rate of UEs with super uplink enabled, because it cannot achieve the expected effects in such cells.
Low-frequency TDD	UL and DL decoupling	**NRDUCellAlgoSwitch.*UlDlDecouplingSwitch***	*UL and DL Decoupling*	When selecting UEs to be handed over, the gNodeB does not select UEs on which SUL has taken effect. This MLB function is not recommended in cells with a high penetration rate of UEs on which SUL has taken effect, because it cannot achieve the expected effects in such cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter LTE: **NSA_LTE_FDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	After the "NSA carrier combination selection based on user experience" function takes effect, radio-resource-based connected mode MLB does not take effect for NSA UEs in the cell.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When selecting UEs to transfer, the gNodeB excludes those with the low latency and high reliability function enabled.

- Impact relationships with other handover functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and frequency-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD	Operator-specific-priority-based inter-frequency handover	**OP_DED_PRI_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and operator-specific-priority-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based inter-frequency handover	**SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both MLB and service-based inter-frequency handover are enabled, the following is true: If a target cell rejects a handover triggered by either function from the source cell, the handover triggered by the other function from the source cell to the target cell will be prohibited within a short period of time.
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	If both coverage-based inter-frequency handover and connected mode MLB are enabled, coverage-based inter-frequency handovers are triggered by event A5, and the value of **NRCellInterFHoMeaGrp**.***InterFreqMlbA4RsrpThld*** is less than or equal to the value of **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***, then the MLB-based handover success rate will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that inter-frequency handover collaboration is enabled, and both radio-resource-based connected mode MLB and experience-based smart carrier selection are enabled. If a handover is triggered for experience-based smart carrier selection, cells in the MLB state will be filtered out, avoiding ping-pong handovers.
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with radio-resource-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage enabled. As such, it is recommended that the inter-frequency handover collaboration be enabled. In this way, cells in the MLB state will be filtered out, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarriersSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and that both radio-resource-based connected mode MLB and multi-frequency beam coordination are enabled. If a handover is triggered for multi-frequency beam coordination, cells in the MLB state will be filtered out, avoiding ping-pong handovers.
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	**ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both radio-resource-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If inter-frequency handovers are triggered based on energy efficiency, cells in the MLB state will not be selected as the target cells, avoiding ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Network slice admission control	**ADMISSION_BASED_ON_NS_ID_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	Assume that MLB is enabled and UEs in a network slice are selected for MLB. If the allowed number of UEs in a target cell is limited, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce the probability of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam**.***ResNotAvailForSliceTimer*** parameter must be set to a value greater than **20**. In this case, no UEs in the network slice will be handed over to the target cell within the specified period.
Low-frequency TDD	Uplink-experience-based inter-frequency handover	**UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink-experience-based inter-frequency handover does not take effect for RedCap UEs that are handed over to the local cell through connected mode MLB.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*CA Experience Improvement*	Assume that both radio-resource-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.

- Impact relationships with other functions are described in the table below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter	*ANR*	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the neighboring cell.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Networking

The serving cell has at least one inter-frequency co-coverage neighboring cell.

During inter-base-station MLB, the base stations involved must be provided by Huawei and equipped with the Xn interface.

An **NRCellFreqRelation** MO must be already configured.

8.3.5 Cells

NRDUCell.***DuplexMode*** must be set to **CELL_TDD**.

8.3.6 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Select this option.
MLB Trigger Mode	**NRCellMlb**.***MlbTriggerMode***	None	Set this parameter to **RADIO_RESOURCE**.

The following parameters are also involved in MLB:

- Parameters such as the MLB evaluation period, MLB direction, UE number threshold and offset, as well as CCE usage threshold must be configured for cells to enter and exit the MLB state. For details, see **Table 8-2**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the MLB handover indicator and no handover indicator must be configured for candidate cell range determination. For details, see **Table 5-6**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the MLB handover admission threshold and the Xn self-setup switch must be configured for inter-cell load information exchange. For details, see **Table 5-7**.
- Same as those for spectral-efficiency-based connected mode MLB. The UE number difference threshold and the parameters related to the air interface capability of a cell must be configured for candidate cell selection. For details, see **Table 6-2**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the UE selection policy, UE selection policy based on radio bearer attributes, and switches and thresholds for each policy must be configured for UE selection. For details, see **Table 5-9**.
- Same as those for UE-number-based connected mode MLB. Parameters such as the frequency selection policy, thresholds and hysteresis for related events, and traffic priorities must be configured for measurement configuration delivery. For details, see **Table 5-10**.
- Same as those for spectral-efficiency-based connected mode MLB. Parameters such as the MLB handover admission threshold, PRB usage threshold for UEs, and strongest-cell handover switch must be configured for the serving cell determination and target cell determination. For details, see **Table 6-4**.

Table 8-2 Parameters involved in the entering and exiting MLB procedure

Parameter Name	Parameter ID	Option	Setting Notes
MLB Judge Period	**NRCellMlb**.***MlbJudgePeriod***	None	Use the recommended value.
MLB Direction	**NRCellMlb**.***MlbDirection***	None	Set this parameter based on the network plan.
Inter-frequency MLB UE Number Threshold	**NRCellMlb**.***InterFreqMlbUeNumThld***	None	Set this parameter based on the network plan.
MLB UE Number Offset	**NRCellMlb**.***MlbUeNumOffset***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-frequency MLB UL UE Number Thld	**NRCellMlb**.***InterFreqMlbUlUeNumThld***	None	Set this parameter based on the network plan.
MLB UL UE Number Offset	**NRCellMlb**.***MlbUlUeNumOffset***	None	Set this parameter based on the network plan.
UE PDCCH Adaptation Max Symbol Number	**NRDUCellPdcch**.***UePdcchAdaptMaxSymNum***	None	Set this parameter based on the network plan.
PDCCH Algorithm Extension Switch	**NRDUCellPdcch**.***PdcchAlgoExtSwitch***	UE_PDCCH_SYM_NUM_ADAPT_SW	Set this parameter based on the network plan.
Uplink PUSCH Algorithm Switch	**NRDUCellPusch**.***UlPuschAlgoSwitch***	DCI_EXTEND_UL_CONGEST_SW	Set this parameter based on the network plan.
MLB DL CCE Usage Threshold	**NRCellMlb**.***MlbDlCceUsageThld***	None	Set this parameter based on the network plan.
MLB UL CCE Usage Threshold	**NRCellMlb**.***MlbUlCceUsageThld***	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Note: For details about how to configure the UE number threshold and offset for inter-frequency MLB, see the following: ● **NRCellMlb**.***InterFreqMlbUeNumThld***: Recommended value = Reference value of the UE number threshold (downlink) x 52/60 ● **NRCellMlb**.***MlbUeNumOffset***: Recommended value = Reference value of the UE number threshold (downlink) x 8/60 ● **NRCellMlb**.***InterFreqMlbUlUeNumThld***: Recommended value = Reference value of the UE number threshold (uplink) x 52/60 ● **NRCellMlb**.***MlbUlUeNumOffset***: Recommended value = Reference value of the UE number threshold (uplink) x 8/60 If the busy-hour PRB usage of every cell that is planned to be involved in MLB is greater than or equal to 60%: ● Reference value of the UE number threshold (downlink) = Number of UEs in the downlink in the cell when **Downlink Resource Block Utilizing Rate (DU)** is equal to 60% ● Reference value of the UE number threshold (uplink) = Number of UEs in the uplink in the cell when **Uplink Resource Block Utilizing Rate (DU)** is equal to 60% If the busy-hour PRB usage of the serving cell involved in MLB is always less than 60%, and there are other cells whose PRB usage is greater than or equal to 60%, then the cell with the highest PRB usage among the other cells is selected as the reference cell. ● Reference value of the UE number threshold (uplink/downlink) = Number of UEs in the reference cell when the PRB usage (uplink/downlink) is 60% x Air interface capability of the serving cell (uplink/downlink)/Air interface capability of the reference cell (uplink/downlink) Specifically: – Uplink air interface capability = **N.ThpVol.UL.Cell**/(**Uplink Resource Block Utilizing Rate (DU)**/100000) – Downlink air interface capability = **N.ThpVol.DL.Cell**/(**Downlink Resource Block Utilizing Rate (DU)**/100000) If the busy-hour PRB usage of every cell is less than 60%, the uplink and downlink UE number thresholds are set to their default values for the cell with the highest uplink and downlink bandwidths. The reference values of the UE number thresholds for cells with other bandwidths are calculated as follows: ● Reference value of the UE number threshold for another cell (uplink) = (Default value of the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUlUeNumOffset*** parameter) x (Uplink bandwidth of the cell/Maximum uplink cell bandwidth) ● Reference value of the UE number threshold for another cell (downlink) = (Default value of the **NRCellMlb**.***InterFreqMlbUeNumThld*** parameter + Default value of the **NRCellMlb**.***MlbUeNumOffset*** parameter) x (Downlink bandwidth of the cell/Maximum downlink cell bandwidth)			

8.4.1.2 Using MML Commands

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-1;
//Setting the cell capability scale factors and the following parameters: MlbTriggerMode and
LoadEvaluationPolicy
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, LoadEvaluationPolicy=DYNAMIC_EVAL,
CellDlCapbScaleFactor=1, CellUlCapbScaleFactor=1;
```

Optimization Command Examples

```
//Setting the MLB evaluation period for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the MlbDirection and MlbHoAdmissionThld parameters
MOD NRCELLMLB: NrCellId=0, MlbDirection=DOWNLINK, MlbHoAdmissionThld=95;
//Setting the InterFreqMlbUeNumThld, MlbUeNumOffset, and UeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUeNumThld=200, MlbUeNumOffset=50, UeNumDiffThld=25;
//Setting the InterFreqMlbUlUeNumThld, MlbUlUeNumOffset, and UlUeNumDiffThld parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20,
UlUeNumDiffThld=15;
//Setting relevant parameters to meet the conditions for the equivalent CCE usage to take effect:
UePdcchAdaptMaxSymNum, UE_PDCCH_SYM_NUM_ADAPT_SW, and DCI_EXTEND_UL_CONGEST_SW.
MOD NRDUCELLPDCCH: NrDuCellId=0, UePdcchAdaptMaxSymNum=3SYM,
PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-0;
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-0;
//Enabling load information exchange with the serving cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
//Disabling MLB for UEs with a specific RFSP
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
//Setting the MlbUeSelectionPolicy parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE,
MlbUeSelectionPolicy=SA_USER-1&NSA_USER-0;
//Setting the ArpThldBasedUeSelPolicyPri and MlbUeSelectionArpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, ArpThldBasedUeSelPolicyPri=210,
MlbUeSelectionArpThld=1;
//Setting the QciThldBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, QciThldBasedUeSelPolicyPri=220;
//Selecting the MLB_TRIG_BY_THLD option
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_TRIG_BY_THLD-1;
//Setting the MlbDetectTime parameter
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDetectTime=5;
//Setting the MlbDlBfrSize and MlbDlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbDlBfrSize=20, MlbDlThptThld=30000;
//Setting the MlbUlBsrDataSize and MlbUlThptThld parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbUlBsrDataSize=10, MlbUlThptThld=30000;
//Setting the QciSwBasedUeSelPolicyPri parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, QciSwBasedUeSelPolicyPri=230;
//Turning on MLB_FILTER_SW for the QCI without MLB involved
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, MlbAlgoSwitch=MLB_FILTER_SW-1;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Setting the PrbUsageThldUeSelPolicyPri parameter
```

```
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, PrbUsageThldUeSelPolicyPri=240;
//Setting the FarPointThldUeSelPolicyPri and MlbUeSelectionRsrpThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, FarPointThldUeSelPolicyPri=250,
MlbUeSelDistancePolicy=SEL_FAR_POINT_UE, MlbUeSelectionRsrpThld=-111;
//Setting the MlbUeSelDlPrbUsageThld and MlbUeSelUlPrbUsageThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, MlbUeSelDlPrbUsageThld=2,
MlbUeSelUlPrbUsageThld=2;
//Setting the MlbUlCceUsageThld and MlbDlCceUsageThld parameters
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE, MlbUlCceUsageThld=30,
MlbDlCceUsageThld=30;
//Setting the FreqSelectionPolicy parameter
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=RADIO_RESOURCE,FreqSelectionPolicy=FREQ_PRIORITY;
//Setting the InterFreqMeasEventConfig parameter for MLB
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A4;
//Setting the InterFreqMlbA4RsrpThld parameter
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqMlbA4RsrpThld=-102;
//Setting the InterFreqA4A5TimeToTrig and InterFreqA4A5Hyst parameters
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
//Setting the offsets to PRB thresholds for UE selection in MLB
MOD NRCELLMLB: NrCellId=0, MlbUeDlPrbHighThldOfs=30, MlbUeDlPrbLowThldOfs=10,
MlbUeUlPrbHighThldOfs=30, MlbUeUlPrbLowThldOfs=10;
//Setting the MLB frequency priority (based on the assumption that SsbFreqPos is set to 8000 in TDD
scenarios)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, MlbFreqPriority=1;
//Turning on ONLY_STRONGEST_CELL_SW
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=ONLY_STRONGEST_CELL_SW-1;
//Turning on INTER_FREQ_HO_SUPPRESS_SW for a specific QCI of UEs so that UEs will not be selected for
handover
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Setting MlbUeNumJudgeFactor
MOD NRCELLMLB: NrCellId=0, MlbUeNumJudgeFactor=50;
//Enabling precise load evaluation when the number of available PRBs in the cell dynamically changes
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoEnhSwitch=PRECISE_LOAD_EVAL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

After this function is enabled, if the cell load meets the conditions for triggering load redistribution described in **8.1.5 Load Redistribution**, the following are true:

- If the **SA_USER** option of the **NRCellMlb.*MlbUeSelectionPolicy*** parameter is selected and the **N.HO.InterFreq.Mlb.PrepAttOut** counter value is not 0, this function has taken effect.

- If the **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Att** counter value is not 0, this function has taken effect.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. If bit 0, bit 1, bit 4, or bit 5 of the binary number converted from the decimal value of **MLB Status** is **1**, radio-resource-based connected mode MLB has taken effect.

8.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful MLB-based inter-frequency outgoing handovers.

Observe the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counter to obtain the number of successful MLB-based inter-frequency PSCell changes in the LTE-NR NSA DC scenario.

Observe the **N.InterFreq.Mlb.Num** counter to obtain the number of times a cell enters the uplink or downlink MLB state.

Observe the **N.InterFreq.Mlb.Dur** counter to obtain the duration in which a cell is in the uplink or downlink MLB state.

9 Intra-Sector UE-Number-based Connected Mode MLB (High-Frequency TDD)

9.1 Principles

In intra-sector UE-number-based connected mode MLB, the number of UEs in connected mode in a cell is used to indicate the cell load. When the base station determines that the loads are unbalanced between intra-sector cells, it will hand over UEs in connected mode from a high-load cell to intra-sector low-load cells to achieve a load balance.

This function balances loads only in the uplink. To enable intra-sector UE-number-based connected mode MLB, perform the following operations:

- Select the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter.
- Set **NRCellMlb**.***MlbTriggerMode*** to **INTRA_SECTOR_CELL_USER_NUM**.
- Set **NRCellMlb**.***MlbDirection*** to **UPLINK**.

Once the preceding configurations have been completed, the setting of the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter determines the scenario in which MLB is supported.

- If this option is deselected, MLB is supported only when all of the following conditions are met:
  - There are multiple co-coverage cells in a sector. A typical scenario is intra-band CA.
  - The number of UEs in the sector is small (less than or equal to 20), and these UEs have the same uplink CA capabilities.
  - There are four cells in the sector, and all these cells have a bandwidth of 100 MHz (specified by the **NRDUCell.*DlBandwidth*** parameter).
  - The cells in the sector are numbered 1 to 4 in ascending order of their frequency numbers, and either of the following conditions is met:
    - Cell 1 and cell 4 are configured as normal cells, and cell 2 and cell 3 are configured as uplink and downlink supplementary cells.

- Cell 2 and cell 4 are configured as normal cells, and cell 1 and cell 3 are configured as uplink and downlink supplementary cells.

A cell is a normal cell if the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **NORMAL_CELL**, and is an uplink and downlink supplementary cell if this parameter is set to **ULDL_SUPPLEMENTARY_CELL**.

When all of these conditions are met, the number of CA UEs that treat a normal cell as their PCell can be transferred to achieve load balancing. Such UEs can be observed using the **N.User.CA.PCell.UL.Avg** counter. When all UEs support uplink 2CC CA, the numbers of uplink CA UEs are balanced among the four cells, as indicated by the sum of the **N.User.CA.PCell.UL.Avg** and **N.User.CA.SCell.UL.Avg** counter values.

- If this option is selected, inter-frequency MLB is supported for mmWave PCells. The gNodeB performs inter-cell MLB among all normal cells in the same sector based on the number of UEs in connected mode in each of these cells. A normal cell is one with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**.

It is recommended that intra-sector UE-number-based connected mode MLB be enabled when all of the following conditions are met:

- There are multiple co-coverage cells in the sector.
- The bandwidths of the cells in the sector are the same.
- The frequency priorities for the cells in the sector are the same.
- UEs are running the same types of services (typically, live streaming services and video surveillance services) in the sector.
- UEs are motionless and evenly distributed.
- The channel conditions of each UE are similar, and there is no weak coverage or strong interference.

9.1.1 Determining Whether to Start MLB

At the end of a configured MLB evaluation period, each cell checks whether the conditions for starting the MLB procedure are met. If so, the MLB procedure is started in the cell, as described in **9.1.2 MLB Procedure**. If not, the cell will check again at the end of the next MLB evaluation period. The conditions for starting the MLB procedure are as follows:

- New UEs access the cell in the MLB evaluation period.
- The number of UEs in connected mode in the sector meets the specific threshold.
  - When inter-frequency MLB for mmWave PCells is disabled, the number of UEs in connected mode in the sector is less than or equal to 20 throughout the period.
  - When inter-frequency MLB for mmWave PCells is enabled, the number of UEs in connected mode in the sector is greater than or equal to the value of MLB uplink UE number threshold and MLB uplink UE number offset throughout the period.

The involved parameters are as follows:

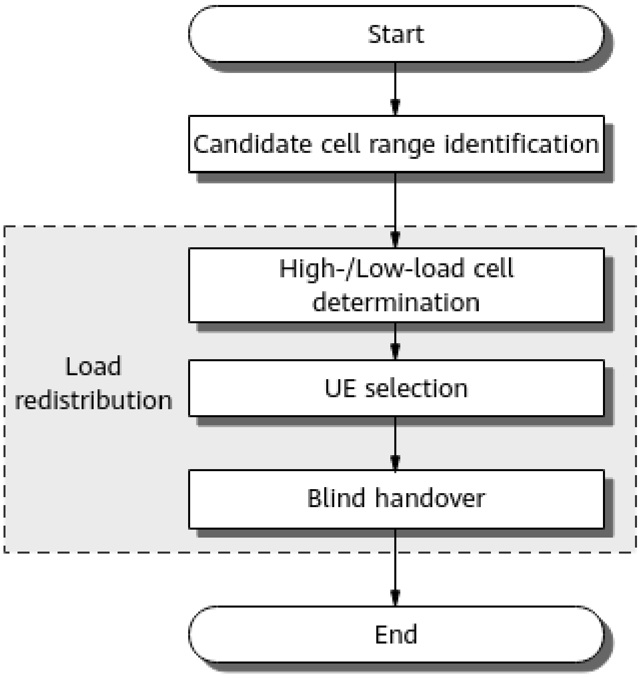
- The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.

- Inter-frequency MLB for mmWave PCells is controlled by the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.
- The MLB uplink UE number threshold is specified by the **NRCellMlb**.***InterFreqMlbUlUeNumThld*** parameter.
- The MLB uplink UE number offset is specified by the **NRCellMlb**.***MlbUlUeNumOffset*** parameter.

9.1.2 MLB Procedure

Figure 9-1 shows the MLB procedure.

Figure 9-1 Procedure for intra-sector UE-number-based connected mode MLB



Candidate Cell Range Identification

The gNodeB selects candidate cells for UE handovers. Any neighboring cell that meets all of the following conditions can be selected as a candidate cell:

- The cell is configured as a neighboring cell of the serving cell through the **NRCellRelation** MO.
- The cell is an inter-frequency cell of the serving cell.
- The cell is an intra-sector cell of the serving cell.
- The cell allows load information exchange. That is, the **NRCellRelation**.***MlbHoFlag*** parameter is set to **TRUE** for the cell.
- The cell allows incoming handovers. That is, the **NRCellRelation**.***NoHoFlag*** parameter is set to **PERMIT_HO** for the cell.
- Any of the following conditions is met:
  – The serving cell is an SA cell, the neighboring cell is an SA cell, and SA UEs can be handed over to the neighboring cell.

– The serving cell is an NSA cell, the neighboring cell is an NSA cell, and NSA UEs can be handed over to the neighboring cell.

– The serving cell is a hybrid NSA-SA cell, and UEs can be handed over to the neighboring cell.

NOTE

- The SA/NSA type of the serving cell and its neighboring cells is specified by the **NRDUCellOp**.***NrNetworkingOption*** parameter. A cell is considered as an SA or NSA cell only when the networking option of all operators of the cell is SA or NSA, respectively.
- The SA/NSA type of an external neighboring cell is specified by the **NRExternalNCell**.***NrNetworkingOption*** parameter. A cell is considered as an SA or NSA cell only when the networking option of all operators of the cell is SA or NSA, respectively.
- Whether a neighboring cell allows incoming handovers of SA UEs and NSA UEs is specified by the **NRCellRelation**.***NoHoFlag*** parameter.

- The cell is in the active state, which is indicated by the **NR Cell State** parameter in the **DSP NRCELL** command output.
- If the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter is selected for the local cell, only neighboring cells for which this option is selected can be selected as candidate cells.

Load Redistribution

In load redistribution, the base station selects some UEs in connected mode in the serving cell in the high-load state and hands over these UEs to intra-sector cells in the low-load state. The detailed process is as follows:

1. The gNodeB determines high- and low-load cells in a sector. Whether a cell is a high- or low-load cell is related to the setting of the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter.
   – If this option is deselected and the UE number difference between cells is greater than or equal to two, the gNodeB regards the cell with more UEs as a high-load cell and the cell with fewer UEs as a low-load cell.
   – If this option is selected and there are more UEs than the target UE number for MLB in a cell, the gNodeB regards the cell as a high-load cell. In all other cases, the gNodeB regards the cell as a low-load cell. The target UE number for MLB is the rounded-up average number of UEs in connected mode in the sector when the cell enters the MLB state.
2. The gNodeB selects UEs to be handed over from a high-load cell in the sector. Each of the selected UEs must meet all of the following conditions:
   – The UE has newly accessed the cell and has camped on the cell for a period of time.
   – The UE is in connected mode.
   – For SA UEs:
     ■ The **SA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter must be selected.

      - The UE treats the serving cell as its PCell.
      - The UE supports intra-FR2 inter-frequency handovers.
   – For NSA UEs:
      - The **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter must be selected.
      - The UE treats the serving cell as its PSCell.

   When inter-frequency MLB for mmWave PCells is enabled, the number of selected UEs must not exceed the maximum number of UEs allowed for load redistribution. If the number of UEs meeting the handover conditions exceeds this upper limit, the gNodeB randomly selects some of them. This upper limit takes the minimum value among the following three:
   – Total number of UEs that have newly accessed the cell and have camped on the cell for a period of time within the MLB evaluation period
   – Total number of UEs in the local cell minus the target UE number for MLB
   – UE number upper limit for connected mode MLB, which is specified by the **NRCellMlb**.***ConnMlbUeNumThld*** parameter
3. The selected UEs are handed over to low-load cells in the sector in a blind manner.

9.2 Network Analysis

9.2.1 Benefits

Intra-sector UE-number-based connected mode MLB increases the resource usage and capacity of cells involved in MLB. The higher the serving cell's load and the greater the inter-cell load difference, the more the gains.

If inter-frequency MLB for mmWave PCells takes effect for a sector for which intra-sector UE-number-based connected mode MLB does not take effect, the number of handovers increases, the load of cells in the same sector decreases, and the resource usage and capacity of the cells involved in MLB increases. These gains are directly proportional to the load of cells in the sector and the inter-cell load difference.

This function is not recommended in the following scenarios:

- eMBB scenarios, FWA scenarios, and other scenarios with a large number of UEs. In these scenarios, enabling this function brings no positive gains and may lead to unnecessary handovers, affecting user experience.
- The uplink traffic load is light and the QoS requirement of the uplink services is not high. Enabling this function in this scenario does not provide positive gains.
- UEs frequently access and exit the network. This function takes effect only for newly admitted UEs. After some online UEs are released, the remaining online UEs are not adjusted. In this case, a cell load balance cannot be achieved.

- UEs are running different types of services.
- UEs are moving and are located in an area with weak coverage or strong interference. If this function is enabled, the handover success rate decreases, the service drop rate increases, the RRC connection reestablishment rate increases, and user experience is affected.

9.2.2 Impacts

The impacts between this function and other functions are described in **9.3.2.3 Function Impacts**.

When a cell is configured as an uplink and downlink supplementary cell (by setting the **NRDUCell**.***SupplementaryCellInd*** parameter to **ULDL_SUPPLEMENTARY_CELL**), the uplink and downlink supplementary cell does not support SSB transmission and cannot be measured by UEs. UEs cannot camp on or access the cell, and therefore values of counters related to the number of online users, network access, mobility are all zero. For details about the uplink and downlink supplementary cell, see *Cell Management*.

UEs are redistributed among cells in a sector. As a result, the loads of some cells increase, and the loads of other cells decrease.

- In some cells, the number of online UEs and the number of activated UEs increase. In other cells, the number of online UEs and the number of activated UEs remain unchanged or decrease.
- The PRB usage of some cells increases, and the PRB usage of other cells remains unchanged or decreases.
- The traffic volume of some cells increases, and the traffic volume of other cells remains unchanged or decreases.
- The throughput of some cells increases, and the throughput of other cells remains unchanged or decreases.
- In general, the average values of the preceding indicators of multiple cells change (either increase or decrease).

When the load of a cell changes, the values of counters related to air interface scheduling also fluctuate, and the performance and experience of every UE also change. For example, when UEs in a large-bandwidth or light-load cell with a relatively large number of UEs are handed over to a small-bandwidth or heavy-load cell with a relatively small number of UEs, throughput of the redistributed UEs and other UEs in the target cell will decrease.

When UEs are redistributed from the serving cell to intra-sector inter-frequency neighboring cells, the UE throughput decreases and the bit error rate (BER) of the UEs increases.

- During UE redistribution in SA networking, the PCells of these UEs switch from the local cell to intra-sector inter-frequency neighboring cells through handovers.
  - The number of inter-frequency handovers (indicated by **N.HO.InterFreq.IntragNB.ExecAttOut**) increases.
  - The number of times random access occurred (indicated by **N.RA.Dedicated.Att**) increases.
  - When the channel conditions of the redistributed UEs are poor:

    - The inter-frequency handover success rate decreases, and the service drop rate and RRC connection reestablishment rate increase.
    - The random access success rate (indicated by the result of **N.RA.Dedicated.Att** divided by **N.RA.Dedicated.Resp**) decreases.

- During UE redistribution in NSA networking, the PSCells of these UEs switch from the local cell to intra-sector inter-frequency neighboring cells.
  - The number of PSCell changes (indicated by **N.NsaDc.IntraSgNB.PSCell.Change.Att**) increases.
  - The number of times random access occurred (indicated by **N.RA.Dedicated.Att** and **N.RA.Contention.Att**) increases.
  - When the channel conditions of the redistributed UEs are poor:
    - The PSCell change success rate (indicated by the result of **N.NsaDc.IntraSgNB.PSCell.Change.Succ** divided by **N.NsaDc.IntraSgNB.PSCell.Change.Att**) decreases, and the service drop rate increases.
    - The random access success rate (indicated by "**N.RA.Dedicated.Resp**/**N.RA.Dedicated.Att**" and "**N.RA.Contention.Resp**/**N.RA.Contention.Att**") decreases.

9.3 Requirements

9.3.1 Licenses

None

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	MLB UE selection policy	**NRCellMlb**.***MlbUeSelectionPolicy***	*Mobility Load Balancing in Connected Mode*	Intra-sector UE-number-based connected mode MLB can be enabled only when at least one of the **SA_USER** and **NSA_USER** options of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected.
High-frequency TDD	Intra-sector UE-number-based connected mode MLB	**NRCellMlb**.***MlbTriggerMode*** set to **INTRA_SECTOR_CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	If the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** is selected, the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter must be selected and the **NRCellMlb**.***MlbTriggerMode*** parameter must be set to **INTRA_SECTOR_CELL_USER_NUM**.

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Intra-sector UE-number-based connected mode MLB does not take effect for inter-FR CA UEs.
High-frequency TDD	Multi-operator sharing	**gNBSharingMode**.***gNBMultiOpSharingMode*** set to **SHARED_FREQ** or **SEPARATED_FREQ**	*Multi-Operator Sharing*	If a UE only supports certain operators in multi-operator sharing scenarios, the UE may fail to be handed over blindly during load redistribution.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations) and DBS5900 LampSite base stations

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For macro base stations, all of the NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

For LampSite base stations, the pRRU5551JR supports this function. For details, see *LampSite pRRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Networking

The **NRDUCell**.***SupplementaryCellInd*** parameter is set to **NORMAL_CELL** for two or more cells in a sector.

The downlink frequencies of all normal cells in the sector are configured in a CA frequency group, which is identified by **gNBCaFrequency**.***CaFreqGroupId***.

Frequency relationships of NR cells have been configured using the **NRCellFreqRelation** MO for all normal cells in the sector.

It is recommended that this function be enabled when all of the following conditions are met:

- There are multiple co-coverage cells in the sector.
- The bandwidths of the cells in the sector are the same.
- The frequency priorities for the cells in the sector are the same.
- UEs are running the same types of services (typically, live streaming services and video surveillance services) in the sector.
- UEs are motionless and evenly distributed.
- The channel conditions of each UE are similar, and there is no weak coverage or strong interference.

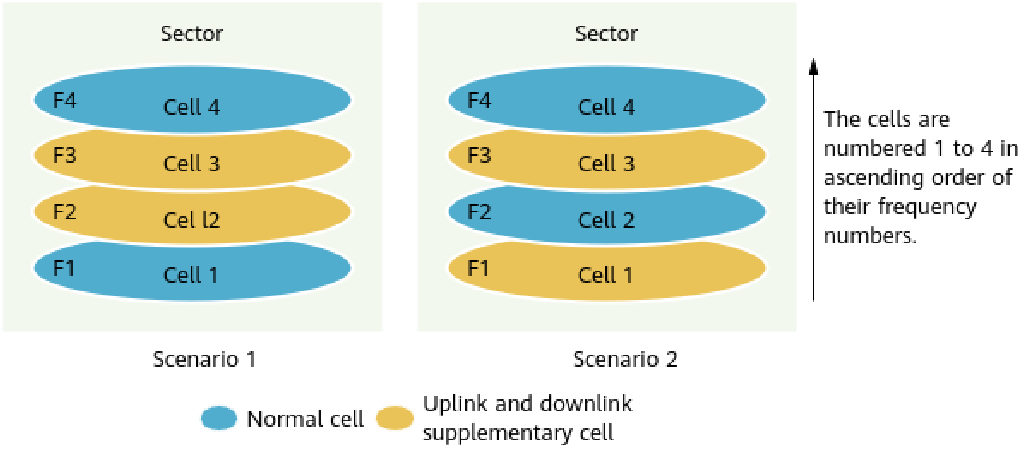
9.3.5 Cells

NRDUCell.***DuplexMode*** must be set to **CELL_TDD**.

When the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter is deselected, this function is supported only in either of the following scenarios, depending on the setting of the **NRDUCell**.***SupplementaryCellInd*** parameter for the supplementary cell of the cell. **Figure 9-2** illustrates these scenarios.

- Scenario 1: There are four cells in a sector, and all these cells have a bandwidth of 100 MHz (the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_100M**). The cells are numbered 1 to 4 in ascending order of their frequency numbers. The **NRDUCell**.***SupplementaryCellInd*** parameter is set to **NORMAL_CELL** for both cell 1 and cell 4 and to **ULDL_SUPPLEMENTARY_CELL** for both cell 2 and cell 3.
- Scenario 2: There are four cells in a sector, and all these cells have a bandwidth of 100 MHz (the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_100M**). The cells are numbered 1 to 4 in ascending order of their frequency numbers. The **NRDUCell**.***SupplementaryCellInd*** parameter is set to **NORMAL_CELL** for both cell 2 and cell 4 and to **ULDL_SUPPLEMENTARY_CELL** for both cell 1 and cell 3.

Figure 9-2 Scenarios supported by intra-sector UE-number-based connected mode MLB



9.3.6 Others

Intra-sector UE-number-based connected mode MLB does not take effect for cells in the energy saving state. You can run the **DSP NRDUCELLPOWERSAVING** command to check whether a cell is in the energy saving state. If the value of **Current State** for any **Power Saving Type** is **Enabled**, the cell is in the energy saving state.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch.** ***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Select this option.
MLB Algorithm Switch	**NRCellAlgoSwitch.** ***MlbAlgoSwitch***	INTER_FREQ_IDLE_MLB_SW	You are advised to deselect this option.

Parameter Name	Parameter ID	Option	Setting Notes
MLB Enhancement Algorithm Switch	**NRCellMlb**.***MlbAlgoEnhSwitch***	MMW_PCELL_INTER_FREQ_MLB_SW	Select this option if inter-frequency MLB for mmWave PCells is required.
MLB Trigger Mode	**NRCellMlb**.***MlbTriggerMode***	None	Set this parameter to **INTRA_SECTOR_CELL_USER_NUM**.
MLB Direction	**NRCellMlb**.***MlbDirection***	None	Set this parameter to **UPLINK**.
MLB Judge Period	**NRCellMlb**.***MlbJudgePeriod***	None	The recommended value of this parameter depends on the setting of the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb**.***MlbAlgoEnhSwitch*** parameter when intra-sector UE-number-based connected mode MLB is enabled. ● If the option is deselected, the recommended value is **1**. ● If the option is selected, the recommended value is **10**.
Downlink NARFCN	**gNBCaFrequency**.***DlNarfcn***	None	All downlink frequencies involved in intra-sector UE-number-based connected mode MLB must be included in the same group, which is identified by the **gNBCaFrequency**.***CaFreqGroupId*** parameter.

Parameter Name	Parameter ID	Option	Setting Notes
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	SA_USER	Set this parameter based on the network plan. Select this option in SA networking or NSA and SA hybrid networking.
MLB UE Selection Policy	**NRCellMlb**.***MlbUeSelectionPolicy***	NSA_USER	Set this parameter based on the network plan. Select this option in NSA networking or NSA and SA hybrid networking.
MLB-Based Handover Flag	**NRCellRelation**.***MlbHoFlag***	None	Set this parameter to **TRUE**.
No Handover Flag	**NRCellRelation**.***NoHoFlag***	None	Set this parameter to **PERMIT_HO**.
Inter-frequency MLB UL UE Number Thld	**NRCellMlb**.***InterFreqMlbUlUeNumThld***	None	Set this parameter based on the network plan.
MLB UL UE Number Offset	**NRCellMlb**.***MlbUlUeNumOffset***	None	Set this parameter based on the network plan.
Connected MLB UE Num Upper Limit	**NRCellMlb**.***ConnMlbUeNumThld***	None	Set this parameter based on the network plan.

NOTE

Every parameter listed in **Table 9-1** must be set to the same value for all cells involved in this function. If the parameter settings are inconsistent, intra-sector UE-number-based connected mode MLB may fail to provide the expected gains.

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the downlink frequency involved in intra-sector UE-number-based connected mode MLB
ADD GNBCAFREQUENCY: CaFreqGroupId=0, DlNarfcn=2055833, SsbFreqPos=22389;
//Setting the MLB algorithm switch for a cell
MOD NRCELLALGOSWITCH: NrCellId=0,
MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-1&INTER_FREQ_IDLE_MLB_SW-0;
//Setting the MlbTriggerMode, MlbDirection, and MlbUeSelectionPolicy parameters for the cell
MOD NRCELLMLB: NrCellId=0, MlbTriggerMode=INTRA_SECTOR_CELL_USER_NUM, MlbDirection=UPLINK,
MlbUeSelectionPolicy=SA_USER-1&NSA_USER-1;
//Turning on the mmWave PCell inter-frequency MLB switch
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=MMW_PCELL_INTER_FREQ_MLB_SW-1;
//Setting the MLB evaluation period for the cell (using the value 10 as an example)
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the InterFreqMlbUlUeNumThld and MlbUlUeNumOffset parameters
MOD NRCELLMLB: NrCellId=0, InterFreqMlbUlUeNumThld=100, MlbUlUeNumOffset=20;
//Setting the ConnMlbUeNumThld parameter
MOD NRCELLMLB: NrCellId=0, ConnMlbUeNumThld=5;
//Enabling load information exchange with the serving cell for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, MlbHoFlag=TRUE,
NoHoFlag=PERMIT_HO;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the mmWave PCell inter-frequency MLB switch
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=MMW_PCELL_INTER_FREQ_MLB_SW-0;
//Turning off the MLB algorithm switch for the cell
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Assume that the cell load meets the conditions for triggering load redistribution described in **9.1.2 MLB Procedure** after this function is enabled:

- In SA networking or NSA and SA hybrid networking, if the **SA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and the **N.HO.InterFreq.Mlb.PrepAttOut** counter value is not 0, this function has taken effect.
- In NSA networking or NSA and SA hybrid networking, if the **NSA_USER** option of the **NRCellMlb**.***MlbUeSelectionPolicy*** parameter is selected and

the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Att** counter value is not 0, this function has taken effect.

9.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful outgoing MLB-based inter-frequency handovers in SA scenarios.

Observe the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counter to obtain the number of successful MLB-based inter-frequency PSCell changes in the NSA DC scenario.

10 UE-Number-based MLB in Frequency Rotation Mode

10.1 Principles

During handovers or blind redirections from low-frequency (FR1) cells to high-frequency (FR2) cells, if one FR2 cell in a sector is selected as the target cell for most UEs, the load of this cell is higher than that of other cells in the sector. As a result, load imbalance occurs between the cells. UE-number-based MLB in frequency rotation mode selects different frequencies in rotation mode for UEs to perform handovers or blind redirections. In this way, UEs are evenly distributed among cells in the same sector after they are transferred from a low-frequency cell to high-frequency cells, achieving inter-cell load balancing.

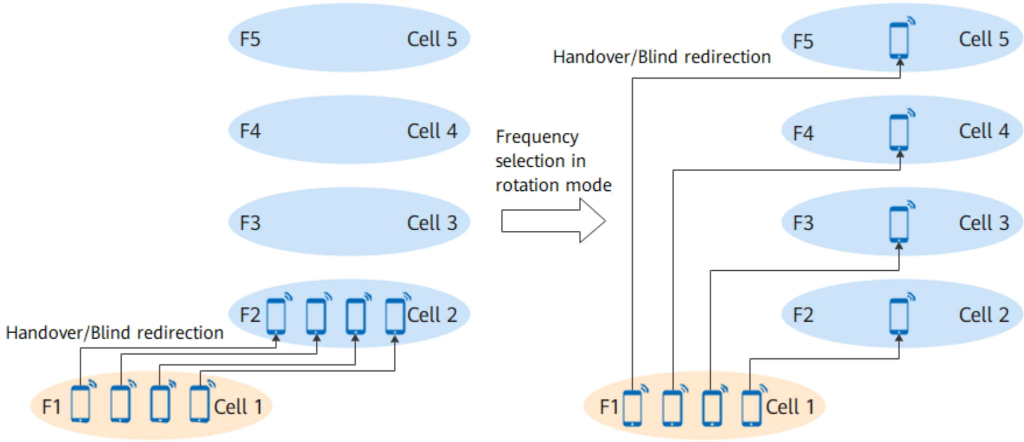
UE-number-based MLB in frequency rotation mode is enabled if the **FR1_TO_FR2_FREQ_RR** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected for an FR1 cell. It is recommended that this function be enabled when the number of FR1-to-FR2 handovers or blind redirections is large. After this function is enabled, when a UE is transferred to a common high-frequency cell (whose **NRDUCell**.***SupplementaryCellInd*** is set to **NORMAL_CELL**), this function selects frequencies in rotation mode.

- During an FR1-to-FR2 handover for a UE, the gNodeB selects a frequency randomly among all neighboring FR2 frequencies with the same priority and delivers this frequency in the measurement parameter configurations for handovers to the UE.
- During an FR1-to-FR2 blind redirection for a UE, the gNodeB selects a frequency randomly among all neighboring FR2 frequencies with the same priority to perform blind redirection.

This function supports only MLB through handovers in NSA networking but not MLB through blind redirections.

Figure 10-1 shows the principles of UE-number-based MLB in frequency rotation mode. F1 indicates the frequency of the FR1 cell, and F2, F3, F4, and F5 indicate the frequencies of FR2 cells.

Figure 10-1 Diagram of UE-number-based MLB in frequency rotation mode



NOTE

- If the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected, cells already in the state of UE-number-based MLB in frequency rotation mode will not be selected as target cells for non-MLB-based inter-frequency handovers. For details, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.
- After UE-number-based MLB in frequency rotation mode is enabled, FR2 frequencies that meet any of the following conditions cannot be measured in inter-frequency measurements, cannot be selected as handover target frequencies, and cannot be configured as neighboring frequencies through the **NRCellFreqRelation** MOs.
  - No cell operates on the frequency.
  - All cells on the frequency cannot provide services.
  - All cells on that frequency are not configured with SSBs.

10.2 Network Analysis

10.2.1 Benefits

UE-number-based MLB in frequency rotation mode balances the number of UEs in connected mode among intra-sector common cells (cells with the **NRDUCell**.***SupplementaryCellInd*** parameter set to **NORMAL_CELL**), thereby balancing the resource usage among these cells. The more the number of UEs involved in FR1-to-FR2 handovers or blind redirections, the greater the preceding gains.

10.2.2 Impacts

The impacts between this function and other functions are described in **10.3.2.3 Function Impacts**.

After UE-number-based MLB in frequency rotation mode is enabled, the values of the following counters for intra-sector FR2 cells will become relatively balanced.

Table 10-1 Counters for intra-sector FR2 cells

Counter ID	Counter Name
1911816771	**N.User.RRCConn.Avg**
1911816773	**N.User.CA.PCell.DL.Avg**
1911816775	**N.User.CA.PCell.UL.Avg**

When high-risk commands such as deactivation and removal are executed in some cells in an FR2 sector, these cells cannot provide services. When UEs are in the coverage of the sector, the following are true after this function is enabled:

- If the gNodeB selects the operating frequencies of these cells, the UEs cannot measure signals of these cells. Therefore, handovers cannot be initiated for the UEs and the UEs cannot be handed over other intra-sector cells that are properly working, either.
- If the gNodeB selects the operating frequencies of these cells, the UEs cannot be transferred to these cells, as well as other intra-sector cells that are properly working, through blind redirections.

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	Connected mode MLB (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) cannot be enabled for cells deployed in IAB mode.

10.3.2.3 Function Impacts

None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter LTE: **NSA_LTE_FDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	After the "NSA carrier combination selection based on user experience" function takes effect, UE-number-based MLB in frequency rotation mode does not take effect for NSA UEs in the cell.

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

10.3.4 Networking

Neighbor relationships with high-frequency cells have been configured for low-frequency cells and at least two high frequencies are configured.

The low-frequency cell is configured with neighboring NR frequencies through the **NRCellFreqRelation** MOs, and these frequencies include high frequencies to which handovers and blind redirections are allowed.

10.3.5 Others

None

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Table 10-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 10-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Inter-Frequency Handover Switch	**NRCellAlgoSwitch.*InterFreqHoSwitch***	FR1_TO_FR2_FREQ_RR	Select this option for a low-frequency cell if neighbor relationships with high-frequency cells of the low-frequency cell are configured.

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on FR1_TO_FR2_FREQ_RR
MOD NRCELLALGOSWITCH: NrCellId=7, InterFreqHoSwitch=FR1_TO_FR2_FREQ_RR-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off FR1_TO_FR2_FREQ_RR
MOD NRCELLALGOSWITCH: NrCellId=7, InterFreqHoSwitch=FR1_TO_FR2_FREQ_RR-0;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

After UE-number-based MLB in frequency rotation mode is enabled and FR1-to-FR2 handovers are triggered, if the values of the following counters become relatively balanced for intra-sector FR2 cells, this function has taken effect.

Table 10-3 MLB-related counters for FR2 cells

Counter ID	Counter Name
1911816771	**N.User.RRCConn.Avg**
1911816773	**N.User.CA.PCell.DL.Avg**
1911816775	**N.User.CA.PCell.UL.Avg**

10.4.3 Network Monitoring

Observe the value of the **N.User.RRCConn.Avg** counter to check whether the distribution of UEs in connected mode in the cells on different frequencies meets the expectation of MLB.

11 Inter-Vendor Connected Mode MLB

11.1 Principles

During inter-base-station MLB, inter-vendor connected mode MLB is required when one of the base stations is provided by another vendor. Inter-vendor connected mode MLB is controlled by the **INTER_VENDOR_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter. Before enabling inter-vendor connected mode MLB, intra-vendor connected mode MLB must have been enabled.

Assume that UE-number-based connected mode MLB is enabled (by selecting the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter and setting the **NRCellMlb**.***MlbTriggerMode*** parameter to **CELL_USER_NUM**). If inter-vendor connected mode MLB is enabled, inter-vendor UE-number-based connected mode MLB is enabled along with it.

It is recommended that inter-vendor connected mode MLB be enabled when all of the following conditions are met:

- The same proportions of small-packet and large-packet services are independently running in inter-frequency co-coverage cells.
- Most UEs are motionless or move slowly.
- An Xn interface has been set up between the base stations, one of which is not provided by Huawei.
- Inter-frequency cells involved in MLB are configured with the same bandwidth.

After inter-vendor connected mode MLB is enabled, if a UE is handed over to a cell served by a base station of another vendor, this function has taken effect.

11.1.1 Entering and Exiting MLB

After inter-vendor UE-number-based connected mode MLB is enabled, the conditions for entering and exiting MLB are the same as those for intra-vendor UE-number-based connected mode MLB. For details, see **5.1.1 Entering and Exiting MLB**.

The MLB evaluation period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter. The MLB direction is specified by the **NRCellMlb**.***MlbDirection*** parameter. Downlink MLB, uplink MLB, as well as downlink and uplink MLB are supported. If the gNodeB determines that a cell meets the conditions for entering or exiting MLB in the current MLB evaluation period, the cell enters or exits MLB in the next MLB evaluation period.

11.1.2 Candidate Cell Range Identification

The gNodeB selects candidate cells for UE handovers. A candidate cell must meet the following conditions:

- For intra-vendor neighboring cells: The conditions for candidate cells are the same as those for intra-vendor UE-number-based connected mode MLB. For details, see **5.1.2 Candidate Cell Range Identification**.
- For inter-vendor neighboring cells: In addition to the conditions for intra-vendor neighboring cells, the remaining capacity of RRC_CONNECTED UEs in inter-vendor neighboring cells is checked. An inter-vendor neighboring cell can be selected as a candidate cell only when the remaining capacity is not 0%.

11.1.3 Inter-Cell Load Information Exchange

The serving cell exchanges load information with candidate cells. The gNodeB triggers intra-base-station load information exchanges between the serving cell and the candidate cells by default. For details, see **5.1.3.1 Intra-gNodeB Inter-Cell Load Information Exchange**.

If there are candidate inter-gNodeB cells, the gNodeB triggers inter-gNodeB inter-cell load information exchanges. The exchange procedure is performed using the method defined in section 8.4.10 "Resource Status Reporting Initiation" in 3GPP TS 38.423 V16.4.0. The exchanged information is as follows:

- For intra-vendor neighboring cells, the exchanged information is as described in **5.1.3.2 Inter-gNodeB Inter-Cell Load Information Exchange**.
- For inter-vendor neighboring cells, the exchanged information is as follows:
  - The maximum number of RRC_CONNECTED UEs supported in a cell
  - The remaining capacity of RRC_CONNECTED UEs in a cell

  When a cell served by a Huawei base station reports load information to a neighboring cell served by another vendor, the principles are as follows:
  - The maximum number of RRC_CONNECTED UEs supported in the cell is fixed to two.
  - The remaining capacity of RRC_CONNECTED UEs in the cell served by a Huawei base station is 0% if the cell is in the MLB state. Otherwise, the remaining capacity is 100%.

11.1.4 Candidate Cell Selection

The gNodeB compares the load information of the serving cell with that of all candidate cells.

- For intra-vendor neighboring cells: The conditions must be met are described in **5.1.4 Candidate Cell Selection**.

- For inter-vendor neighboring cell: If the remaining capacity of RRC_CONNECTED UEs in an inter-vendor neighboring cell is not 0%, the inter-vendor neighboring cell can be selected as a candidate cell.

11.1.5 Load Redistribution

Load redistribution is a process in which some UEs in connected mode in the serving cell are handed over to candidate cells that meet certain signal quality requirements.

11.1.5.1 UE Selection

The procedure for selecting a UE to be handed over is as follows:

1. According to basic conditions that UEs must meet, initial selection is performed. The procedure for inter-vendor connected mode MLB is the same as that for intra-vendor UE-number-based connected mode MLB. For details about the basic conditions that UEs must meet, see **5.1.5.1.1 Conditions Used for Initial Selection**.
2. The priority of a UE is determined according to the UE selection policies based on radio bearer attributes. The procedure for inter-vendor connected mode MLB is the same as that for intra-vendor UE-number-based connected mode MLB. For details about UE priorities, see **5.1.5.1.2 UE Priority Determination**.
3. The calculation for the upper limit (N_{max}) of the total number of UEs involved in load redistribution within each MLB evaluation period is basically the same as that for UE-number-based connected mode MLB. For details about the calculation, see **5.1.5.1.3 Upper Limit Calculation for the Total Number of UEs Involved in Load Redistribution**.

   The difference is that if there are inter-vendor neighboring cells in the candidate cells, during the calculation of N_{max}, the minimum number of UEs in the downlink in the candidate cell and minimum number of UEs in the uplink in the candidate cell are fixed to zero.

   NOTE

   In inter-vendor connected mode MLB, during the calculation for the upper limit of the total number of UEs involved in load redistribution, the minimum number of UEs in the downlink in the candidate cell and minimum number of UEs in the uplink in the candidate cell are fixed to zero. As a result, the calculation result may be larger than expected, causing too many UEs to be transferred from the serving cell.
4. The initially-selected UEs are then selected in descending order of UE priorities for measurement-based handovers, and the number of selected UEs cannot exceed N_{max}. If UEs have the same priority, the gNodeB randomly selects one from them.

11.1.5.2 Measurement-based Handover

Measurement-based handovers are performed for a selected UE. The procedure for inter-vendor connected mode MLB is the same as that for intra-vendor UE-number-based connected mode MLB. For details, see **5.1.5.2 Measurement-based Handover**.

NOTE

If the target cell is an inter-vendor neighboring cell, the configuration of the inter-vendor neighboring cell may cause incoming handover failures during target cell determination.

11.2 Network Analysis

11.2.1 Benefits

The benefits of inter-vendor connected mode MLB is the same as those of intra-vendor UE-number-based connected mode MLB. For details, see **5.2.1 Benefits**.

After inter-vendor connected mode MLB is enabled, UEs in the serving cell can be handed over to inter-vendor base stations. This adds paths for data split by MLB. As a result, the values of the **N.HO.InterFreq.Mlb.ExecSuccOut** and **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counters may increase, and the values of the **N.InterFreq.Mlb.Num** and **N.InterFreq.Mlb.Dur** counters may fluctuate.

11.2.2 Impacts

The impacts between this function and other functions are described in **11.3.2.3 Function Impacts**.

The network impacts of inter-vendor connected mode MLB are the same as those of intra-vendor UE-number-based connected mode MLB. For details, see **5.2.2 Impacts**.

11.3 Requirements

11.3.1 Licenses

None

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	If inter-vendor UE-number-based connected mode MLB is enabled (by selecting the **INTER_VENDOR_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter), the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter must be selected and the **NRCellMlb**.***MlbTriggerMode*** parameter must be set to **CELL_USER_NUM**.

11.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions of inter-vendor connected mode MLB are the same as those of intra-vendor UE-number-based connected mode MLB. For details, see **5.3.2.2 Mutually Exclusive Functions**.

11.3.2.3 Function Impacts

The function impacts of inter-vendor connected mode MLB are the same as those of intra-vendor UE-number-based connected mode MLB. For details, see **5.3.2.3 Function Impacts**.

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

11.3.4 Networking

Inter-vendor connected mode MLB requires that an Xn interface has been set up between the base stations and one of them is not provided by Huawei.

The networking requirements for inter-vendor connected mode MLB are the same as those for intra-vendor UE-number-based connected mode MLB. For details, see **5.3.4 Networking**.

11.3.5 Others

None

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

Table 11-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 11-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
MLB Algorithm Switch	**NRCellAlgoSwitch.** ***MlbAlgoSwitch***	INTER_VENDOR_MLB_SW	It is recommended that this option be selected when there are inter-vendor neighboring cells.

Other parameters used for inter-vendor connected mode MLB are the same as those for intra-vendor UE-number-based connected mode MLB. For details, see **5.4.1.1 Data Preparation**.

11.4.1.2 Using MML Commands

Before using MML commands, refer to **11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling inter-vendor connected mode MLB when there are neighboring cells served by non-Huawei base stations
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_VENDOR_MLB_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inter-vendor connected mode MLB
MOD NRCELLALGOSWITCH: NrCellId=0, MlbAlgoSwitch=INTER_VENDOR_MLB_SW-0;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

Activation verification for inter-vendor connected mode MLB is the same as that for intra-vendor UE-number-based connected mode MLB. For details, see **5.4.2 Activation Verification**.

11.4.3 Network Monitoring

Observe the **N.HO.InterFreq.Mlb.ExecSuccOut** counter to obtain the number of successful MLB-based inter-frequency outgoing handovers.

Observe the **N.NsaDc.InterFreq.Mlb.PSCell.Change.Succ** counter to obtain the number of successful MLB-based inter-frequency PSCell changes in the LTE-NR NSA DC scenario.

Observe the **N.InterFreq.Mlb.Num** counter to obtain the number of times a cell enters the uplink or downlink MLB state.

Observe the **N.InterFreq.Mlb.Dur** counter to obtain the duration in which a cell is in the uplink or downlink MLB state.

On the MAE-Access, start a signaling trace management task with the parameters for **MLB Monitoring** under **Cell Performance Monitoring** specified to monitor MLB. Observe **MLB HO Out Succ User Num** to obtain the number of UEs that are successfully handed over from a cell.

12 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

13 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

14 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

15 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.423: "NG-RAN; Xn Application Protocol (XnAP)"
- 3GPP TS 38.314: "NR; Layer 2 measurements"
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *VoNR*
- *Super Uplink*
- *UL and DL Decoupling*
- *EPC-based NSA Experience Improvement*
- *Hyper Cell*
- *Multi-Operator Sharing*
- *Cell Combination*
- *Mobility Load Balancing in Idle Mode*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *NSA Mobility Management*
- *ANR*
- *Network Slicing*
- *URLLC*
- *Multi-Frequency Smart Selection*
- *Cell Management*
- *Energy Conservation and Emission Reduction*
- *Uplink Scheduling*
- *NG and Xn Self-Management*
- *License Control Item Lists*
- *RedCap*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*